

Bose condensation in gluon plasma

Thesis presented for the degree of
Master of Science in Theoretical Physics
in the Department of Physics
University of Cape Town
2025

by Gary G. Louw
supervised by André Peshier

Declaration of Authorship

I, Gary Grant Louw, declare that this thesis is my own work. Contributions originating from other people have been cited and referenced.

Signature:

Date: 13 May 2025

Abstract

To explore the possibility of a gluon condensate forming in heavy ion collisions, we study the quantum relativistic Boltzmann equation with glasma-inspired initial conditions. We focus on the standard case of a homogeneous and isotropic gluon plasma. Generalising an existing numerical approach for binary interactions in the scalar ϕ^4 -theory to QCD, we develop an efficient framework to estimate equilibration and condensation timescales (to be compared to those for number-changing processes due to inelastic interactions). Our scheme improves on existing test particle simulations and Fokker-Planck approaches based on the small-scattering-angle approximation of the Boltzmann equation. By comparing the latter to our Boltzmann model allows the specification of an auxiliary UV cut-off which hitherto was known only parametrically. Uncertainties are thus reduced to details of QCD screening, which we parametrise with a screening mass and investigate with two different regularisations of the square amplitude. Preliminary results suggest that gluon condensation in heavy ion collisions is less likely than expected from Fokker-Planck approaches, which appear to underestimate equilibration and condensation timescales.

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Introduction

The Bose-Einstein phase transition was first described in 1924 by Albert Einstein (with credit to prior calculations by Satyendra Nath Bose). A Bose condensate was first produced in a laboratory setting in 1995 when Eric Cornell and Carl Wiemann of the Joint Institute of Laboratory Astrophysics cooled a sample of rubidium gas near to absolute zero [1]. Below a threshold temperature, they observed a macroscopic occupation of the gas' ground state. Later that year, Wolfgang Ketterle independently condensed sodium atoms at low temperature [2].

Practical demonstrations have continued to successfully prepare Bose condensates at very low temperature. However, condensates may also exist in gases with large energy densities (provided the system is suitably dense). The gluonic subsystem of the quark-gluon plasma (QGP) is such a candidate; the QGP has very high equilibrium temperature (roughly one trillion kelvin) and may foster the development of a gluon condensate in the early stages of equilibration [3].

During the moments before a relativistic heavy ion collision, the participating hadrons are strongly Lorentz contracted along the collision axis. In the centre-of-momentum frame, the collision partners appear as highly relativistic 'sheets' of colour-charged fields. The state of the system at this time is described by the Colour Glass Condensate (CGC) effective field theory [4]. These incoming sheets of 'colour glass' then interpenetrate one another and form a dense, gluon-dominated state of matter called 'glasma'. If inelastic processes are negligible, a gluon condensate may form temporarily during the evolution from 'glasma-like' initial conditions towards a thermalised QGP.

In this dissertation, we study an idealised model of the gluonic subsystem and assess the possibility of condensation. The framework of *kinetic theory* is used to study the evolution of homogeneous and isotropic gluon plasmas undergoing elastic collisions. Quarks are neglected for simplicity. The system's evolution is then understood by solving the quantum relativistic Boltzmann equation. We also compare our approach to the widely-used 'soft-scattering' limit of the evolution equation, which has the structure of a Fokker-Planck equation.

A central question in both models is *how quickly* condensation occurs (if at all). Inelastic processes tend to reduce the number density of the, initially, very dense glasma. If the elastic collisions studied in our model produce a condensate more quickly than inelastic collisions depopulate the low-momentum modes, then we can motivate the existence of a transient condensate during equilibration. To this end, we calculate equilibration and condensation timescales as the central observables of this thesis.

The QGP is a topic of major interest in modern physics research. Whether gluon condensation occurs in the QGP is a significant question. In particle physics, the presence of a gluon condensate during the QGP's thermalisation may explain the abundance of certain decay products detected in collider experiments [5]. In the field of astrophysics, gluon condensates may influence the internal dynamics of ultra-compact objects like neutron stars [6].

Core assumptions

We study gluon plasmas whose interactions are described by a tree-level improved square amplitude, where small t and u divergences are regulated using a screening mass. The energy of the plasma is approximated as the sum of the one-particle energies of the gluons. To simplify the structure of the evolution equation, we assume the system is homogeneous and isotropic. Inelastic processes and processes coupling to quarks are neglected. The gluons are described with positive kinematic mass. We consider only moderate values for the coupling $0.1 \lesssim \alpha \lesssim 0.5$.

Notation and conventions

We use natural units, $\hbar = c = k_B = 1$, where these constants are the reduced Planck constant, the speed of light in vacuum, and Boltzmann's constant (respectively). The gluon degeneracy factor, $d_g = 2(N_c^2 - 1)$ where $N_c = 3$ is the number of colours, is ignored but may be easily re-inserted at the ends of calculations. The strong coupling is symbolised with α .

4-Vectors are written in uppercase while 3-vectors are written in lowercase and bold. For example, a 4-momentum K has spatial components $\mathbf{k}$. The invariant product of two 4-vectors A and B is written like $A \cdot B$. The distinction between $A \cdot B$ and $\mathbf{a} \cdot \mathbf{b}$ should be obvious. The metric tensor used to define the invariant product of two 4-vectors is the 'mostly-minus' Minkowski metric for flat spacetime

$$\eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}_{\mu\nu}.$$

The 4-momentum of a particle is usually represented with a K , where the corresponding 3-momentum is $\mathbf{k}$. The particle's energy is then $E_k = \sqrt{\mathbf{k}^2 + m^2}$ where m is the *mass* of the particle. The norms of 3-momenta are abbreviated $|\mathbf{k}| = k$.

Like discussed in the [introduction](#), the scattering events studied are strictly $2 \rightarrow 2$. The labelling convention for incoming and outgoing momenta is shown in Figure 1.

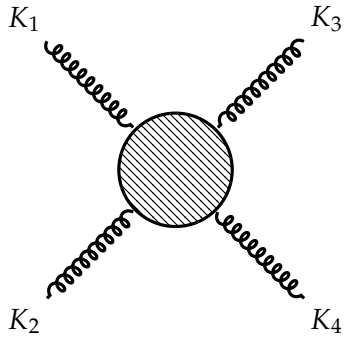


Figure 1: The incoming momenta are given indices 1 and 2; the outgoing momenta are given indices 3 and 4. The blob obscures details of the interaction for generality.

The square amplitude for collisions is averaged over the spin and colour for particles 2, 3 and 4 and summed over spin and colour for particle 1. The Mandelstam variables are defined

$$s = (K_1 + K_2)^2 = (K_3 + K_4)^2,$$

$$t = (K_1 - K_3)^2 = (K_2 - K_4)^2,$$

$$u = (K_1 - K_4)^2 = (K_2 - K_3)^2.$$

Occasionally, functions are written with suppressed arguments to allow for more compact expression of equations. For example, a function $f(k)$ may be written as just f when appearing inside a momentum integral: $\int_k f(k) \mapsto \int_k f$.

1 Kinetic theory

The starting point of this thesis is the Boltzmann equation; an evolution equation which describes the transport of particles (in a many-particle system; e.g. a plasma) in phase space.

1.1 The basic structure

The most basic object which features in the Boltzmann equation is the *one-particle distribution function* (or simply, “the distribution function”) which we commonly denote with f . The distribution function measures the mean density of particles inside an infinitesimal phase space cell of volume $d^3x d^3k / (2\pi)^3$ which is centred at coordinates (x, k) at time t ,

$$f(x, k; t) = (2\pi)^3 \frac{dN(x, k; t)}{d^3x d^3k}. \quad (1.1)$$

The distribution function can be defined, in more fundamental terms, as a scaled marginal distribution of the density of pure states in $6N$ -dimensional phase space [7]. In the same framework, one can develop a hierarchy of equations which describe the time-derivatives of the one-particle distribution function (1.1) and its multi-particle analogues. These equations are known as the Bogoliubov-Born-Green-Kirkwood-Yvon (BBGKY) hierarchy. By neglecting two-particle correlations, the hierarchy is truncated and yields an evolution equation for the distribution function, which has the form of a transport equation; the Boltzmann equation. Instead of reproducing the 152 year-old calculation here, we just share the result,

$$\mathcal{D}_t f = C_{[f]}. \quad (1.2)$$

(1.2) is a compact expression of the Boltzmann equation. Featured on the lefthand side is the total time derivative of f (sometimes called the *convective* derivative). Simple calculus dictates that

$$\mathcal{D}_t = \partial_t + v \cdot \text{grad}_x + F \cdot \text{grad}_k,$$

where v is the flow velocity, F is any external force acting on the plasma, grad_x is a position-gradient and grad_k is a momentum-gradient. For simplicity, we study a plasma which is homogeneous and which isn't subject to external forces; starting with these assumptions, the position- and momentum-gradients fall out of the convective derivative which leaves only a partial derivative with respect to time. Assuming in addition momentum isotropy, the distribution function will depend only on the *norm* of the momentum, $f(k; t) \mapsto f(k; t)$.

On the righthand side of (1.2) is the so-called collision term, $C_{[f]}$. Without the collision term, (1.2) implies that (like the pure state density) the distribution function evolves like an incompressible fluid. That is, $\mathcal{D}_t f = 0$ describes only *free-streaming* whilst the collision term $C_{[f]}$ accounts for inter-particle interactions (i.e. a redistribution of particle momenta around position x). Because, in our case, the streaming terms are zero, the principal equation of this thesis may be written

$$\partial_t f = C_{[f]}. \quad (1.3)$$

If the distribution function is stationary (i.e. $\partial_t f = 0$) the collision term must vanish in (1.3). However, the collision term being zero doesn't imply that the gas is collision-less; the rates of scattering into and out of each mode may be non-zero but balanced. How such an arrangement may arise is detailed in the [section 1.2](#).

1.2 The collision term

Boltzmann's original collision term was constructed to describe dilute gases at low energies. Here, we use a generalised form which incorporates quantum and relativistic effects. To lighten notation, the invariant momentum integral for the i^{th} on-shell particle is abbreviated like

$$\int_i = \int \frac{d^3 k_i}{(2\pi)^3} \frac{1}{2E_i}.$$

The integral over two-body phase space is then defined

$$\int d\Gamma = \int_{234} (2\pi)^4 \delta^{(4)}(K_1 + K_2 - K_3 - K_4), \quad (1.4)$$

where the δ -distribution encodes 4-momentum conservation at the microscopic level. As described in [section 1.3](#), energy-momentum conservation at the level of collisions implies conservation for the whole system. The quantum-relativistic Boltzmann collision term is compactly expressed

$$C_{[f]}(E_1) = \frac{1}{4E_1} \int d\Gamma |\mathcal{M}|^2 \mathcal{F}_{[f]}, \quad (1.5)$$

where $|\mathcal{M}|^2$ is the square-amplitude for the scattering process under consideration and

$$\mathcal{F}_{[f]} = \bar{f}_1 \bar{f}_2 f_3 f_4 - f_1 f_2 \bar{f}_3 \bar{f}_4, \quad (1.6)$$

where $f_i = f(k_i)$ and $\bar{f} = 1 + f$. $\mathcal{F}_{[f]}$ is a statistical factor which quantifies the occupation of in- and outgoing momentum states in a $2 \rightarrow 2$ interaction. The statistical factor is a difference between a gain term, $\bar{f}_1 \bar{f}_2 f_3 f_4$, and a loss term, $f_1 f_2 \bar{f}_3 \bar{f}_4$. Looking at the latter, $f_1 f_2$ is the product of the (uncorrelated) momentum densities of the in-states. Extra factors of $\bar{f}_3 \bar{f}_4$ account for Bose enhancement of the out-states. The gain term has the same structure, but represents the reverse process $k_3, k_4 \rightarrow k_1, k_2$.

When the gas is dilute ($f \ll 1$) the leading contribution to the statistical factor are Boltzmann's classical gain and loss terms, $f_3 f_4 - f_1 f_2$. Hence, Boltzmann's original collision integral is obtainable as a dilute limit of (1.5). The classical gain and loss terms are products of the (uncorrelated) densities of the in-states and are independent of the densities of the outgoing states. If quantum statistics are included, the occupations of final states influence the likelihood of in-scatterings, hence the extra factors of $\bar{f}$ in (1.6)¹.

Having laid bare the full structure of the Boltzmann equation, we see it's a non-linear, integro-differential equation; $\partial_t f$ is given by an integral involving powers of f . Non-linearity of the Boltzmann equation enters at the level of the statistical factor $\mathcal{F}_{[f]}$ which, when integrated over two-body phase space, couples the evolution of the distribution function at a momentum k_1 to the values of the distribution function *at all other momenta*. Hence, it is usually impossible to solve the Boltzmann equation analytically and motivates a numerical solution or further simplification of the collision term (e.g. the 'relaxation time approximation' described in [Appendix A](#)).

¹For bosons, $\bar{f} = 1 + f$ represents Bose enhancement. Otherwise, $\bar{f} = 1 - f$ would represent Pauli blocking for fermions.

1.3 Collisional invariants

The collision integral (1.5) and its components exhibit a number of symmetries. These encode, at the microscopic level, conservation laws associated with $2 \rightarrow 2$ scattering. In this section, the symmetries of the collision term (1.5) are connected to macroscopic conservation laws.

In-vacuum, $|\mathcal{M}|^2$ is naturally a function of the Mandelstam variables. Hence, $|\mathcal{M}|^2$ is time-reversal symmetric. To understand this statement, one could expand s, t and u as functions of the 3-momenta k_i . In each case, the Mandelstam variable is constructed from terms like k_i^2 or $k_i \cdot k_j$. Therefore, to swap between the kinematics $k_1, k_2 \rightarrow k_3, k_4$ and $-k_3, -k_4 \rightarrow -k_1, -k_2$ leaves the square amplitude unchanged. Moreover, $|\mathcal{M}|^2$ is parity invariant for the same reason. Naturally, $|\mathcal{M}|^2$ is unitary and is unchanged under a transposition of in- and out-states. In summary [8]

$$|\mathcal{M}|^2(k_1, k_2, k_3, k_4) = |\mathcal{M}|^2(-k_3, -k_4, -k_1, -k_2), \quad (\text{time-reversal symmetry})$$

$$|\mathcal{M}|^2(k_1, k_2, k_3, k_4) = |\mathcal{M}|^2(-k_1, -k_2, -k_3, -k_4), \quad (\text{parity invariance})$$

$$|\mathcal{M}|^2(k_1, k_2, k_3, k_4) = |\mathcal{M}|^2(k_3, k_4, k_1, k_2). \quad (\text{unitarity})$$

The statistical factor appearing in (1.5) does not possess the same symmetries as $|\mathcal{M}|^2$. In an isotropic plasma, $\mathcal{F}_{[f]}$ is only parity invariant. Under time-reversal or a transposition of in- and out-states, $\mathcal{F}_{[f]}$ picks up a minus sign. Otherwise, the integral measure of (1.5) contains a δ -distribution which fixes 4-momentum conservation at the level of individual collisions. Presumably, 4-momentum conservation at the microscopic level should translate to macroscopic energy-momentum conservation (and expectation which will be verified shortly).

The microscopic symmetries of the Boltzmann equation, will now be connected to macroscopic conserved quantities (i.e. quantities conserved at the level of the whole system). The macroscopic expectation value of an observable O (per unit volume) is calculated as a weighted average over all modes localised around a point x_1

$$\langle O \rangle(x_1; t) = \int \frac{d^3 k_1}{(2\pi)^3} O(x_1, k_1; t) f(x_1, k_1; t).$$

In our framework, the observable O is ‘macroscopically conserved’ if $\mathcal{D}_t \langle O \rangle = 0$. With the prevailing assumptions of homogeneity and isotropy, $f(x_1, k_1; t) \mapsto f(k_1; t)$. For the purposes of this dissertation, one may also assume that O is explicitly time independent and, at most, a function of the momentum. Then,

$$\mathcal{D}_t \langle O \rangle = \int \frac{d^3 k_1}{(2\pi)^3} O(k_1) \partial_t f(k_1; t) = \int \frac{d^3 k_1}{(2\pi)^3} O(k_1) C_{[f]}(k_1). \quad (1.7)$$

In general, $\partial_t f = C_{[f]} - \mathbf{v} \cdot \mathbf{grad}_x f - \mathbf{F} \cdot \mathbf{grad}_k f$. However, the streaming terms in our model (1.3) are zero. The inclusion of non-trivial streaming terms (as would be relevant for an inhomogeneous or anisotropic plasma) does not affect the categorisation of conserved quantities. In that case, the macroscopic expectation value $\langle \dots \rangle$ would be defined as an integral over the full phase space instead of just the momentum components. The streaming contributions could then be integrated by parts and found to contribute nothing at infinity. Hence, only the collisional contribution to $\partial_t f$ determines whether a quantity is, in total, conserved [7, 9].

After substituting (1.5) for $C_{[f]}$ in (1.7), the criterion for $\langle O \rangle$ being conserved is therefore

$$0 = \frac{1}{2} \int_{1234} (2\pi)^4 \delta^{(3)}(K_1 + K_2 - K_3 - K_4) O(k_1) |\mathcal{M}|^2 \mathcal{F}_{[f]}. \quad (1.8)$$

where $|\mathcal{M}|^2$ and $\mathcal{F}_{[f]}$ are functions of all participating momenta but the notation has been suppressed. To gain more insight, we expand the statistical factor into a difference of gain and loss terms. Focus, for example, on the loss term. Because momenta 1, 2, 3 and 4 are integration variables in (1.8), they can be relabelled for convenience. Switching in- and out-states, $k_1, k_2 \leftrightarrow k_3, k_4$ turns the loss term into a gain term, $f_1 f_2 \bar{f}_3 \bar{f}_4 \mapsto \bar{f}_1 \bar{f}_2 f_3 f_4$ and leaves $|\mathcal{M}|^2$ invariant (due to unitarity). By doubling the result and relabelling $k_1 \mapsto k_2$ and $k_3 \mapsto k_4$ in one copy, one extracts the following

$$0 = \int_{1234} (2\pi)^4 \delta^{(4)}(K_1 + K_2 - K_3 - K_4) \{O(k_1) + O(k_2) - O(k_3) - O(k_4)\} |\mathcal{M}|^2 \bar{f}_1 \bar{f}_2 f_3 f_4.$$

Evidently, the above equality holds true when

$$0 = O(k_1) + O(k_2) - O(k_3) - O(k_4) \quad (1.9)$$

for k_i which satisfy momentum conservation. Setting $O(k) \propto |k|^0$ (i.e. a constant) effectively counts the number of particles in the in- or out-state (e.g. $|k_1|^0 + |k_2|^0 = 2$) and clearly satisfies (1.9). The macroscopic expectation $\langle |k|^0 \rangle$ is the *number density*, n , of the plasma. Setting $O(k) = K^\mu$, demonstrates that each component of the system's total 4-momentum is conserved. The temporal component, $\langle K^0 \rangle = \langle \sqrt{m^2 + k^2} \rangle$ is the *energy density*, ε , of the plasma. These results are expected; n -conservation is implied by the $2 \rightarrow 2$ nature of the model while ε -conservation is universal among closed systems.

One of the great strengths of kinetic theory is its ability to connect the microscopic behaviour of a system to its bulk properties. In this short section, simple symmetries of the collision term were linked to macroscopic conservation of particle number and 4-momentum. Altogether, the collision term (1.5) generates a time-evolution which is naturally consistent with expected conservation laws

$$n_{\text{in}} = n_{\text{eq}}, \quad \varepsilon_{\text{in}} = \varepsilon_{\text{eq}}, \quad (1.10)$$

which are equations of ‘initial’ and ‘equilibrium’ number or energy densities.

1.4 Equilibrium

Essential to the construction of the collision term (1.5) is Boltzmann's *Stoßzahlansatz* [7]. The *Stoßzahlansatz*, otherwise known as the “assumption of molecular chaos”, is the assertion that (before a collision) the two-particle distribution function factorises into a product of two one-particle distribution functions, $f_2(k_1, k_2) = f_1(k_1)f_1(k_2)$. The *Stoßzahlansatz* therefore allows the collision term to be expressed as a functional of f_1 (the one-particle distribution function) and none of the higher-order distribution functions (thereby truncating the BBGKY hierarchy) [7].

A particular consequence of molecular chaos is Boltzmann's so-called ‘*H*-theorem’, which states that solutions of the Boltzmann equation have monotonically increasing entropy [7, 9]. The *H*-theorem therefore implies that evolution generated by the Boltzmann equation is irreversible. By conforming to the *H*-theorem, solutions of the Boltzmann equation are compatible with the (more general) Second Law of Thermodynamics.

To illustrate Boltzmann's *H*-theorem, we first define the entropy (density), s , of the plasma. The entropy must be defined in a way which is calculable off-equilibrium [10]. Consider the gluons

within a small position cell d^3x which is centred on some x . Foliate the local momentum space into thin spherical shells with radii k_i and thicknesses dk . The volume of each shell in phase space is approximately $4\pi k_i^2 dk d^3x$. Then, partition the phase space into one-particle states with volume $h^3 = (2\pi)^3$. Each spherical shell contains roughly $k_i^2 dk d^3x / 2\pi^2$ one-particle states. The number of ways to distribute n_i gluons among the i^{th} shell's $v_i = k_i^2 dk d^3x / 2\pi^2$ one-particle states is

$$\Omega_i = \binom{n_i + v_i - 1}{n_i} = \frac{(n_i + v_i + 1)!}{n_i!(v_i + 1)!}.$$

Realising that $f_i = n_i/v_i$, one can use Stirling's approximation to calculate $\ln \Omega_i$, which is the entropy contribution of the i^{th} momentum shell

$$\ln \Omega_i = \frac{k_i^2 dk d^3x}{2\pi^2} \{ \bar{f}_i \ln \bar{f}_i - f_i \ln f_i \}$$

By passing to the continuum, it's easy to realise that the total entropy of the plasma, s , is expressible

$$s = \int \frac{d^3k}{(2\pi)^3} \{ \bar{f} \ln \bar{f} - f \ln f \}, \quad (1.11)$$

The H -theorem is then demonstrated by computing $\partial_t s$

$$\partial_t s = \int \frac{d^3k_1}{(2\pi)^3} \ln(\bar{f}_1/f_1) C_{[f]} = \frac{1}{2} \int_{1234} (2\pi)^4 \delta^{(4)}(K_1 + K_2 - K_3 - K_4) \ln(\bar{f}_1/f_1) |\mathcal{M}|^2 \mathcal{F}_{[f]}.$$

Imitating the derivation of (1.9), the statistical factor is separated into gain and loss terms. Integration variables are then relabelled $k_1, k_2 \leftrightarrow k_3, k_4$ and the result is doubled (with $k_1 \mapsto k_2$ and $k_3 \mapsto k_4$ in one copy) to find

$$\partial_t s = \frac{1}{4} \int_{1234} (2\pi)^4 \delta^{(4)}(K_1 + K_2 - K_3 - K_4) \ln \left(\frac{\bar{f}_1 \bar{f}_2 \bar{f}_3 \bar{f}_4}{f_1 f_2 f_3 f_4} \right) |\mathcal{M}|^2 \bar{f}_1 \bar{f}_2 \bar{f}_3 \bar{f}_4. \quad (1.12)$$

Following the lead of [9], define $x = \bar{f}_1 \bar{f}_2 \bar{f}_3 \bar{f}_4 / f_1 f_2 f_3 f_4$. By subtracting half the zeroth moment of the collision term (which is zero) from (1.12), one obtains [9]

$$\partial_t s = \frac{1}{4} \int_{1234} (2\pi)^4 \delta^{(4)}(K_1 + K_2 - K_3 - K_4) \{ x \ln x - x + 1 \} |\mathcal{M}|^2 f_1 f_2 \bar{f}_3 \bar{f}_4,$$

where the function of x in braces is non-negative. Since every other component of the integrand is also non-negative, the H -theorem follows

$$\partial_t s \geq 0. \quad (1.13)$$

When $\partial_t s = 0$, the logarithmic expression in (1.12), $\ln(\bar{f}_1 \bar{f}_2 \bar{f}_3 \bar{f}_4 / f_1 f_2 f_3 f_4)$, must vanish on momenta satisfying $k_1 + k_2 = k_3 + k_4$. That is [7, 8, 9]

$$\ln(\bar{f}_1/f_1) + \ln(\bar{f}_2/f_2) = \ln(\bar{f}_3/f_3) + \ln(\bar{f}_4/f_4). \quad (1.14)$$

An isotropic solution to (1.14) is the well-known Bose distribution [7, 8, 9]

$$b(k; \mu, T) = \frac{1}{e^{(E_k - \mu)/T} - 1}, \quad (1.15)$$

which is parametrised by a chemical potential, μ , and a temperature, T , whose values are uniquely determined by the conservation laws (1.10). Altogether, the H -theorem (1.13) and conservation laws (1.10) imply that f evolves towards a unique *equilibrium* state satisfying (1.14). The constraint (1.14) is equivalent to $\mathcal{F}_{[f]} = 0$, a sufficient condition for the collision term (1.5) to vanish; $C_{[f]} = 0$. Therefore, the system is generally driven towards a stationary f , cf. (1.3), with maximal entropy.

1.5 Bose condensation

In our framework, n and ε can be calculated as the zeroth and first moments of the distribution function

$$n_{[f]} = \int \frac{d^3k}{(2\pi)^3} f, \quad \varepsilon_{[f]} = \int \frac{d^3k}{(2\pi)^3} E_k f. \quad (1.16)$$

The moments of a Bose distribution (1.15) have the general structure

$$n_{\text{eq}}(m, \mu, T) = \frac{T^3}{2\pi^2} \int_0^\infty \frac{dx x^2}{e^{\sqrt{x^2 + \theta^2}/z} - 1}, \quad \varepsilon_{\text{eq}}(m, \mu, T) = \frac{T^4}{2\pi^2} \int_0^\infty \frac{dx x^2 \sqrt{x^2 + \theta^2}}{e^{\sqrt{x^2 + \theta^2}/z} - 1}, \quad (1.17)$$

where $\theta = m/T$ and $z = \exp[\mu/T]$ is known as the ‘fugacity’ of the gas. In the massless case, the Bose integrals in (1.17) can be written as polylogarithms depending only on z [8]. However, the approximation $\theta \ll 1$ is rarely accurate for the systems we study. Generally, there is no simple reduction of the integrals in (1.17) which is compatible with an arbitrary mass. Hence, the moments (1.17) are kept in their basic form, with n_{eq} and ε_{eq} being computed numerically as functions of μ and T at a fixed mass, $m = \underline{m}$. Therefore, the conservation laws (1.10), $n_{\text{in}} = n_{\text{eq}}(\underline{m}, \mu, T)$ and $\varepsilon_{\text{in}} = \varepsilon_{\text{eq}}(\underline{m}, \mu, T)$, are a system of two equations which jointly determine the equilibrium parameters, μ and T .

To facilitate our calculations, we define the ‘weight’ of the distribution $\kappa = n^4/\varepsilon^3$ [3]. The weight is the simplest dimensionless quantity one can create using n and ε , and is obviously conserved (due to individual number and energy conservation). Then, we construct a function $\gamma_{\underline{m}}$ which computes κ at equilibrium as a function of θ and z (alternatively, μ and T) at a fixed mass $m = \underline{m}$

$$\gamma_{\underline{m}}(\theta, z) = \frac{n_{\text{eq}}(\underline{m}, \theta, z)^4}{\varepsilon_{\text{eq}}(\underline{m}, \theta, z)^3}.$$

The equilibrium parameters are then extracted by solving the constraint equations $n_{\text{eq}}(\underline{m}, \theta, z) = n_{\text{in}}$ and $\gamma_{\underline{m}}(\theta, z) = \kappa_{\text{in}}$. One may solve the original conservation laws (1.10) instead, but the constraint curves generated in the θz -plane by those two equations are often near-parallel in the neighbourhood of intersection (which can interfere with root-finding). On the other hand, the curve generated by κ -conservation is usually near-perpendicular to those associated with n - or ε -conservation. Another reason we construct κ , to be discussed shortly, is that κ is a simple indicator of Bose condensation.

By imposing κ - and n -conservation, contours of $\gamma_{\underline{m}}$ and n_{eq} are selected in the θz -plane. Generally, these contours may be parametrised as monotonically increasing functions of θ or z and are described by the differentials

$$\left. \frac{dz}{d\theta} \right|_n = - \frac{\partial_\theta n_{\text{eq}}}{\partial_z n_{\text{eq}}}, \quad \left. \frac{dz}{d\theta} \right|_\kappa = - \frac{4\partial_\theta n_{\text{eq}} - 3n_{\text{eq}}\partial_\theta \ln \varepsilon_{\text{eq}}}{4\partial_z n_{\text{eq}} - 3n_{\text{eq}}\partial_z \ln \varepsilon_{\text{eq}}}.$$

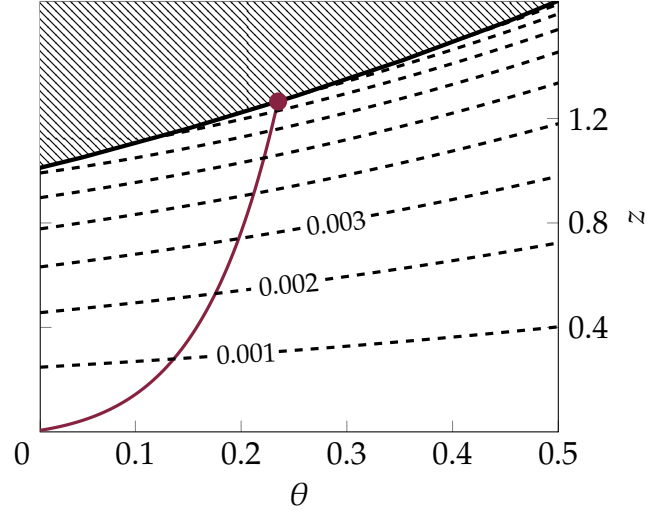
As one might recognise, the partial derivatives of (1.17) against θ are negative while those against z are positive. Further inspection reveals that $|\partial_z \varepsilon_{\text{eq}}| < |\partial_\theta \varepsilon_{\text{eq}}|$. The implication is that

$$\left. \frac{dz}{d\theta} \right|_n > \left. \frac{dz}{d\theta} \right|_\kappa \geq 0$$

for all θ . The curve generated by n -conservation therefore intersects successive $\gamma_{\underline{m}}$ contours corresponding to increasing values for κ . Hence, the solutions for θ and z (at fixed $\underline{m}$ and n_{in}) are

monotonically increasing functions of κ . Because the distribution function remains non-negative only when $\mu \leq \underline{m}$, the n_{eq} contour must terminate somewhere along the line $z = \exp \theta$. This termination point implies a maximal value for κ at fixed $\underline{m}$ and n_{in} (illustrated in Figure 2).

Figure 2: The dashed lines are contours of $\gamma_{\underline{m}}$. A contour of n_{eq} is shown in red. The equilibrium parameters are determined by the intersection of $\gamma_{\underline{m}}$ and n_{eq} contours. The bound $z \leq \exp \theta$ is demarcated by a solid line and the region $z > \exp \theta$ is shaded.



However, there is no *a priori* motivation that κ_{in} is within the range of $\gamma_{\underline{m}}$ on the n_{eq} contour. In fact, [3] argues that in a typical collider experiment, $\kappa_{\text{in}} > \max \gamma_{\underline{m}}$ (in this sense, κ_{in} is ‘supercritical’). Plasmas with supercritical κ_{in} simply contain more gluons than can be accommodated in the excited modes of a Bose distribution (1.15). In these cases, the system of n - and ε -conservation laws are well-posed only if we permit ‘surplus’ gluons to occupy a condensate. In the infinite volume limit, the zero mode is commonly represented as a δ -distribution with support at $\mathbf{k} = \mathbf{0}$

$$b(\mathbf{k}; n_c, T) = \frac{1}{e^{(E_{\mathbf{k}} - m)/T} - 1} + (2\pi)^3 n_c \delta^{(3)}(\mathbf{k}),$$

where the chemical potential is fixed $\mu = m$ and the gluon condensate density, n_c , is the active ‘number parameter’. At arbitrary times (e.g. off-equilibrium) one may calculate n_c by integrating the distribution function over an infinitesimal ball which is centred at the origin of momentum space

$$n_c = \int_0^{\epsilon \rightarrow 0} dk k^2 \int \frac{d\Omega}{(2\pi)^3} f(\mathbf{k}), \quad (1.18)$$

which remains well-defined as long as $f \propto 1/k^{p < 3}$ as $k \rightarrow 0$.

If κ is supercritical, the equilibrium moments are modified

$$\tilde{n}_{\text{eq}}(m, n_c, T) = n_{\text{eq}}(m, m, T) + n_c, \quad \tilde{\varepsilon}_{\text{eq}}(m, n_c, T) = \varepsilon_{\text{eq}}(m, m, T) + m n_c, \quad (1.19)$$

where $\tilde{n}_{\text{eq}}$ and $\tilde{\varepsilon}_{\text{eq}}$ are the modified moments expressed in terms of the original moments (1.17). Exchanging (1.17) and (1.19) means that n_c has replaced μ as an unknown in the conservation laws (1.10). One can imagine (1.19) as continuing the red curve in Figure 2 along a third n_c -axis for supercritical κ . Supercritical solutions would then exist on a curve (defined by n - or ε -conservation) residing inside the surface defined by $z = \exp \theta$ and $n_c > 0$. The solution for n_c and T would then be determined by the intersection of the extended n - or ε -curve and a modified $\tilde{\gamma}_{\underline{m}}$ contour (with $\tilde{\gamma}_{\underline{m}}$ being constructed using the moments (1.19)).

We therefore distinguish three kinds of system: *underpopulated*, *critically populated* and *overpopulated* systems [8]. Underpopulated systems have subcritical κ and obtain a chemical potential $\mu < \underline{m}$. Critically populated systems have a critical κ and obtain a maximal chemical potential, $\mu = \underline{m}$, but develop no condensate. On the other hand, overpopulated systems have supercritical κ and develop a condensate. Hence, the formation of a Bose condensate is justified if the system is initially overpopulated and inelastic processes (which may reduce n) remain negligible.

1.6 Colour glass condensate initial conditions

The initial conditions studied in this thesis are glasma-inspired. They are characterised by uniform support below some threshold momentum Q [8, 11, 12],

$$f_{\text{in}}(k; Q, \chi) = \chi \Theta(1 - k/Q). \quad (1.20)$$

The quantity χ is a ‘population parameter’. We see that n , ε and hence κ scale linearly with χ for systems with such an initial condition. With Q being the only salient energy scale, n and ε must scale like Q^3 and Q^4 respectively (up to corrections of order m/Q). To circumvent artefacts in the numerics, the hard step in (1.20) is softened by using a sigmoid-type distribution

$$f_{\text{in}}(k; Q, \chi, \sigma) = \frac{\chi}{e^{(k-Q)/\sigma} + 1}, \quad (1.21)$$

where the auxiliary parameter σ ‘smooths’ the step at $k = Q$. Without a positive width ($\sigma > 0$), it becomes difficult to accurately resolve the step in (1.20) in a discrete representation of momentum space (i.e. on a finite ‘grid’). By changing grid resolution, values of (1.20) near the threshold $k \simeq Q$ may ‘jump’ between 0 and χ . On the other hand, discrete representations of (1.21) respond more stably to changes in the grid properties. In our numerics, $Q = 1$ and is used as the basic energy scale (realistically, $Q \simeq 2 \text{ GeV}$ [12]). Unless otherwise stated, $\sigma/Q = 0.05$ throughout this thesis.

The moments of (1.21) are

$$n_{\text{in}}(Q, \chi, \sigma) = \frac{\chi}{2\pi^2} \int_0^\infty \frac{dk k^2}{e^{(k-Q)/\sigma} + 1}, \quad \varepsilon_{\text{in}}(m, Q, \chi, \sigma) = \frac{\chi}{2\pi^2} \int_0^\infty \frac{dk k^2 \sqrt{k^2 + m^2}}{e^{(k-Q)/\sigma} + 1}. \quad (1.22)$$

Like the Bose integrals (1.17), there are no simplifications of (1.22) more practical than their integral expressions. Therefore, the moments (1.22) are left in their basic form. Understanding that energies are measured in units of Q , the Q dependence of (1.22) is ignored by mapping $Q \mapsto 1$.

With the initial conditions specified, we find (as functions of m) the values of χ which define critical population. One can calculate χ and T at criticality (here called χ_c and T_c respectively) by setting $\mu = \underline{m}$ in (1.17) and using the complementary (1.22) to set up the conservation laws

$$n_{\text{in}}(\chi_c, \underline{\sigma}) = n_{\text{eq}}(\underline{m}, T_c), \quad \varepsilon_{\text{in}}(\underline{m}, \chi_c, \underline{\sigma}) = \varepsilon_{\text{eq}}(\underline{m}, T_c),$$

where the underlined quantities are all fixed, $m = \underline{m}$, $\mu = \underline{m}$ and $\sigma = \underline{\sigma}$. The result is two constraint equations which jointly determine χ_c and T_c . Figure 3 shows the critical parameters as functions of m and for some different choices of σ . As seen in Figure 3 the effects of varying σ are residual.

According to Figure 3, χ_c increases monotonically with m . Moreover, the response of χ_c to changes in m is significant (for the range of masses sampled in Figure 3, χ_c is more than quadrupled). On the other hand, T_c varies only mildly over the range of sampled masses. Unexpectedly, the variation of T_c against m is non-monotonic. For the few values of σ sampled in Figure 3, T_c is maximised somewhere in the region $0.2 \lesssim m/Q \lesssim 0.4$.

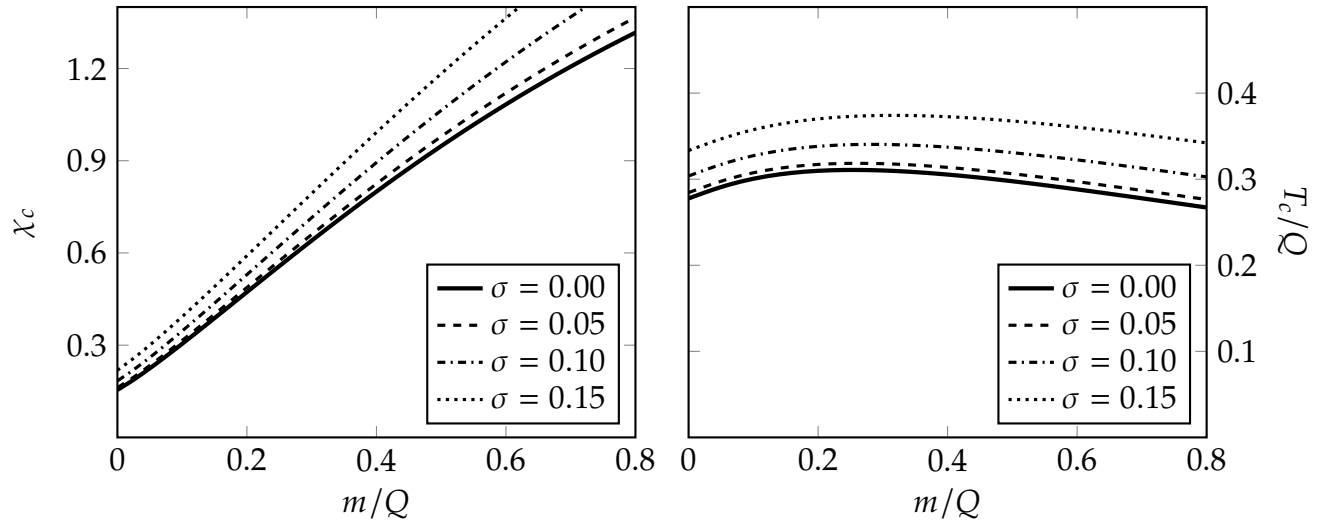


Figure 3: $\chi_c(m)$ and $T_c(m)$ for different σ .

By fixing m and σ , the population parameter χ is the only remaining degree of freedom in the initial condition. Under these conditions, χ spans an effectively one-parameter family of initial and equilibrium states. When $\chi < \chi_c$, the system is underpopulated. When $\chi = \chi_c$, the system is critically populated. When $\chi > \chi_c$, the system is overpopulated. Knowing $\chi_c(m)$ from Figure 3, it's a simple matter to define initial conditions which are suitably under- or overpopulated. Figure 4 shows the equilibrium parameters as functions of χ for particular choices of m and σ .

When $\chi > \chi_c$, the condensate density n_c becomes the active equilibrium parameter associated with n -conservation. Simultaneously, the chemical potential is fixed $\mu = m$. When $\chi \leq \chi_c$ μ is the active number parameter and $n_c = 0$. General features of the curves in Figure 4 don't change for different choices of m and σ . For example, the equilibrium parameters always seem to have a discontinuous first derivative at $\chi = \chi_c$. The temperature always seems to increase more strongly as a function of χ when $\chi > \chi_c$ and the chemical potential is always maximised ($\mu = m$) when $\chi \geq \chi_c$ (likewise, the condensate density is always minimised, $n_c = 0$, when $\chi \leq \chi_c$).

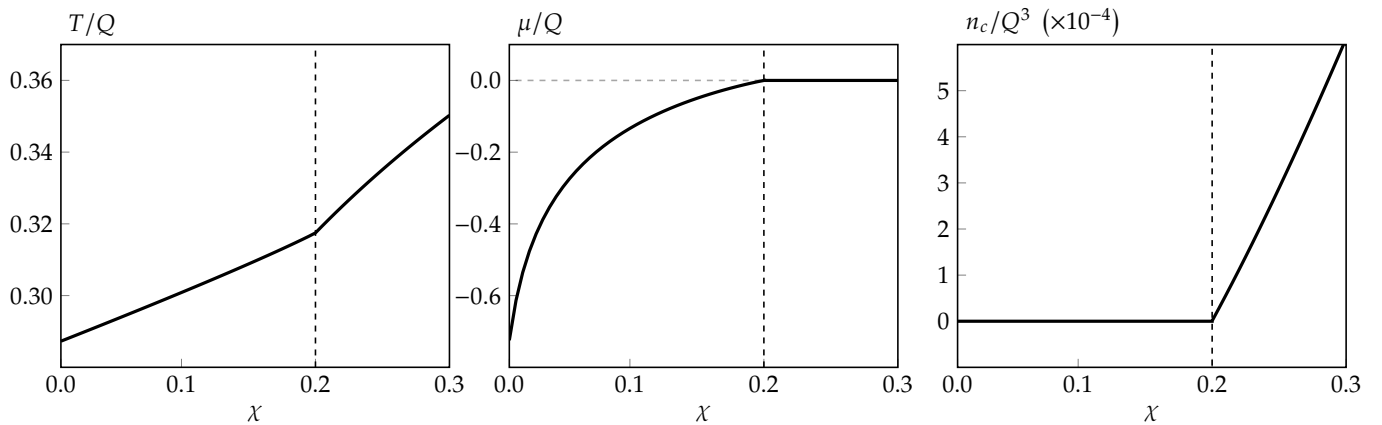


Figure 4: Equilibrium parameters as functions of χ . Results are calculated for $m = 0$ and $\sigma/Q = 0.125$. In this case, $\chi_c \approx 0.2$. A vertical dashed line demarcates $\chi = \chi_c$.

1.7 Solving the Boltzmann equation

Due to the global nature of the collision (1.5) term discussed in section 1.2, the evolution equation (1.3) is impossible to solve analytically. However, it's possible in our model to find seven of the nine integrals in (1.5) in closed-form. The remaining two integrals of (1.5) and the final integration of (1.3) with respect to time are approximated with a simple discretisation. The method for simplifying the collision term (1.5) presented in this section is due to [8, 13].

Firstly, consider the collision term (1.5) and separate the integral over Γ into on-shell momentum integrals for particles 2, 3 and 4 in the fashion of (1.4). Conventionally, the integral over k_4 is calculated first and is done by making the on-shell criterion explicit

$$\int_4 = \int \frac{d^4 K_4}{(2\pi)^4} 2\pi \delta(K_4^2 - m^2) \Theta(E_4),$$

which implies

$$\int_4 (2\pi)^4 \delta^{(4)}(K_1 + K_2 - K_3 - K_4) |\mathcal{M}|^2 \mathcal{F}_{[f]} = 2\pi \delta(\underline{K}_4^2 - m^2) \Theta(E_4) |\underline{\mathcal{M}}|^2 \underline{\mathcal{F}}_{[f]},$$

where the underlining of a symbol means that its value or, in the case of a function, the value of one of its arguments has been constrained by two-body kinematics. For example, $\underline{K}_4 = K_1 + K_2 - K_3$ and $\underline{\mathcal{F}}_{[f]} = \mathcal{F}_{[f]}(k_1, k_2, k_3, \underline{k}_4)$. Remaining are the integrals over k_2 and k_3 .

Due to isotropy, the statistical factor carries no angular dependence. Indeed, any angular dependence in the integrand of (1.5) resides in the square amplitude and manifests through combinations of Mandelstam variables. With this in mind, we pull $\mathcal{F}_{[f]}$ outside the integrals over Ω_2 and Ω_3 and integrate $|\underline{\mathcal{M}}|^2$ against $d\Omega_2$ and $d\Omega_3$. To facilitate the angular integrals, a coordinate scheme is specified in Figure 5.

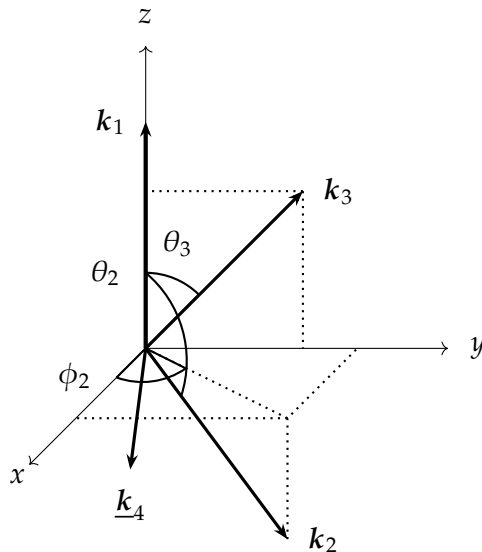


Figure 5: To facilitate the integrations over k_2 and k_3 , we install a coordinate system where the z -axis is aligned with k_1 and the y - z plane is arranged to contain k_3 . The 3-momenta k_1 , k_2 , k_3 and $\underline{k}_4$ (for a specific $2 \rightarrow 2$ scattering) are shown against the described Cartesian system.

The components for k_1 , k_2 and k_3 in the coordinate frame of Figure 5 are

$$k_1 = k_1(0, 0, 1), \quad k_2 = k_2(\sin \phi_2 \sin \theta_2, \cos \phi_2 \sin \theta_2, \cos \theta_2), \quad k_3 = k_3(0, \sin \theta_3, \cos \theta_3).$$

With $\underline{K}_4 = K_1 + K_2 - K_3$, it's simple to show that $\underline{K}_4^2 - m^2 = s + t - 2m^2 - 2K_2 \cdot K_3$ where s and t are Mandelstam variables. Using the coordinate convention explained in Figure 5, one can easily calculate the product $K_2 \cdot K_3$ and hence rewrite the constraint $\underline{K}_4^2 - m^2 = A + B \cos \phi_2$ with

$$A = s + t - 2(m^2 + E_2 E_3 - k_2 k_3 \cos \theta_2 \cos \theta_3), \quad B = 2k_2 k_3 \sin \theta_2 \sin \theta_3.$$

At this point, we've reduced (1.3) to the following form

$$C_{[f]}(E_1) = \frac{1}{16(2\pi)^4 E_1} \int_0^\infty dk_2 \frac{k_2^2}{E_2} \int_0^\infty dk_3 \frac{k_3^2}{E_3} \mathcal{F}_{[f]} \int_{-1}^1 dx_2 \int_{-1}^1 dx_3 \int_0^{2\pi} d\phi_2 |\underline{\mathcal{M}}|^2 \delta(A + B \cos \phi_2) \Theta(\underline{E}_4),$$

where $x_i = \cos \theta_i$ and the integral over ϕ_3 was trivial. Elementary techniques can be used to compute the integral over ϕ_2 ,

$$\int_0^{2\pi} d\phi_2 |\underline{\mathcal{M}}|^2 \delta(A + B \cos \phi_2) = \frac{\Theta(B^2 - A^2)}{\sqrt{B^2 - A^2}} \sum_{\pm} |\underline{\mathcal{M}}|^2|_{\phi_{\pm}},$$

with $\phi_{\pm} = \pi \pm \arccos(A/B)$. In vacuum, the square amplitude is naturally a function of the Mandelstam variables s , t and u . Therefore, $|\underline{\mathcal{M}}|^2$ carries no azimuthal dependence and the sum over $\phi_{\pm}$ simply creates a factor of two. Given that $|\underline{\mathcal{M}}|^2$ is a function of the Mandelstam variables, it's convenient to exchange the integrals over $\cos \theta_2$ and $\cos \theta_3$ for integrals over s and t respectively. To expedite the forthcoming integrals over s and t , it's useful to define

$$Q = k_1^2(B^2 - A^2).$$

By changing variables $\cos \theta_2 \mapsto s$ and $\cos \theta_3 \mapsto t$ (incurring a Jacobian $1/4k_1^2 k_2 k_3$) one finds

$$C_{[f]}(E_1) = \frac{1}{32(2\pi)^4 k_1 E_1} \int_0^\infty dk_2 \frac{k_2}{E_2} \int_0^\infty dk_3 \frac{k_3}{E_3} \mathcal{F}_{[f]} \int_{4m^2}^\infty ds \int_{-\infty}^0 dt \frac{\Theta(Q)}{\sqrt{Q}} |\underline{\mathcal{M}}|^2,$$

where the bounds on s and t can be further constrained by $\Theta(Q)$. To fix the bounds on the Mandelstam integrals, we first note that Q is a quadratic form in t , $Q = a_t t^2 + b_t t + c_t$, with coefficients

$$\begin{aligned} a_t &= s - (E_3 + E_4)^2, \\ b_t &= s^2 - 2s(E_1 E_4 + \underline{E}_2 E_3 + 2m^2) + 4m^2(E_3 + E_4)^2, \\ c_t &= -s(s - 4m^2)(E_1 - E_3)^2. \end{aligned} \tag{1.23}$$

Clearly, $a_t \leq 0$. Since Q is concave in t , the factor of $\Theta(Q)$ constrains $t_- \leq t \leq t_+$ where $t_{\pm}$ are the roots of Q (which are easily computed using the quadratic equation). For $C_{[f]}$ to remain real-valued, the discriminant $b_t^2 - 4a_t c_t$ must be non-negative. Therefore, s is bounded like

$$s_{\pm} = 2m^2 + 2 \begin{cases} E_1 E_2 \pm k_1 k_2 & \text{if } E_1 E_2 < E_3 \underline{E}_4 \\ E_3 \underline{E}_4 \pm k_3 \underline{k}_4 & \text{if } E_3 \underline{E}_4 < E_1 E_2 \end{cases}. \tag{1.24}$$

Apprised of the bounds on s and t , the *kernel* is defined to be the double integral

$$\mathcal{K}(k_1, k_2, k_3, \underline{k}_4) = \int_{s_-}^{s_+} ds \int_{t_-}^{t_+} dt \frac{1}{\sqrt{Q}} |\underline{\mathcal{M}}|^2, \quad (1.25)$$

where, by specifying $s_{\pm}$ and $t_{\pm}$, we can drop the factor of $\Theta(Q)$.² For the choices of $|\underline{\mathcal{M}}|^2$ studied in this thesis, the kernel can be found in closed-form.

Remaining are the integrals over k_2 and k_3 . For reasons which are made clear in [Appendix B](#), we change variables $k_i \mapsto E_i$ (with the on-shell condition implying $k dk = E dE$). So that we speak of outgoing particles alone, we also trade the integral over E_2 for an integral over E_4 (now treating $\underline{K}_2$ as the constrained 4-momentum and K_4 as free). Finally, (1.5) is reduced

$$C_{[f]}(E_1) = \frac{1}{32(2\pi)^4 k_1 E_1} \iint_{\mathcal{R}} dE_3 dE_4 \mathcal{K}(k_1, \underline{k}_2, k_3, k_4) \mathcal{F}_{[f]}(k_1, \underline{k}_2, k_3, k_4) \quad (1.26)$$

where $\mathcal{R}$ is a region in the $E_3 E_4$ -plane defined by $E_3, E_4, \underline{E}_2 \geq m$. In summary, the originally *nine*-dimensional collision integral (1.5) was simplified into a *two*-dimensional convolution of a kernel (1.25) and the statistical factor (1.6). Four integrals were done gratis by exploiting overall 4-momentum conservation; one azimuthal integral was made trivial by a convenient choice of coordinates and two of the remaining four integrals (over s and t) are doable by hand in our model. Remaining in (1.5) are just two integrals over the kinematically viable outgoing energies.

1.8 Coupling to the condensate

As implied by (1.18), the distribution function is naturally divisible into a sum of two terms: a ‘regular’ part accounting for modes with $\mathbf{k} \neq \mathbf{0}$ and a singular part which accounts for the condensate (represented in the infinite volume limit with a δ -distribution). For example, an arbitrary distribution function F is expressed

$$F(\mathbf{k}) = f(\mathbf{k}) + (2\pi)^3 n_c \delta^{(3)}(\mathbf{k}), \quad (1.27)$$

where f is the regular part of F . In order to make the evolution equation (1.3) compatible with the representation of the distribution function in the infinite volume limit (1.27), the evolution equation (1.3) must be separated into two (coupled) equations for the regular part of the distribution function and the condensate density.

The procedure for ‘coupling to the condensate’, as described in [8, 14], is straightforward: one substitutes (1.27) into the collision term (1.5) and then reorganises the result in powers of n_c

$$\mathcal{F}_{[F]} = \mathcal{F}_{[f]} + (2\pi)^3 n_c \left\{ \delta^{(3)}(\mathbf{k}_1) \mathcal{F}_{[f]}^{\circ 234} + \delta^{(3)}(\mathbf{k}_2) \mathcal{F}_{[f]}^{1 \circ 34} + \delta^{(3)}(\mathbf{k}_3) \mathcal{F}_{[f]}^{12 \circ 4} + \delta^{(3)}(\mathbf{k}_4) \mathcal{F}_{[f]}^{123 \circ} \right\} + \mathcal{O}(n_c^2). \quad (1.28)$$

where $\mathcal{F}_{[f]}$ represents the gain and loss terms for processes involving only excited gluons. When convolved with $|\underline{\mathcal{M}}|^2$, the resulting collision term accounts for scattering events where all participating momenta are non-zero. We call this the ‘regular collision integral’ and name it $C_{[f]}$.

²Note that there are multiple disconnected regions in the st -plane where $Q \geq 0$. However, (1.24) selects the region which is physically relevant. It’s well-known that $s + t + u = 4m^2$. Because u is always negative, physical values for s and t must satisfy $s + t > 4m^2$. The criteria (1.24) define the region in the st -plane where $s + t > 4m^2$ and $Q \geq 0$ (assuming $t_- \leq t \leq t_+$).

The terms $\mathcal{O}(n_c)$ in (1.28) are statistical factors for processes where exactly one of the participating particles is from the condensate. They are

$$\begin{aligned}\mathcal{F}_{[f]}^{\circ 234} &= \bar{f}_2 f_3 f_4 - f_2 \bar{f}_3 \bar{f}_4, & \mathcal{F}_{[f]}^{1\circ 34} &= \bar{f}_1 f_3 f_4 - f_1 \bar{f}_3 \bar{f}_4, \\ \mathcal{F}_{[f]}^{12\circ 4} &= \bar{f}_1 \bar{f}_2 f_4 - f_1 f_2 \bar{f}_4, & \mathcal{F}_{[f]}^{123\circ} &= \bar{f}_1 \bar{f}_2 f_3 - f_1 f_2 \bar{f}_3.\end{aligned}\tag{1.29}$$

$\mathcal{F}_{[f]}^{12\circ 4}$ and $\mathcal{F}_{[f]}^{123\circ}$ are the same up to a relabelling of momenta $3 \leftrightarrow 4$ and therefore produce equal collision terms when convolved with $|\mathcal{M}|^2$. The δ -distributions which appear in products with the condensate factors (1.29) greatly simplify the integration over Γ when finding the ‘condensate collision terms’. The resulting condensate integrals are

$$C_{[f]}^{\circ 234}(\mathbf{k}_1) = (2\pi)^3 \delta^{(3)}(\mathbf{k}_1) \frac{1}{4E_1} \int d\Gamma |\mathcal{M}|^2 \mathcal{F}_{[f]}^{\circ 234},$$

$$C_{[f]}^{1\circ 34}(\mathbf{k}_1) = \frac{1}{8mE_1} \int_{34} (2\pi)^4 \delta^{(4)}(K_1 + K_2 - K_3 - K_4) |\mathcal{M}|^2 \mathcal{F}_{[f]}^{1\circ 34} \Big|_{k_2=0'} \tag{1.30}$$

$$C_{[f]}^{12\circ 4}(\mathbf{k}_1) = \frac{1}{8mE_1} \int_{24} (2\pi)^4 \delta^{(4)}(K_1 + K_2 - K_3 - K_4) |\mathcal{M}|^2 \mathcal{F}_{[f]}^{12\circ 4} \Big|_{k_3=0'} \tag{1.31}$$

$$C_{[f]}^{123\circ}(\mathbf{k}_1) = \frac{1}{8mE_1} \int_{23} (2\pi)^4 \delta^{(4)}(K_1 + K_2 - K_3 - K_4) |\mathcal{M}|^2 \mathcal{F}_{[f]}^{123\circ} \Big|_{k_4=0'}.$$

Because $C_{[f]}^{12\circ 4} = C_{[f]}^{123\circ}$, the total collision term may be written

$$C_{[F]} = C_{[f]} + n_c \left\{ C_{[f]}^{\circ 234} + C_{[f]}^{1\circ 34} + C_{[f]}^{12\circ 4} + C_{[f]}^{123\circ} \right\} = C_{[f]} + n_c \left\{ C_{[f]}^{\circ 234} + C_{[f]}^{1\circ 34} + 2C_{[f]}^{12\circ 4} \right\}. \tag{1.32}$$

Notably, the terms $\mathcal{O}(n_c^2)$ in (1.28) don’t contribute to the collision term (1.32). Any term which is $\mathcal{O}(n_c^2)$ contains at least two δ -distributions with support on the zeros of *different* momenta. Hence, these terms are non-zero only when two or more of the participating particles are in the condensate. However, the collision term must vanish whenever more than one particle is in the condensate; either because the implied kinematics are forbidden by momentum conservation or the in- and out-states are kinematically identical (implying a perfect cancellation of gain and loss terms). Therefore, all $\mathcal{O}(n_c^2)$ contributions vanish.

Finally, the conservation laws (1.10) imply an integral relation between $C_{[f]}^{\circ 234}$ and the other condensate integrals. Recall that one of the collisional invariants is the total number density of the plasma, n . Integrating (1.32) over $\mathbf{k}_1$, we find a condition for n -conservation

$$0 = \int \frac{d^3 k_1}{(2\pi)^3} C_{[f]} + n_c \int \frac{d^3 k_1}{(2\pi)^3} \left\{ C_{[f]}^{\circ 234} + C_{[f]}^{1\circ 34} + 2C_{[f]}^{12\circ 4} \right\}. \tag{1.33}$$

Recall that $C_{[f]}$ accounts for scattering events involving only excited modes. Hence, the integral of $C_{[f]}$ must be the resulting change in the number density of the regular part of the distribution function. Because the scattering is $2 \rightarrow 2$, the first term in (1.33) necessarily vanishes. The only

non-vanishing terms in (1.33) are linear in n_c . Recall the definition of the condensate (1.18). By integrating the evolution equation $\partial_t F = C_{[F]}$ over an infinitesimal ball in k -space (and by using (1.32)) we produce the following evolution equation for n_c

$$\partial_t n_c = \int_0^{\epsilon \rightarrow 0} dk k^2 \int \frac{d\Omega}{(2\pi)^3} C_{[f]}^{\circ 234} = \frac{n_c}{4m} \int d\Gamma |\underline{\mathcal{M}}|^2 \underline{\mathcal{F}}_{[f]}^{\circ 234}. \quad (1.34)$$

Therefore $C_{[f]}^{\circ 234}$ is the only collision term which generates a non-zero contribution when integrated over the ‘condensate ball’. Having classified $C_{[f]}^{\circ 234}$ as the collision term which explicitly tracks changes in n_c due to processes involving a condensate particle, we may then identify $C_{[f]}^{1\circ 34}$, $C_{[f]}^{12\circ 4}$ and $C_{[f]}^{123\circ}$ as the terms which describe changes in the regular part of the distribution function due to processes involving a condensate particle. Following from the conservation of total number density, changes in the regular distribution can be related to changes in the condensate via

$$\partial_t n_c = -n_c \int \frac{d^3 k_1}{(2\pi)^3} \left\{ C_{[f]}^{1\circ 34} + 2C_{[f]}^{12\circ 4} \right\},$$

which comes from (1.34) and (1.33) [8, 14]. Unlike the regular collision integral, the condensate integrals are evaluated fairly effortlessly. Of the nine integrals which make up each collision term, three are done automatically via the extra δ -distribution coming from the zero mode; four are done via the δ -distribution which fixes overall conservation of 4-momentum; one can be made trivial through a convenient choice of coordinates whilst the final integral is not generally expressible in closed-form. Starting from (1.30) and (1.31), we can show that for a general $|\mathcal{M}|^2$

$$C_{[f]}^{1\circ 34} = \frac{1}{64\pi m E_1 k_1} \int_m^{E_1} dE_3 |\underline{\mathcal{M}}|^2 \underline{\mathcal{F}}_{[f]}^{1\circ 34}, \quad C_{[f]}^{12\circ 4} = \frac{1}{64\pi m E_1 k_1} \int_{E_1}^{\infty} dE_4 |\underline{\mathcal{M}}|^2 \underline{\mathcal{F}}_{[f]}^{12\circ 4}, \quad (1.35)$$

a result already known in [8, 14]. Important to note is that, in either copy of $|\mathcal{M}|^2$ in (1.35), the Mandelstam variables s , t and u are fixed by kinematics. When $k_2 = \mathbf{0}$ they are

$$\underline{s} = 2m(m + E_1), \quad \underline{t} = 2m(E_3 - E_1), \quad \underline{u} = 2m(m - E_3),$$

and when $k_3 = \mathbf{0}$ they are

$$\underline{s} = 2m(m + E_4), \quad \underline{t} = 2m(m - E_1), \quad \underline{u} = 2m(E_1 - E_4).$$

In summary, a complete description of the system’s evolution is obtainable by solving the two coupled evolution equations

$$\partial_t f = C_{[f]} + n_c \{ C_{[f]}^{1\circ 34} + 2C_{[f]}^{12\circ 4} \}, \quad (1.36)$$

$$\partial_t n_c = -n_c \int \frac{d^3 k_1}{(2\pi)^3} \left\{ C_{[f]}^{1\circ 34} + 2C_{[f]}^{12\circ 4} \right\}, \quad (1.37)$$

where (1.36) describes the evolution of the regular distribution and (1.37) controls changes in the condensate. The numerical procedure used to solve (1.36) and (1.37) is described in Appendix B.

1.9 The square amplitude

At tree-level and in vacuum, the square amplitude for $2 \rightarrow 2$ gluon scattering is [15]

$$|\mathcal{M}|^2 = 72(4\pi\alpha)^2 \left[3 - \frac{tu}{s^2} - \frac{su}{t^2} - \frac{st}{u^2} \right], \quad (1.38)$$

where α is the strong coupling. The diagrams contributing to (1.38) are shown in Figure 6.

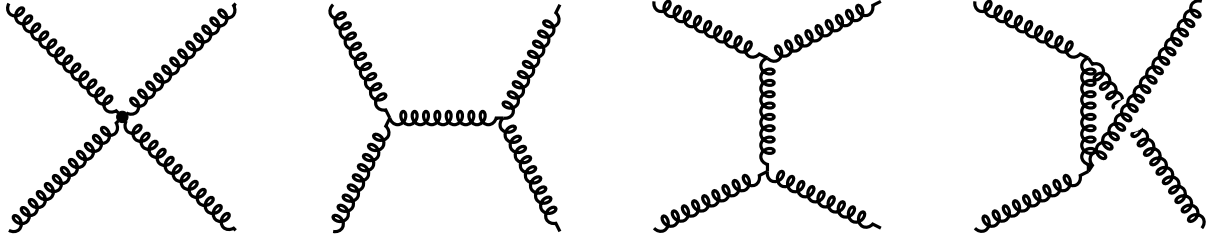


Figure 6: Feynman diagrams contributing to (1.38). From left-to-right, these are the 4-point, s -, t - and u -channel processes.

Evidently, (1.38) diverges quadratically in s , t and u . Therefore, $|\mathcal{M}|^2 \rightarrow \infty$ for kinematics where $s, t, u \rightarrow 0$. In its current form, $|\mathcal{M}|^2$ is not integrable in the sense of (1.5). Consequently, $C_{[f]} \rightarrow \infty$ and equilibration occurs instantaneously (an obviously nonsensical result). For $C_{[f]}$ to remain well-defined, $|\mathcal{M}|^2$ as it appears in (1.38) must be regulated.

By incorporating single-loop (in-medium) corrections, t and u are modified like $t \mapsto t - \mu^2$ and $u \mapsto u - \mu^2$ where $\mu > 0$ is a *screening mass*. In medium, the external gluons obtain a *kinematic mass*, $m > 0$ ³, which is parametrically similar to μ (at equilibrium, m and μ are both proportional to $\alpha^{1/2}T$). Neglecting single-loop corrections for s , it obtains a simple bound $s \geq 4m^2$ (which immediately regulates the $1/s^2$ contribution in (1.38)).

By dropping sub-leading terms, $|\mathcal{M}|^2$ is approximately

$$|\mathcal{M}|^2 \simeq 72(4\pi\alpha)^2 s^2 \left[\frac{1}{t^2} + \frac{1}{u^2} \right], \quad (1.39)$$

where the $1/t^2$ and $1/u^2$ divergences emphasise small-angle scatterings due to *soft* momentum exchanges (where $|t| \ll s$ and $|u| \sim s$ or vice-versa). There are two techniques of regulating (1.39) which admit a finite collision term (1.5). Firstly, define $\mathbf{q} = \mathbf{k}_3 - \mathbf{k}_1$ and notice that t is small when $\mathbf{q}$ is small. Expanding the statistical factor in powers of small $\mathbf{q}$, one finds

$$\mathcal{F}_{[f]} = \mathbf{q} \cdot \{f_2 \bar{f}_2 \mathbf{grad}_k f_1 - f_1 \bar{f}_1 \mathbf{grad}_k f_2\} + O(q^2). \quad (1.40)$$

Likewise, by expanding the energy conservation criterion in powers of small $\mathbf{q}$ we see

$$E_1 + E_2 - E_3 - E_4 = 0 \mapsto \mathbf{q} \cdot (\mathbf{v}_1 - \mathbf{v}_2) + O(q^2) = 0,$$

where $\mathbf{v}_i$ is the speed of particle i , $\mathbf{v}_i = \mathbf{k}_i/E_i$. By exchanging the momentum gradients in (1.40) for energy derivatives, $\mathbf{grad}_k f = \mathbf{v} df/dE$, and setting $\mathbf{v}_2 \mapsto \mathbf{v}_1$ in line with energy conservation

$$\mathcal{F}_{[f]} \simeq (\mathbf{q} \cdot \mathbf{v}_1) \{f_2 \bar{f}_2 f'_1 - f_1 \bar{f}_1 f'_2\},$$

³Without a finite kinematic mass, $m > 0$, we couldn't expect Bose condensation to occur. Consider, for example, a two-body collision of massless particles where $k_3 = 0$. By equating s before and after the collision, one finds a problematic equation $0 = k_1 k_2 (1 - \cos \theta)$ where θ is the angle of $\mathbf{k}_1$ relative to $\mathbf{k}_2$. The immediate implication is that condensation may only occur when incoming particles are exactly collinear (an arrangement occurring with zero probability).

where $f'_i = f'(E_i)$ and all terms $O(q^2)$ have been dropped. However, $\mathbf{q} \cdot \mathbf{v}_1 \propto \cos \theta_3$ which, in turn, is proportional to t . Therefore, whenever t is ‘small’ the statistical factor becomes linear in t and

$$|\mathcal{M}|^2 \mathcal{F}_{[f]} \simeq \frac{72(4\pi\alpha)^2 s^2}{t} \{f_2 \bar{f}_2 f'_1 - f_1 \bar{f}_1 f'_2\},$$

where contributions $O(t^0)$ are ignored. Similarly, if one defines $\mathbf{q}' = \mathbf{k}_4 - \mathbf{k}_1$ and expands the statistical factor in powers of small $\mathbf{q}'$, the integrand of the collision term (1.5) is found to be $1/u$ -divergent. Therefore, for the collision term (1.5) to remain well-defined at least one factor of t and u must be regulated in (1.39) through the introduction of a screening mass. The two possible regularisations of (1.39) are, firstly, the *weakly-screened* square amplitude [16]

$$|\mathcal{M}_W|^2 = 72(4\pi\alpha)^2 s^2 \left[\frac{1}{t(t - \mu^2)} + \frac{1}{u(u - \mu^2)} \right], \quad (1.41)$$

and, secondly, the *strongly-screened* square amplitude [11]

$$|\mathcal{M}_S|^2 = 72(4\pi\alpha)^2 s^2 \left[\frac{1}{(t - \mu^2)^2} + \frac{1}{(u - \mu^2)^2} \right]. \quad (1.42)$$

The kernel (1.25) associated with the weakly-screened square amplitude (1.41), which we call the w -kernel, is an already-known result [8]. Appendix C gives an overview of how to simplify the w -kernel from (1.41) and (1.25). On the other hand, the kernel associated with the strongly-screened square amplitude (1.42), which we call the s -kernel, is a novel result of this thesis. The s -kernel is derived in Appendix D. Clearly, $|\mathcal{M}_W|^2 \geq |\mathcal{M}_S|^2$ which holds true for the corresponding kernels and collision terms. Later, we verify that equilibration is generally faster when using (1.41) than it is when using (1.42).

1.10 The small-scattering-angle approximation

An alternative to Boltzmann’s collision integral (1.5) which is often used in the literature [11, 17, 18] is the so-called ‘small-scattering-angle’ approximation of the collision term. As discussed in the previous section, the square amplitude (1.39) is largest when t or u are small (i.e. the scattering angle is small). By expanding the integrand of Boltzmann’s collision integral (1.5) in powers of small $\mathbf{q} = \mathbf{k}_3 - \mathbf{k}_1$ and neglecting terms $O(q^2)$, one finds that the collision term is expressible [11]

$$\partial_t f = -\text{div}_k \mathcal{J}_{[f]}, \quad (1.43)$$

where div_k is a momentum-divergence and $\mathcal{J}_{[f]}$ is a particle current. The current appearing in (1.43) is a functional of f and has the form [11]

$$\mathcal{J}_{[f]}(k_1) = \frac{18\alpha^2}{\pi} \int dk_2 k_2^2 \{f_2 \bar{f}_2 f'_1 - f_1 \bar{f}_1 f'_2\} \mathcal{K}_{\text{FP}}(k_1, k_2; \mu, \Lambda_{\text{FP}}), \quad (1.44)$$

where $\mathcal{K}_{\text{FP}}$ is a ‘Fokker-Planck kernel’ and Λ_{FP} is an auxiliary parameter which emerges as an artefact of the small- $\mathbf{q}$ expansion. Immediately, we see that (1.44) lends the continuity equation (1.43) the structure of a Fokker-Planck equation [9, 11]. Typically, the integral over k_2 in (1.44) cannot be found in closed form and must be computed numerically. An advantage of the Fokker-Planck

approximation over the Boltzmann model is that one extra integral of the collision term has been simplified analytically, which makes numerical implementations of (1.43) relatively fast⁴.

Like most of the literature [19, 17, 18], we consider the approximation of small scattering angle which one derives using (1.42) as the square amplitude. To derive the continuity equation (1.43) we followed the procedure used in [11]. Although the structure we obtain for the current (1.44) is the same as [11], we reduce a different kernel $\mathcal{K}_{\text{FP}}$. A summary of the small- q expansion used in [11] is given in Appendix E. In the same appendix is our result for the Fokker-Planck kernel.

A feature of the small-scattering-angle approximation which is often overlooked in the literature is that the Fokker-Planck kernel is logarithmically divergent [11]

$$\mathcal{K}_{\text{FP}}(k_1, k_2; \mu, \Lambda_{\text{FP}}) \propto \mathcal{L}(\mu, \Lambda_{\text{FP}}),$$

where $\mathcal{L}$ is the so-called ‘Coulomb logarithm’ and is defined

$$\mathcal{L}(\mu, \Lambda_{\text{FP}}) = \log(\Lambda_{\text{FP}}/\mu). \quad (1.45)$$

The Coulomb logarithm (1.45) diverges in both of its arguments, μ and Λ_{FP} . The ‘dual’ logarithmic divergence of $\mathcal{K}_{\text{FP}}$ is demonstrated numerically in Figure 7.

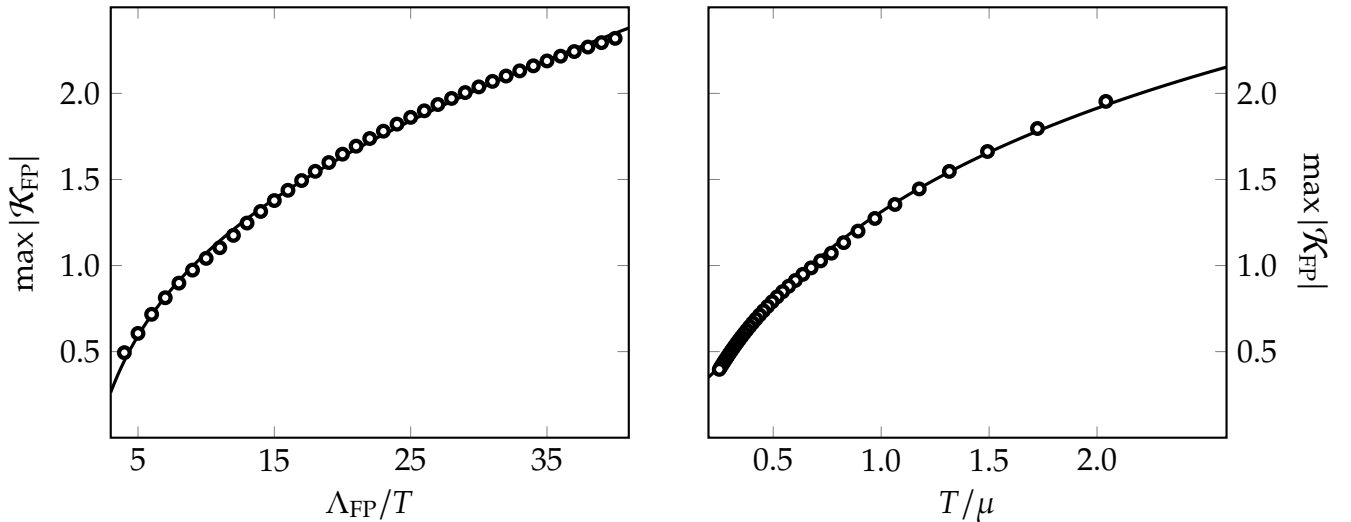


Figure 7: Magnitude of the Fokker-Planck kernel for different values of Λ_{FP}/T and T/μ (calculated with $m/Q = 0.3$). T is the temperature obtained by an initial condition with $\chi = 0.1$. The circular markers are data and the solid lines are logarithmic fits. On the left, $\mu/Q \approx 0.3$. On the right, $\Lambda_{\text{FP}}/Q \approx 3$.

However, as discussed in the section 1.11, the value for μ is fixed by the initial condition of the plasma (with some uncertainty). As far as we understand, there is no motivation in the literature for the value of Λ_{FP} and hence for the Coulomb logarithm. A major endeavour of this thesis is to calibrate the small-scattering-angle approximation of the collision term (1.43) against Boltzmann’s collision integral (1.5). We do so by motivating a value for the ultraviolet cut-off, Λ_{FP} , which then fixes the Coulomb logarithm (1.45). These analyses are presented in the next chapter.

⁴In a discrete approximation with M grid points, cf. Appendix E, the time to compute Boltzmann’s collision integral scales like $\mathcal{O}(M^3)$ whereas the time to compute the Fokker-Planck collision term, $-\text{div}_k \mathcal{J}_{[f]}$, scales like $\mathcal{O}(M^2)$.

1.11 Some comments on the screening mass

Other treatments including the QCD square amplitude [8, 16, 11] regulate (1.39) using a so-called ‘Debye mass’, m_D . According to well-known hard thermal loop calculations, m_D^2 is proportional to an integral of df/dE [3, 19, 20] over momentum space

$$m_D^2 = 48\pi\alpha \int \frac{d^3k}{(2\pi)^3} \frac{f}{E}, \quad (1.46)$$

which is a ‘minus first’ moment of the distribution function. The screening mass, μ , and Debye mass, m_D , are parametrically similar in that they are both proportional to $\alpha^{1/2}T$ at equilibrium. However, there is some uncertainty surrounding reasonable values for the ‘screening ratio’

$$\xi = \mu/m_D, \quad (1.47)$$

and how these relate to properties of the plasma. In our calculations, we take a simple approach and assume that ξ is of order one and constant throughout the evolution. The particular value we use for m_D is computed using (1.46) using the relevant *initial* condition for f . A known result in thermal field theory is that $m^2 \simeq m_D^2/2$ (where m is what we call the ‘kinematic mass’ of the gluons) [20]. Once m_D has been estimated for the initial condition, we adjust m to approximate $m/m_D \simeq 1/\sqrt{2}$.

Because $O(k) = 1/E_k$ fails to satisfy (1.9) in general, the ‘minus first’ moment of f is not a conserved quantity. Therefore, the screening mass may change throughout the evolution. To implement a dynamical screening mass represents a significant complication in our model; both practically and conceptually. For simplicity, we calculate μ for the initial condition (using (1.46)) and keep it fixed for all time. Appendix F justifies the approximation of μ as being constant (for a representative evolution, the variation in μ is less than 10% between initial and final states).

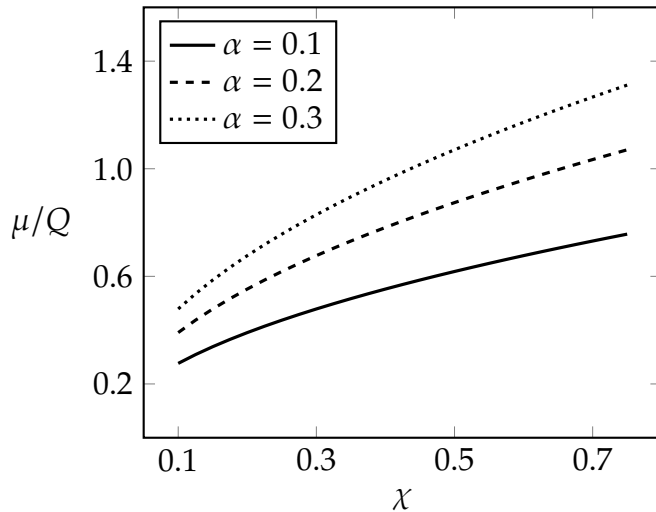


Figure 8: μ as a function of χ for some different choices of the coupling. All data were generated for an initial condition of the form (1.21) using $m/Q = 0.25$, $\sigma/Q = 0.05$ and $\xi = 1$. There is no condensate seed.

2 Solving the evolution equation

In this chapter, we solve the evolution equation using the Boltzmann (1.5) and Fokker-Planck (1.43) collision terms. Others have studied the Boltzmann equation in a leading-order ϕ^4 -theory [8, 14] and verified condensation in overpopulated systems. Condensation has also been confirmed in the small-scattering-angle approximation of the collision term [11, 12] using a strongly-screened QCD amplitude (1.42) and also for a simple RTA [12]. Test particle approaches [21, 16] have also supported the formation of a condensate using a weakly-screened QCD amplitude (1.41).

Here, we present results for both weakly- and strongly-screened QCD amplitudes in a Boltzmann model (1.5). The Fokker-Planck collision term is calibrated against the Boltzmann collision term through an adjustment of the Coulomb logarithm. Results from the Fokker-Planck and Boltzmann models are then compared. The ‘observables’ we study are briefly summarised in section 2.1.

2.1 Observables

The most basic object in our problem is the one-particle distribution function. By solving the evolution equation, we obtain the occupancies of all modes as functions of time. A sometimes expedient alternative to f is its so-called ‘Bose transform’, $\mathcal{B}_{[f]}$. The Bose transform of f is defined

$$\mathcal{B}_{[f]} = \ln(\bar{f}/f), \quad (2.1)$$

and hence maps Bose distributions (1.15) to straight lines

$$\mathcal{B}_{\text{eq}}(E) = (E - \mu)/T. \quad (2.2)$$

Bose transforms simplify calculation of the realised equilibrium parameters; they are extractable from a linear fit to $\mathcal{B}_{[f]}$. From the viewpoint of (2.2), the temperature T is a ‘slope-parameter’ and the chemical potential μ is an ‘intercept-parameter’. Critical and overpopulated systems are singular at $E = m$ and satisfy $\mathcal{B}_{\text{eq}}(m) = 0$.

Bose transforms also allow exact interpolation or extrapolation of Bose functions from a discrete set of function values. A piecewise linear interpolation of $\mathcal{B}_{[f]}$ translates to an interpolation of f with piecewise Bose functions, called ‘Bose-splines’ or simply ‘ b -lines’ [12]. If f is itself a Bose function, the linear interpolation of $\mathcal{B}_{[f]}$ generates an *exact* b -line of f .

Another quantity of interest is the entropy of the system (1.11). As discussed in Appendix A, results obtained using the RTA collision term suggest the entropy evolution is approximately exponential near equilibrium. Defining $\Delta s_\tau = s_\infty - s_\tau$, one finds

$$\Delta s_\tau = \Delta s_{\tau_0} e^{-(\tau - \tau_0)/\tau_s}, \quad (2.3)$$

where $\tau_0 > 0$ is a lower bound on the times τ where the exponential model (2.3) is valid and τ_s is the ‘entropy relaxation time’. In the coming analyses, we compute τ_s as a timescale for equilibration. Even though particles in the condensate contribute nothing to the entropy (1.11), a point of interest is whether there are signatures of the condensate dynamics in s_τ (a feature identified in [16]).

As far as the condensate is concerned, we obtain n_c as a function of time when solving the evolution equation. Using these data, we determine if overpopulated systems develop a macroscopic condensate. Implementations of the Boltzmann equation in a leading-order ϕ^4 -theory [8, 14] as well as diffusion models using a strongly-screened QCD amplitude [11, 12] suggest that condensation, if it occurs, happens after an initial ‘refractory period’ with negligible change in n_c .

Then, at a characteristic onset time, the condensate density rapidly grows to its equilibrium value. We compute the time of onset, τ_o , and the time of maximum flux, τ_c as timescales for condensation.

Other interesting quantities are the ‘differential entropy’, $\sigma_k = ds/dk$, and the ‘differential number’, $\nu_k = dn/dk$. Knowing σ_k and ν_k as functions of time indicates how particles and entropy are reorganised between different momenta during the evolution. Also relevant is the collision term, $C_{[f]}$, which underpins all results in this thesis.

2.2 Solutions of the Boltzmann equation

Firstly, we solve the Boltzmann equation for representative under- and overpopulated initial conditions. In each case, we check the sensibility of our results and compare solutions which are generated using the w - and s -kernels respectively. Results are generated on an equidistant energy grid with cut-off Λ and resolution Δ_E . Details of the numerics are provided in [Appendix B](#).

In [Figure 9](#), the regular parts of f and $C_{[f]}$ are shown at various stages of the equilibration of a typical underpopulated system (in this example using the w -kernel). Clearly, the collision term emphasises scatterings into the infrared. As time proceeds, the collision term diminishes in magnitude whilst maintaining most of its support at ‘small’ momenta. The form of $C_{[f]}$ is plain to see in the evolution of f ; the early stages of equilibration are characterised by a rapid redistribution of ‘hard’ gluons into the ‘soft’ modes. The later stages of equilibration are relatively uninteresting.

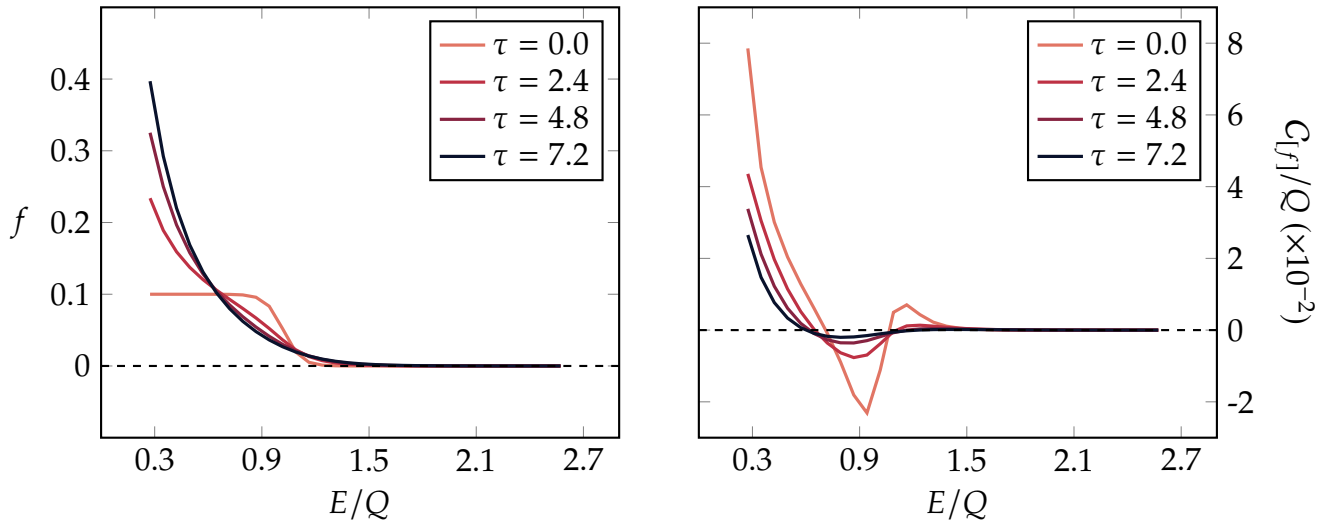


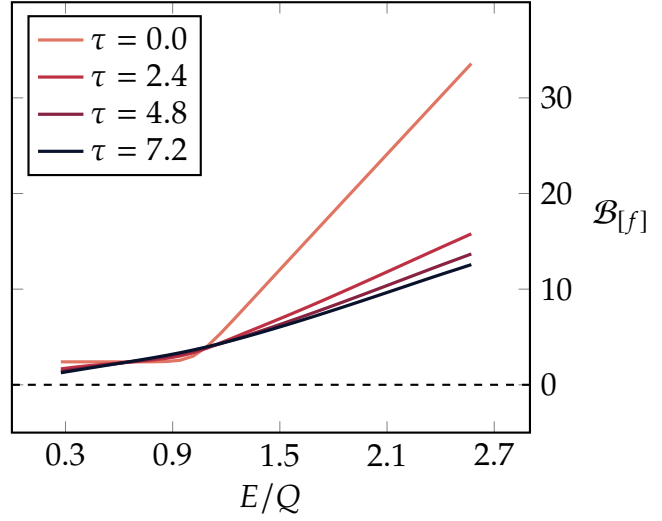
Figure 9: Evolution generated by the w -kernel. Shown are f and $C_{[f]}$ at different times for an initial condition with $\alpha = 0.3$, $\chi = 0.1$, $n_c(0)/Q^3 = 10^{-5}$, $m/Q = 0.2$ and $\mu/Q = 0.33$. Results were computed with an energy resolution $\Delta_E/Q \approx 0.07$ and cut-off $\Lambda/Q \approx 2.6$.

The large time limit of f shown in [Figure 9](#) seems like a Bose function. By computing the Bose transform (2.1) of f , we verify the equilibrium state obtained in our numerical solution is indeed a Bose function (the Bose transforms of f tend to a straight line) and that the equilibrium parameters realised ‘on-grid’ agree with those predicted by the conservation laws (1.10) (in this case, $\mu/Q \approx 0.03$ and $T/Q \approx 0.25$). The Bose transforms of the f in [Figure 9](#) are shown in [Figure 10](#).

Although hard to see in [Figure 9](#), the Bose transforms shown in [Figure 10](#) make clear that the mean densities of gluons in the modes $k \gtrsim Q$ increase between initial and final states (but only

marginally). As expected for an underpopulated system, $\mathcal{B}_{[f]} > 0$ as $k \rightarrow 0$. Otherwise, the behaviour for $k \lesssim Q$ is already clear in Figure 9.

Figure 10: $\mathcal{B}_{[f]}$ for the f shown in Figure 9. Keep in mind that $\mathcal{B}_{[f]}$ is large when f is small and vice versa.



The entropy and differential entropy for the system originally described in Figure 9 are shown in Figure 11. As expected, the numerical solution for f conforms to Boltzmann's H -theorem (the entropy increases monotonically with time). Recall that the smallest energy for which we track the regular part of the distribution function is not m but rather $m + \Delta_E$ (cf. Appendix B). Therefore, the non-zero σ_k seen in the infrared in Figure 11 is actually due to excited (but very soft) modes. The expected behaviour, $\sigma_k \rightarrow 0$ as $k \rightarrow 0$, is not excluded by Figure 11. The entropy relaxation time computed from the s -evolution in this example is $\tau_s \simeq 3.5$.

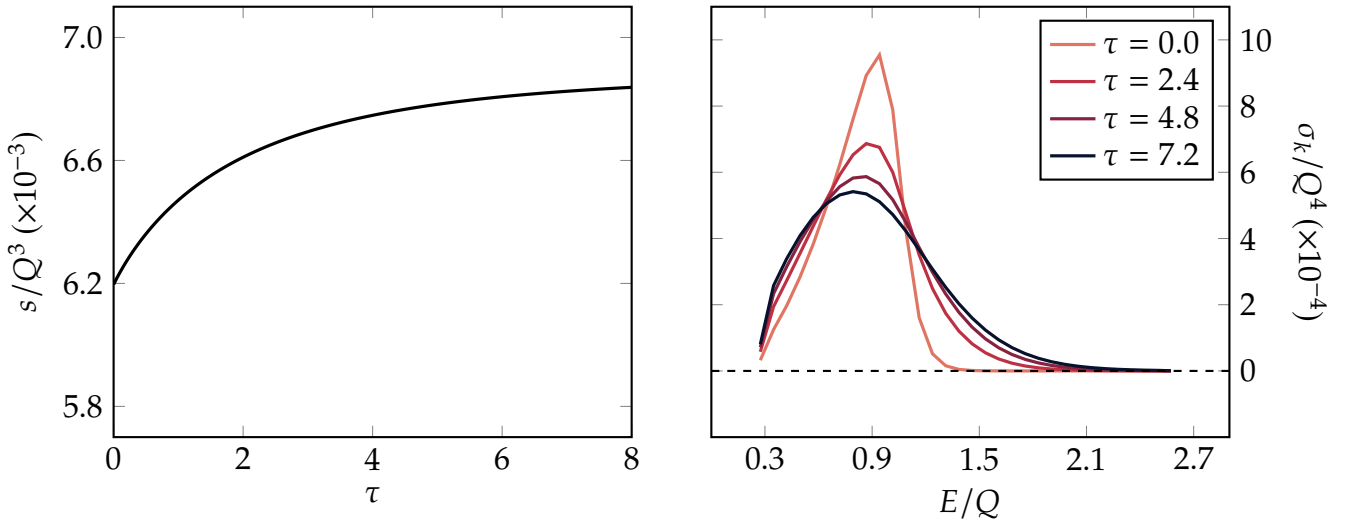


Figure 11: s and σ_k for the system initially described in Figure 9.

Finally, Figure 12 shows n_c and ν_k for the system originally described in Figure 9. Figure 12 verifies that condensate dynamics of underpopulated systems are trivial; any non-zero condensate seed is quickly depleted as condensate gluons are ‘up-scattered’ into excited modes. The differentials ν_k and σ_k are strikingly similar. A quantity of interest is, therefore, the average entropy

per gluon with given momentum σ_k/ν_k . A quick calculation confirms $\sigma/\nu = \ln(\bar{f}/f) + (1/f) \ln \bar{f}$, which is simply the Bose transform of f up to a correction $(1/f) \ln \bar{f}$, which in this case is small.

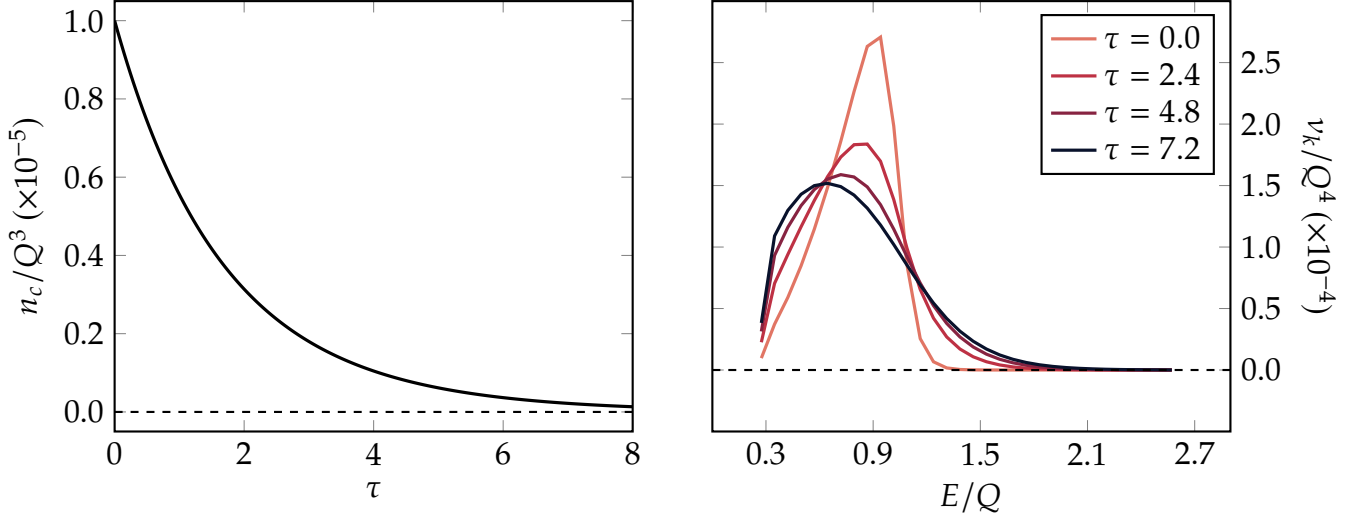


Figure 12: n_c and ν_k for the system initially described in Figure 9.

Figures 9, 10, 11 and 12 show results for an underpopulated initial condition which were all generated using the w -kernel. Results generated for the same initial condition, but using the s -kernel, are qualitatively the same up to a lengthening of the equilibration timescales (for example, the equivalent entropy relaxation time in the case of strong screening is $\tau_s \simeq 5.2$). Timescales for equilibration, and how they compare between the two kernels, is a topic explored later in this section.

We now consider a representative overpopulated initial condition. Figure 13 shows the entropy and condensate dynamics of an overpopulated system, where the evolution is generated using the s -kernel. Regarding the condensate dynamics, Figure 13 confirms the formation of a macroscopic condensate, preceded by a refractory period with little change in n_c . The dynamics generated using the w -kernel are similar up to an overall acceleration of the dynamics relative to the s -kernel. The n_c -evolutions obtained for both kernels are therefore consistent with the leading-order ϕ^4 -model [8, 14] and test-particle simulations using the weakly-screened QCD amplitude [16, 21].

Also shown in Figure 13 is the number density of soft gluons, ' n_{soft} '. During the early stages of equilibration, gluons typically accumulate in the infrared. The increased occupancies of the soft modes enhance the rates of scattering into and out of the condensate, which are temporarily held in a state of detailed balance. Once n_{soft} exceeds a threshold value, this detailed balance is abruptly violated and soft gluons flood into the zero mode. The onset time for the example shown in Figure 13 is $\tau_o \simeq 20$ while the time of maximum flux into the condensate is $\tau_c \simeq 26$.

The entropy evolution shown in Figure 13 again conforms to Boltzmann's H -theorem (s increases monotonically with time). Interestingly, there is a signature of the n_c -dynamics in s . Like the s -dynamics for an underpopulated system, the s shown in Figure 13 grows without delay as excited gluons are re-arranged among the $k > 0$ modes. However, s begins to plateau as excited gluons (now mostly equilibrated) slowly gather in the infrared. Once condensation occurs, $\partial_t s$ is enhanced by a reorganisation of gluons in the $k > 0$ modes which is catalysed by condensate scatterings. The $k = 0$ mode carries no entropy, so the abrupt increase in s must originate in the regular part of f . The 'bump' in s around the time of condensation is corroborated in [16].

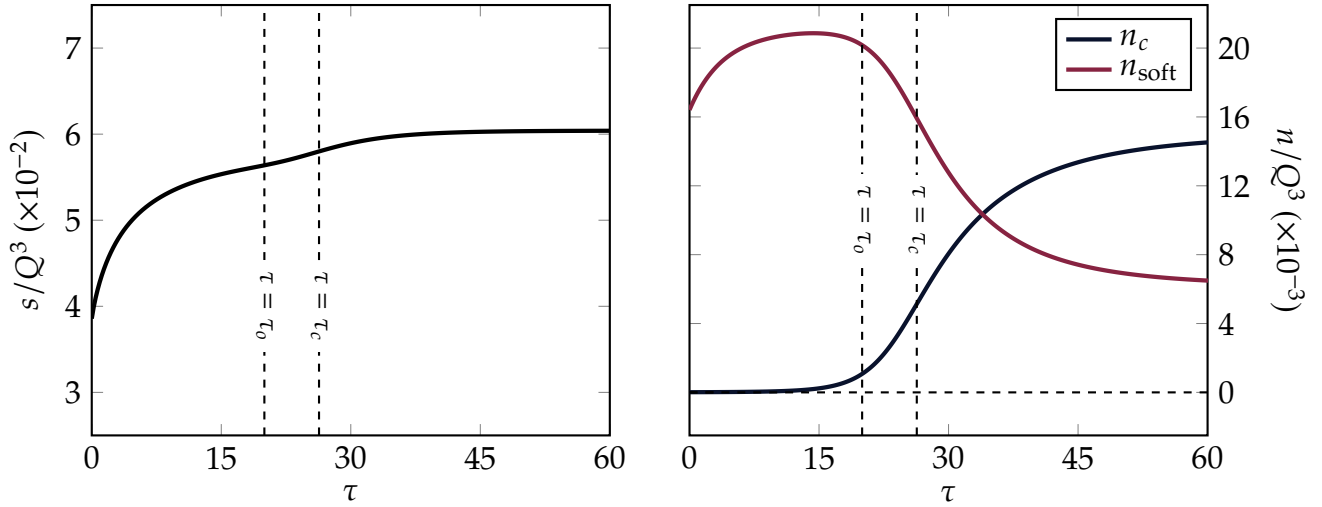


Figure 13: s , n_c and n_{soft} for an initial condition with $\alpha = 0.3$, $\chi = 2$, $n_c(0)/Q^3 = 10^{-5}$, $m/Q = 0.2$ and $\mu/Q \simeq 1.4$. n_{soft} is the number density of gluons with ‘soft’ (but non-zero) momenta $0 < k \lesssim 0.6Q$. Computed using the s -kernel with an energy resolution $\Delta_E/Q \simeq 0.14$ and cut-off $\Lambda/Q \simeq 4.7$.

Entropy relaxation in the Boltzmann model

Firstly, we compute entropy relaxation times (2.3) for different initial conditions. Three ensembles of initial conditions are studied, where each ensemble varies one of α , χ or m/Q . The s -relaxation times are calculated using both the w - and s -kernels for each initial condition. The choice of kernel is indicated by a superscript in the tables of results (e.g. $\tau_s^{[w]}$ for the w -kernel and $\tau_s^{[s]}$ for the s -kernel). The s -relaxation time quantifies how quickly the regular part of the distribution function thermalises. The goals of this section are, firstly, to estimate τ_s in weakly- and strongly-screened gluon plasmas and, secondly, to determine how τ_s scales with α , χ and m/Q .

The s -evolutions of overpopulated systems contain additional structure around the time of condensation (cf. Figure 13) and are not well-fit by an exponential model (2.3). By contrast, the s -evolutions of underpopulated systems are sufficiently exponential to validate (2.3) (cf. Figure 11). Hence, we only sample underpopulated initial conditions when calculating τ_s . Because the condensate of an underpopulated system is quickly depleted (cf. Figure 12), the influence of condensate scatterings on the regular distribution, f , are strongly suppressed compared to regular-regular scatterings (cf. (1.36)). Therefore, condensate dynamics are ignored by setting $n_c(0) = 0$.

The first family of initial conditions are identical up to the value of the coupling (parameter choices documented in Table 1). The resulting τ_s are shown in Table 1. We see the expected behaviour in τ_s ; the entropy relaxation time is a monotonically *decreasing* function of the coupling. The coupling enters $|\mathcal{M}|^2$ in two ways. Firstly, there is a leading α^2 -dependence in the square amplitude (1.39). Otherwise, μ (1.46) (which enters the screened versions of $|\mathcal{M}|^2$; (1.41) and (1.42)) is a function of α . The dependence of τ_s on μ is subtle. Like the Fokker-Planck kernel discussed in section 1.10, the w - and s -kernels used in the Boltzmann model have logarithmic μ -dependence, $\mathcal{K} \propto \ln(Q/\mu)$. The expected logarithmic scaling, $\alpha^2 \tau_s \propto 1/\ln(Q/\mu)$, is verified for both the w - and s -kernels. The results in Table 1 are visualised in Figure 14.

α	χ	$n_c(0)/Q^3$	m/Q	μ/Q	$\tau_s^{[w]}$	$\tau_s^{[s]}$
0.1	0.1	0	0.2	0.19	17.34	40.73
0.2	0.1	0	0.2	0.26	4.79	14.31
0.3	0.1	0	0.2	0.32	2.27	7.95
0.4	0.1	0	0.2	0.37	1.34	5.29
0.5	0.1	0	0.2	0.42	0.89	3.87

Table 1: (α -scaling) The s -relaxation times calculated for the first ensemble of systems. The screening mass is determined using (1.46).

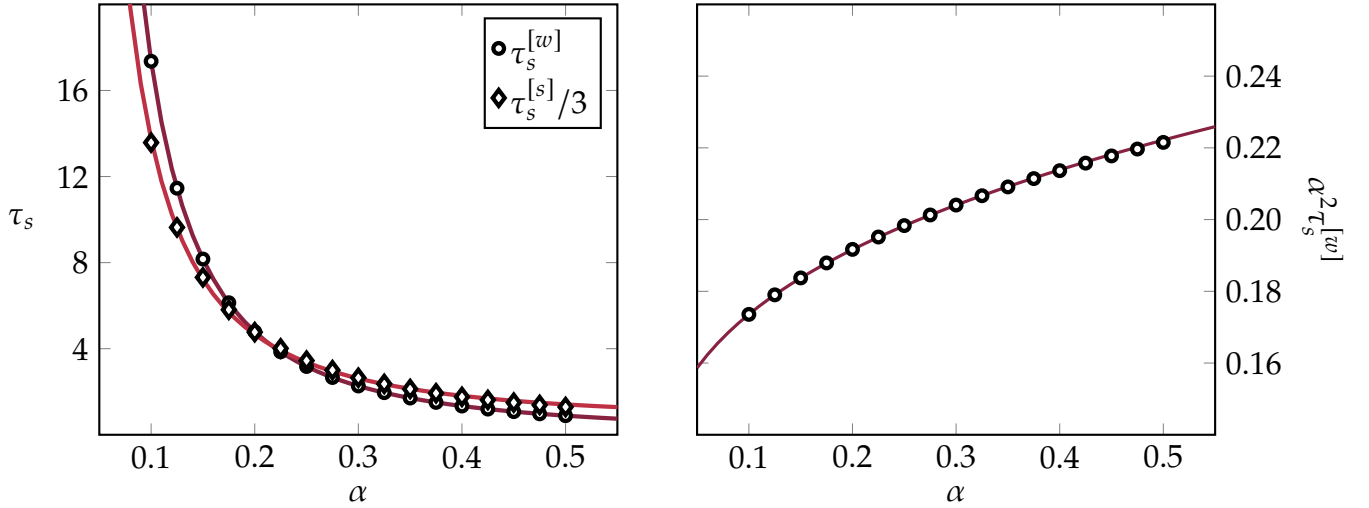


Figure 14: On the left is $\tau_s(\alpha)$. The markers are data from Table 1 and the solid lines are fits to the data, $\alpha^2 \tau_s = a/\ln(b/\alpha)$. For the w -kernel, $a \simeq 1.3$ and $b/Q \simeq 160$. For the s -kernel, $a \simeq 1.1$ and $b/Q \simeq 1.4$. On the right, $\alpha^2 \tau_s^{[w]}(\alpha)$ clarifies the logarithmic dependence on μ .

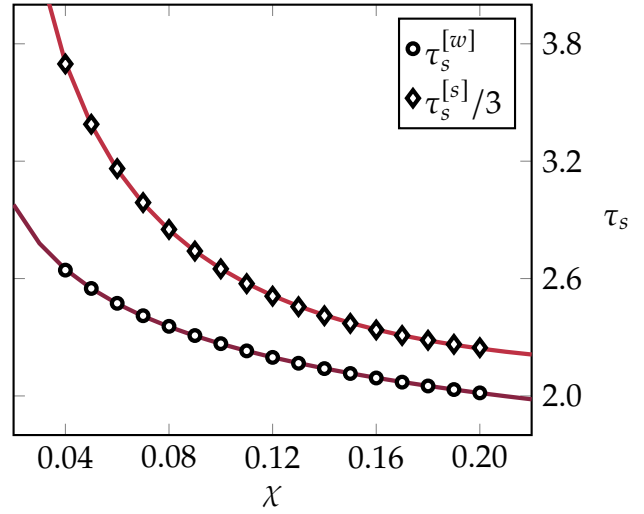
In the second ensemble of initial conditions, the population parameter χ is varied. Table 2 shows the τ_s calculated for different χ . For both kernels, τ_s is monotonically *decreasing* against χ . Like the coupling, χ influences the timescales logarithmically through the screening mass (which is determined using an initial condition defined by χ). The population parameter also affects the statistical factor through f (at $\tau = 0$, $\mathcal{F}_{[f]} \propto \chi^3$). The results of Table 2 are visualised in Figure 15.

In Figure 15, the s -relaxation times are fit using a combination of logarithmic and power-law behaviours, $\tau_s \propto 1/\chi^p \ln(1/\chi)$. The logarithmic part of the scaling relation is due to μ 's dependence on χ . As χ increases, so does the screening mass computed using (1.46) with $f = f_{\text{in}}$ (1.21). Increasing χ therefore strengthens screening effects in the plasma which tend to lengthen equilibration times. However, the increased density of collision partners at larger χ accelerates equilibration (a feature captured with the $1/\chi^p$ component of the scaling relation, where $0.1 \lesssim p \lesssim 2$ depending on parameter choices). As demonstrated in Figure 15, the acceleration due to an increased abundance of collision partners dominates over stronger screening effects; $\tau_s(\chi)$ is monotonically *decreasing*.

α	χ	$n_c(0)/Q^3$	m/Q	μ/Q	$\tau_s^{[w]}$	$\tau_s^{[s]}$
0.3	0.04	0	0.2	0.21	2.64	11.09
0.3	0.08	0	0.2	0.29	2.36	8.55
0.3	0.12	0	0.2	0.35	2.20	7.53
0.3	0.16	0	0.2	0.41	2.10	7.01
0.3	0.20	0	0.2	0.46	2.02	6.74

Table 2: (χ -scaling) The s -relaxation times calculated for the second ensemble of systems. The screening mass μ is determined using (1.46).

Figure 15: Results for $\tau_s(\chi)$. The markers are data from Table 2. The solid lines are fits, $\tau_s = a/\chi^p \ln(b/\chi)$. For the w -kernel, $p \simeq 0.2$. For the s -kernel, $p \simeq 0.7$.



Finally, we investigate how τ_s relates to m/Q . In Table 3 are data generated treating m/Q as an independent variable, which are then visualised in Figure 16. Evidently, τ_s is monotonically *decreasing* against m/Q . Note that changing m while holding χ fixed leaves the initial distribution (1.21) unchanged ((1.21) takes as its argument the momentum k and not the energy E_k). By increasing m without a corresponding change in f_{in} , the mean energy of collisions $\langle \sqrt{s} \rangle$ is greater. This is reflected by an increase in the square amplitude (1.39) which scales like s^2 (becoming arbitrarily large as $m \rightarrow \infty$ ⁵). Therefore, increasing m without simultaneously changing the initial distribution or screening mass inflates the square amplitude, resulting in an accelerated evolution (at least at early times).

Otherwise, the dependence of τ_s on m/Q seems moderate; increasing m/Q by a factor of five results in a reduction of τ_s by roughly one third for the w -kernel and a reduction of almost half for the s -kernel. For other choices of parameters (e.g. by changing χ or ξ) the variation of τ_s against m/Q can be made milder or stronger. A future survey would do well to adjust μ/Q in lockstep with m/Q , or to express f_{in} as a distribution in energies.

⁵The Mandelstam variables t and u , on the other hand, are bounded as functions of m .

α	χ	$n_c(0)/Q^3$	m/Q	μ/Q	$\tau_s^{[w]}$	$\tau_s^{[s]}$
0.3	0.1	0	0.1	0.33	2.37	8.86
0.3	0.1	0	0.2	0.32	2.27	7.95
0.3	0.1	0	0.3	0.32	2.15	6.87
0.3	0.1	0	0.4	0.30	2.02	5.75
0.3	0.1	0	0.5	0.29	1.87	4.66

Table 3: (m/Q -scaling) The s -relaxation times calculated for the third ensemble of systems. The screening mass μ is determined using (1.46).

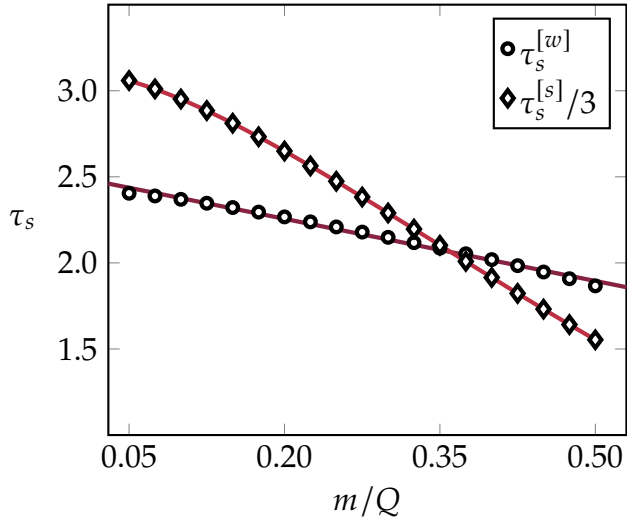


Figure 16: Results for $\tau_s(m/Q)$. The markers are data from Table 3. The solid lines are linear fits to the data.

As time proceeds, the system evolves towards an equilibrium state described by a Bose distribution, whose Bose transform is a straight line (2.2). As a measure of convergence, we interpolate the discrete f with b -lines and compute the dispersion in slope parameters (which must vanish at equilibrium, but is otherwise positive). The b -lines interpolating f between E_i and E_{i+1} are defined

$$f^i(E) = \frac{1}{e^{a_i E + b_i} - 1}, \quad (2.4)$$

where a_i and b_i are locally-determined b -line parameters. To be clear, E_i and E_{i+1} are consecutive points on the energy grid. In the space of Bose transforms, (2.4) corresponds to a piecewise linear interpolation which makes calculation of the a_i s and b_i s elementary. For a temperature to be well-defined, the dispersion in the a_i s must be zero (or at least within bounds). We show an example evolution of the a_i parameters in Figure 17. In the same figure is the evolution of the difference between maximum and minimum slope parameters,

$$\Delta a = \max\{a_i\} - \min\{a_i\}.$$

The straightening of the Bose transform (as described by Δa) seems to proceed exponentially after the early stages of equilibration.

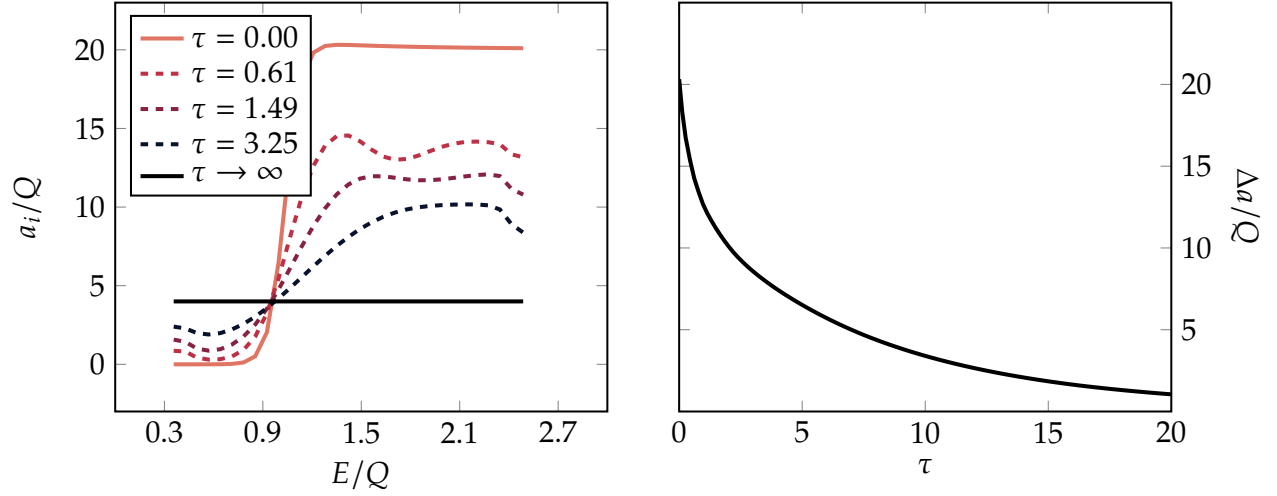


Figure 17: a_i and Δa . Results are computed for an initial condition with $\alpha = 0.1$, $\chi = 0.1$, $n_c(0) = 0$ and $m/Q = 0.25$. The energy grid has resolution $\Delta E/Q \simeq 0.08$ and cut-off $\Lambda/Q \simeq 2.5$.

By fitting $\Delta a \propto \exp[-\tau/\tau_l]$, we extract ‘linearisation times’, τ_l , as alternative quantifiers for the pace of equilibration. Like for the s -relaxation time, we determine the scaling behaviour of τ_l against changes in α , χ and m/Q . Using the same initial conditions described in Tables 1, 2 and 3, we find the scaling relations for τ_l are qualitatively the same as those for τ_s (and for both kernels). Altogether, τ_l is generally proportional to τ_s with $\tau_l/\tau_s \simeq 2$. Understanding that our results come with some uncertainty, we can at least say that τ_s and τ_l are the same order of magnitude.

This section’s results are summarised in the form of an ‘empirical’ relation which describes how the s -relaxation time varies against changes in the coupling, population parameter or gluon mass. Assuming the variations are small (roughly separable)

$$\tau_s \propto \frac{\omega(m/Q)}{\alpha^2 \ln(Q/\mu) \chi^p}, \quad (2.5)$$

where ω is some monotonically decreasing function of m/Q and p is a number of order one. The s -relaxation times generated by the w - and s -kernels were found to have roughly the same scaling behaviours, up to an overall constant. Therefore, the ‘rule-of-thumb’ scaling relation (2.5) is applicable to both kernels.

The effect of the screening ratio

Before discussing condensation timescales, we quantify the influence of the screening ratio (1.47) on the evolution of underpopulated systems. To gauge the effect of ξ without too much effort, we estimate the s -relaxation time for a given underpopulated initial condition using different values for ξ . We probe $0.2 \leq \xi \leq 2$. The initial conditions we simulate are documented in Table 4 and the evolution is calculated on a grid with $\Delta_E \approx 0.07$ and $\Lambda/Q \approx 2.5$. Results are generated using the s -kernel and are shown in Table 4.

α	χ	$n_c(0)/Q^3$	m/Q	ξ	$\tau_s^{[s]}$
0.3	0.1	0	0.25	0.2	2.02
0.3	0.1	0	0.25	0.5	3.32
0.3	0.1	0	0.25	0.8	4.42
0.3	0.1	0	0.25	1.1	5.43
0.3	0.1	0	0.25	1.4	6.37
0.3	0.1	0	0.25	1.7	7.24
0.3	0.1	0	0.25	2.0	8.01

Table 4: (ξ -scaling) s -relaxation times associated with a given initial condition for different ξ .

As we see in Table 4, the entropy relaxation time (and presumably all other timescales) lengthen for increasing ξ , an expected result. The screening mass is directly proportional to ξ (which means that varying ξ like is done in Table 4 just varies the Coulomb logarithm appearing in as a pre-factor to the kernel). Over the range of sampled ξ , the minimum and maximum τ_s differ by a factor of roughly four. At present, there are no estimates for a ‘plausible’ ξ . Specifying ξ remains the largest source of uncertainty in our calculations. For definiteness, we keep $\xi = 1$ in our calculations.

Condensation timescales in the Boltzmann model

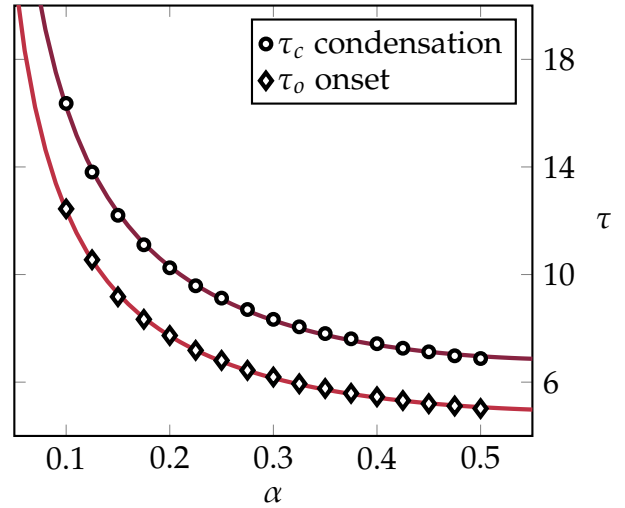
We now investigate the condensate dynamics of overpopulated systems. The central observables in the forthcoming analyses are the condensation time, τ_c , which we define as the time of maximum flux into the condensate, and the onset time, τ_o , which we define as the time when f at the first grid-point starts to decrease. Like for the s -relaxation time, we determine the scaling behaviours of these timescales against α , χ and m/Q (and compare results for both kernels). An additional parameter relevant to overpopulated systems is the ‘condensate seed’, $n_c(\tau = 0)$. We also test how the seed value affects condensation timescales. In every example, $M = 32$ and $\Lambda = 10T_\infty$.

In Table 5, are τ_c and τ_o for the first ensemble of systems (which vary the coupling). The condensation and onset times are monotonically decreasing against α . We model the α -scaling behaviour of the condensation timescales in a similar manner as for τ_s ; τ_c and τ_o are proportional to $1/\alpha^p \ln(Q/\mu)$ where p may differ from two. Figure 18 shows results of Table 5 associated with the s -kernel. The w -kernel’s results are not shown in a figure, but their α -scaling is qualitatively the same as for the s -kernel.

α	χ	$n_c(0)/Q^3$	m/Q	μ/Q	$\tau_c^{[w]}$	$\tau_c^{[s]}$	$\tau_o^{[w]}$	$\tau_o^{[s]}$
0.1	2	10^{-5}	0.8	0.67	4.92	16.36	3.82	12.44
0.2	2	10^{-5}	0.8	0.94	2.06	10.25	1.59	7.73
0.3	2	10^{-5}	0.8	1.16	1.28	8.33	0.98	6.19
0.4	2	10^{-5}	0.8	1.34	0.91	7.43	0.71	5.46
0.5	2	10^{-5}	0.8	1.49	0.71	6.88	0.55	5.02

Table 5: (α -scaling) Here are the condensation and onset times calculated for the first ensemble of systems. The screening mass μ is determined using (1.46).

Figure 18: Condensation timescales τ_c and τ_o . The markers are data from Table 5 (corresponding to the s -kernel). The solid lines are fits of the form $\alpha^p \tau = a/\ln(b/\alpha)$. For τ_c , we find $a \simeq 3.7$, $b \simeq 1.5$ and $p \simeq 1.1$. For τ_o , we find $a \simeq 2.7$, $b \simeq 1.6$ and $p \simeq 1.1$. The fits for τ_c and τ_o seem to differ only in their overall magnitude, a .



Before investigating the χ -scaling, we clarify some assumptions about the variation of τ_c and τ_o against χ . As already known, if $\chi < \chi_c$ the system is underpopulated and no condensation occurs. If $\chi = \chi_c$, the system is critically populated and, still, no condensation occurs. However, if $\chi > \chi_c$, a Bose condensate will form in finite time. Important in constructing our fits of τ_c and τ_o against χ is an understanding of the limiting behaviour $\chi \rightarrow \chi_c$ ‘from above’. Because no condensation happens when $\chi = \chi_c$, a first guess is that $\tau_o \rightarrow \infty$ as $\chi \rightarrow \chi_c$. To this end, we fit $\tau_o \propto 1/(\chi - \chi_c)^p$ for some exponent p . Results are shown in Table 6 and data produced using the s -kernel are shown in Figure 19.

According to Figure 19, the condensation and onset times scale roughly like $1/(\chi - \chi_c)$. Recall that, for overpopulated systems, the asymptotic condensate density scales like $n_c(\tau \rightarrow \infty) \sim \chi - \chi_c$. In other words, the condensation time is inversely proportional to the number of ‘surplus’ gluons in the system, $\tau_c \sim 1/n_c(\tau \rightarrow \infty)$. This result is sensible; the more gluons that are packed into the $k > 0$ modes, the more quickly a critical number of them can accumulate in the infrared and trigger condensation. This statement is informed by our finding that $\tau_s \sim 1/\chi^p$; the rate at which gluons are reorganised in the $k > 0$ modes is roughly proportional to the number of collision partners.

α	χ	$n_c(0)/Q^3$	m/Q	μ/Q	$\tau_c^{[w]}$	$\tau_c^{[s]}$	$\tau_o^{[w]}$	$\tau_o^{[s]}$
0.3	1.50	10^{-5}	0.8	1.01	1.88	10.48	1.45	7.65
0.3	1.75	10^{-5}	0.8	1.08	1.51	9.16	1.16	6.77
0.3	2.00	10^{-5}	0.8	1.16	1.28	8.33	0.98	6.19
0.3	2.25	10^{-5}	0.8	1.22	1.10	7.75	0.85	5.75
0.3	2.50	10^{-5}	0.8	1.29	0.98	7.34	0.76	5.46

Table 6: (χ -scaling) Condensation and onset times for the second ensemble. Screening masses are determined using (1.46).

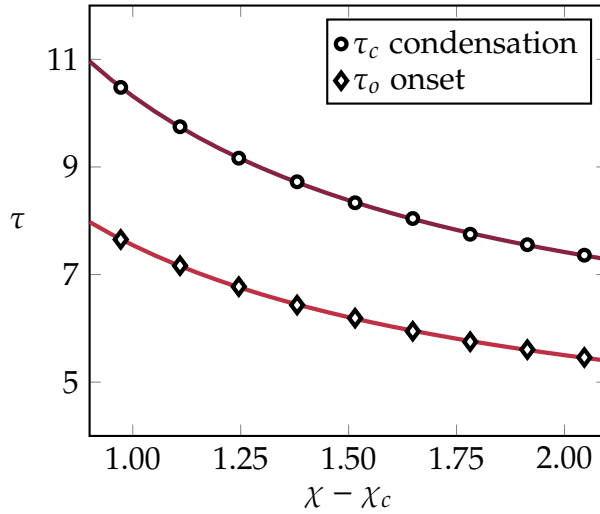


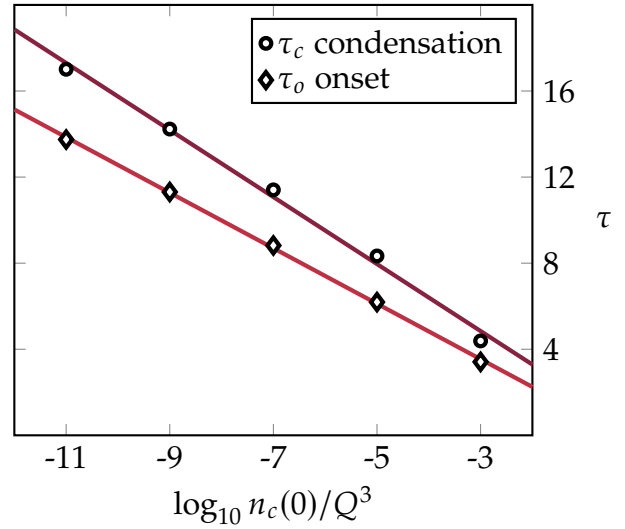
Figure 19: τ_o and τ_c as functions of χ . The markers are data from Table 6 (corresponding to the s -kernel). The solid lines are fits $\tau = a/(\chi - \chi_c)^p + c$. For τ_c , $a \simeq 5.7$, $p \simeq 1.0$ and $c \simeq 4.6$. For τ_o , $a \simeq 4.6$, $p \simeq 0.9$ and $c \simeq 3.0$. In both fits, p is order one, which suggests $p \simeq 1$ is a good estimate to keep in our empirical relation. Scaling behaviour for the w -kernel is the same.

Data for the third ensemble of initial conditions, which vary the $n_c(0)$, are shown in Table 7. The condensate seed is varied across eight orders of magnitude. Evidently, the condensation timescales are not very sensitive to changes in $n_c(0)$; recognisable changes in τ_c and τ_o only occur when $n_c(0)$ is varied across many orders of magnitude. Condensation timescales are typically smaller for larger values of the $n_c(0)$. This behaviour is sensible because $\partial_t n_c \propto n_c$ (1.37). By increasing $n_c(0)$, the effect of the condensate integrals in the early stages of the evolution are amplified. That the influence of $n_c(0)$ isn't stronger is probably due to an initial dominance of processes involving only excited gluons. Our results in this regard seem to corroborate the findings of [14], which varied the seed value in a leading-order ϕ^4 -theory.

α	χ	$n_c(0)/Q^3$	m/Q	μ/Q	$\tau_c^{[w]}$	$\tau_c^{[s]}$	$\tau_o^{[w]}$	$\tau_o^{[s]}$
0.3	2	10^{-11}	0.8	1.16	2.25	17.01	1.93	13.74
0.3	2	10^{-9}	0.8	1.16	1.95	14.23	1.64	11.31
0.3	2	10^{-7}	0.8	1.16	1.62	11.40	1.32	8.82
0.3	2	10^{-5}	0.8	1.16	1.27	8.33	0.98	6.19
0.3	2	10^{-3}	0.8	1.18	0.77	4.39	0.58	3.41

Table 7: ($n_c(0)$ -scaling) The τ_c and τ_o calculated for the third ensemble of systems. The screening mass μ is determined using (1.46).

Figure 20: τ_o and τ_c as functions of $\log_{10} n_c(0)/Q^3$. The markers are data from Table 7 generated using the s -kernel. The solid lines are fits, $\tau = a \log_{10} n_c(0)/Q^3 + b$. For τ_c , $a \simeq -1.6$ and $b \simeq 0.2$. For τ_o , $a \simeq -1.3$ and $b \simeq -0.3$. Scaling behaviour for the w -kernel is the same.



Data for the final ensemble of initial conditions, which varies m/Q , are shown in Table 8. Results for the s -kernel are then visualised in Figure 21. We see that τ_c and τ_o scale in roughly inverse proportion to m/Q . A potential explanation is that the kernels of the condensate integrals (D.2) and (D.3) contain leading factors which scale like m^2 . Against the overall $\sim 1/m$ behaviour of the condensate integrals in (1.35), this amounts to a net $\sim m$ proportionality in $C_{[f]}^{1\circ 34}$ and $C_{[f]}^{12\circ 4}$, further implying that $\partial_t n_c \propto m$ in (1.37). This result is compatible with the observation that condensation cannot occur for massless bosons.

α	χ	$n_c(0)/Q^3$	m/Q	μ/Q	$\tau_c^{[w]}$	$\tau_c^{[s]}$	$\tau_o^{[w]}$	$\tau_o^{[s]}$
0.3	2	10^{-5}	0.5	1.29	1.70	13.89	1.32	10.28
0.3	2	10^{-5}	0.6	1.24	1.54	11.60	1.20	8.57
0.3	2	10^{-5}	0.7	1.20	1.40	9.78	1.09	7.26
0.3	2	10^{-5}	0.8	1.16	1.28	8.33	0.98	6.19
0.3	2	10^{-5}	0.9	1.12	1.15	7.21	0.89	5.36

Table 8: (m/Q -scaling) The τ_c and τ_o calculated for the fourth ensemble of systems. The screening mass μ is determined using (1.46).

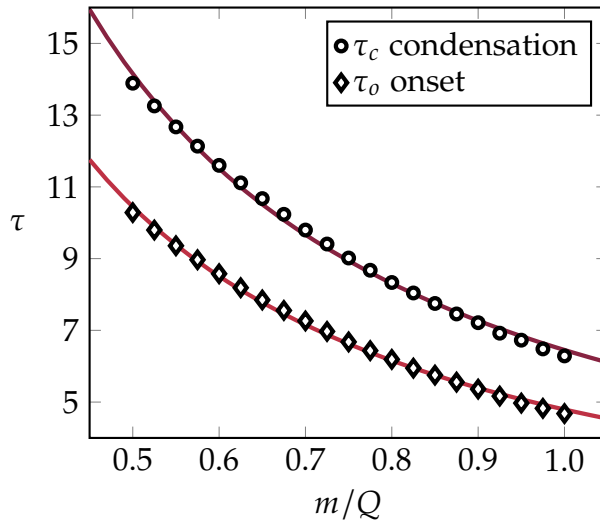


Figure 21: τ_c and τ_o as functions of m/Q . The markers are data from Table 8, corresponding to the s -kernel. The solid lines are fits, $\tau = a/(m/Q)^p$. For τ_c , $a \simeq 6.5$ and $p \simeq 1.1$. For τ_o , $a \simeq 4.8$ and $p \simeq 1.1$. Since p is order one for both kernels, we approximate $p \simeq 1$ in our empirical relation.

Like was done for the s -relaxation time, we summarise the results of this section in the form of an ‘empirical’ relation which describes how τ_c and τ_o respond to changes in the coupling, population parameter, condensate seed and gluon mass. Of course, we don’t exclude the possibility that the full dependence of the condensate timescales on these parameters is non-separable. Assuming that variations in these parameters are small

$$\tau_{\text{zero}} \propto \frac{Q/m}{\alpha^p \ln(Q/\mu)(\chi - \chi_c)}, \quad (2.6)$$

where all exponents appearing in power laws have been rounded to the nearest integer. The dependence of τ_{zero} on the condensate seed has been excluded from (2.6) as we consider its effect to be negligible (at least against small variations in $n_c(0)/Q^3$). Importantly, the exponent p is of order one and, for broader surveys of the parameters, can range between $1 \lesssim p \lesssim 2$ depending on whether we use the w - or s -kernel.

2.3 Calibrating the Coulomb logarithm

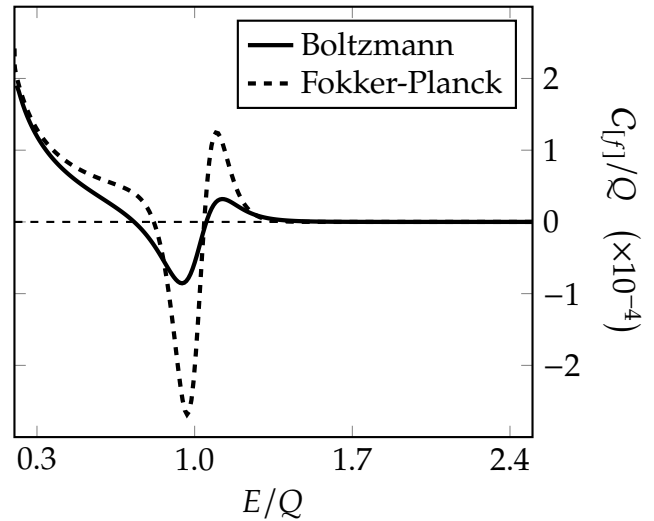
As demonstrated in Figure 7, the Fokker-Planck kernel is logarithmically divergent. The kernel (and hence, the collision term) are proportional to a Coulomb logarithm (1.45), which is a logarithm of the ratio of hard and soft momentum scales; $\mathcal{L} = \ln(\Lambda_{\text{FP}}/\mu)$. As discussed in section 1.11, the screening mass is fixed by the initial condition of the plasma, up to our choice of ‘screening ratio’, ξ . Therefore, μ is quantified with ‘controllable’ uncertainty.

The value of Λ_{FP} is more difficult to constrain. The Boltzmann collision term is ultraviolet convergent, which justifies our use of a finite energy grid in section 2.2. By contrast, the Fokker-Planck model requires the artificial insertion of an ultraviolet cut-off, Λ_{FP} , due to the small-momentum expansion discussed in Appendix E. Blaizot et al. argue in [11] that $\Lambda_{\text{FP}} \propto T$, but set $\Lambda_{\text{FP}} = 4Q$ in their numerics. Be that as it may, there is an additional uncertainty (besides that from the infrared cut-off) in the Fokker-Planck approximation which arises from specifying Λ_{FP} . We resolve the uncertainty in Λ_{FP} by comparison to our Boltzmann approach.

Importantly, the Fokker-Planck and Boltzmann models are only expected to be similar in the limit of small α . The leading factors of α^2 in both kernels don’t affect calibration. The α -dependence of μ , on the other hand, is significant. Because the Fokker-Planck kernel is, quite literally, the leading-log approximation of the full Boltzmann kernel, the two ought to agree only when the Coulomb logarithm is large (i.e. $\mu \sim \alpha^{1/2}$ is small). The small- α limit therefore serves as an ‘island of safety’ when calibrating Λ_{FP} .

As an example, Figure 22 shows the strongly-screened Boltzmann collision term alongside the Fokker-Planck collision term, for a given f and with ‘small’ coupling, $\alpha = 10^{-2}$. Firstly, we see that the collision terms in the Boltzmann and Fokker-Planck models have the same overall structure; the locations of roots, stationary points and infrared behaviour are similar. However, there is an obvious discrepancy between their magnitudes (for the represented choice of Λ_{FP}/T).

Figure 22: Collision terms in the Boltzmann and Fokker-Planck models for a given f . The regular part of the test distribution is prescribed $f(k) = 0.1/(\exp[20(k-1)] + 1)$ and there is no condensate. The coupling is small, $\alpha = 10^{-2}$. The screening mass is determined using (1.46) and $m/Q = 0.2$. The collision terms are computed on grids with resolution $\Delta E/Q \simeq 10^{-2}$ and cut-off $\Lambda = 10T$.



Because most dynamics occur during the early stages of equilibration, we use the initial rate of entropy production, $\dot{s}_0$, as a measure of agreement between the two models. The initial rate of entropy production in the Fokker-Planck model, $\dot{s}_0^{\text{FP}}$, is calculated as a function of Λ_{FP}/T . The difference between $\dot{s}_0$ in the Boltzmann model, $\dot{s}_0^{\text{BE}}$, and in the Fokker-Planck model is then constructed $\Delta\dot{s}_0 = \dot{s}_0^{\text{BE}} - \dot{s}_0^{\text{FP}}$. The optimal value for Λ_{FP}/T minimises $|\Delta\dot{s}_0|$. Figure 23 demonstrates this procedure for a representative initial condition, with optimal $\Lambda_{\text{FP}}/T \simeq 4.8$.

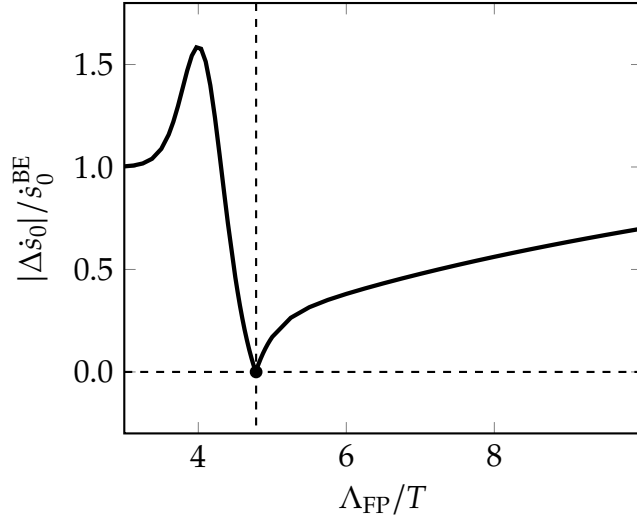


Figure 23: $|\dot{s}_0^{\text{BE}} - \dot{s}_0^{\text{FP}}|/\dot{s}_0^{\text{BE}}$. These are calculated for the same f_0 described in Figure 22 with the same grid settings. The vertical dashed line indicates the value of Λ_{FP}/T where $\dot{s}_0^{\text{FP}}$ coincides with $\dot{s}_0^{\text{BE}}$. We find the best agreement when $\Lambda_{\text{FP}}/T \simeq 4.8$.

We expect to see small variations in the optimal value for Λ_{FP}/T for different initial conditions. However, these changes are somewhat ‘small’. For a representative underpopulated initial condition ($\chi = 0.1$), the optimal cut-off is estimated $\Lambda_{\text{FP}}/T \simeq 4.8$. For a typical overpopulated initial condition ($\chi = 2$) the optimal cut-off is estimated $\Lambda_{\text{FP}}/T \simeq 5.0$ (a variation less than 10% between the two cases). We emphasise that these cut-off ratios are computed in the limit of weak coupling.

2.4 Solutions of the Fokker-Planck equation

Now that the Fokker-Planck and Boltzmann models have been matched by tuning the Coulomb logarithm, the solutions produced by either model are systematically compared. We calculate s -relaxation times for the same initial conditions tested in section 2.2 and present a cursory discussion of condensation in the Fokker-Planck approximation. Using the Boltzmann results as a baseline, we provide quantitative estimates for the reliability of the soft-scattering approximation.

Entropy relaxation in the Fokker-Planck approximation

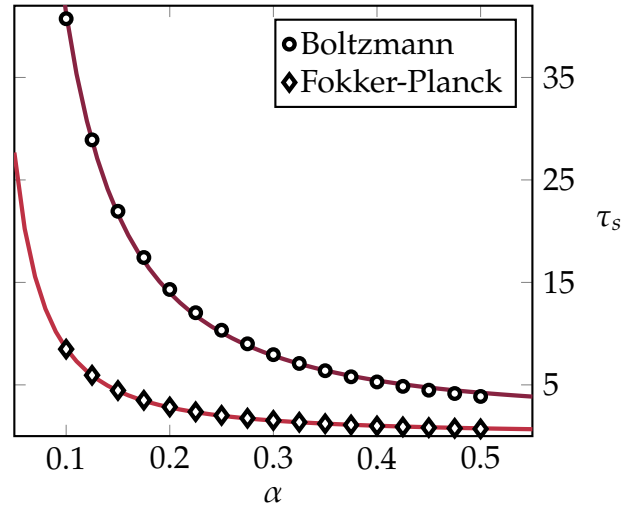
In this subsection, s -relaxation times are calculated for three ensembles of initial conditions. These are the same initial conditions studied in section 2.2 and each ensemble varies exactly one of α , χ or m/Q . The Fokker-Planck approximation studied in this thesis (due to [11]) was developed using a strongly-screened square amplitude (1.42), so results from the strongly-screened Boltzmann model are used for comparison. s -relaxation times in the Fokker-Planck approximation are denoted τ_s^{FP} while those in the Boltzmann model are denoted τ_s^{BE} . We develop scaling relations for τ_s^{FP} and compare our results to those for τ_s^{BE} .

Table 9 shows the s -relaxation times computed for the first ensemble of initial conditions (which vary the coupling). These results suggest that τ_s in the Fokker-Planck model is generally smaller than in the Boltzmann model. Despite the observed ‘speed-up’ in the evolution, which in this case is by roughly a factor of five, we are still able to fit an inverse-logarithmic model to the data generated by the Fokker-Planck evolution, which shows the α -scaling is qualitatively the same as for τ_s^{BE} .

Table 9: (α -scaling) Entropy relaxation times for the first ensemble of systems. The screening mass is determined using (1.46).

α	χ	$n_c(0)/Q^3$	m/Q	μ/Q	τ_s^{FP}	τ_s^{BE}
0.1	0.1	0	0.2	0.19	8.49	40.73
0.2	0.1	0	0.2	0.26	2.84	14.31
0.3	0.1	0	0.2	0.32	1.53	7.95
0.4	0.1	0	0.2	0.37	0.99	5.29
0.5	0.1	0	0.2	0.42	0.71	3.87

Figure 24: $\tau_s(\alpha)$ for Boltzmann and Fokker-Planck models. The markers are data from Table 9. The solid lines are fits, $\alpha^2 \tau_s = a/\ln(b/\alpha)$. For τ_s^{FP} , $a \simeq 0.25$ and $b \simeq 1.9$. For τ_s^{BE} , $a \simeq 1.1$ and $b \simeq 1.4$.



Next, we consider the variation of the entropy relaxation time against the population parameter. Table 10 shows the entropy relaxation times computed for the second ensemble of initial conditions, which vary χ , and whose results are shown in Figure 25. Again, we see that the soft-scattering limit of the transport equation consistently reduces the equilibration timescale (again, by a factor of roughly five). At the same time, the χ -scaling relation for τ_s^{FP} remains qualitatively the same as for τ_s^{BE} ; in the Fokker-Planck approximation, we recover $\tau_s \propto 1/\chi^p \ln(1/\chi)$ with p of order one.

Table 10: (χ -scaling) Entropy relaxation times for the second ensemble of systems. The screening mass is determined using (1.46).

α	χ	$n_c(0)/Q^3$	m/Q	μ/Q	τ_s^{FP}	τ_s^{BE}
0.3	0.04	0	0.2	0.21	2.05	11.09
0.3	0.08	0	0.2	0.29	1.64	8.55
0.3	0.12	0	0.2	0.35	1.45	7.53
0.3	0.16	0	0.2	0.41	1.35	7.01
0.3	0.20	0	0.2	0.46	1.29	6.74

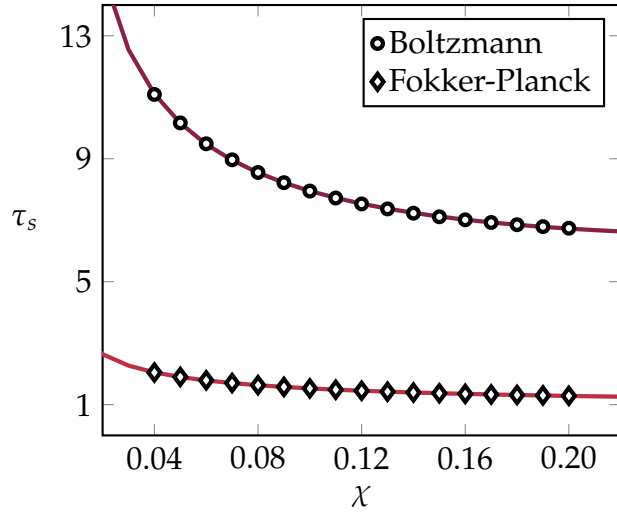


Figure 25: $\tau_s(\chi)$ for the Boltzmann and Fokker-Planck models. The markers are data from Table 10. The solid lines are fits, $\tau_s = a/\chi^p \ln(b/\chi)$. For τ_s^{FP} , $a \simeq 1.4$, $b \simeq 3.0$ and $p \simeq 0.58$. For τ_s^{BE} , $a \simeq 4.0$, $b = 1.3$ and $p \simeq 0.70$.

Finally, entropy relaxation times are computed for an ensemble of initial conditions which vary m/Q . Results for this ensemble are shown in Table 11 and are illustrated in Figure 26. Consistent with the other results of this section, timescales are again reduced by roughly a factor of five in the soft-scattering limit. Otherwise, the m/Q -scaling relation has the same basic form as it does in the Boltzmann model (τ_s remains monotone decreasing against m/Q).

α	χ	$n_c(0)/Q^3$	m/Q	μ/Q	τ_s^{FP}	τ_s^{BE}
0.3	0.1	0	0.1	0.33	1.66	8.86
0.3	0.1	0	0.2	0.32	1.53	7.95
0.3	0.1	0	0.3	0.32	1.37	6.87
0.3	0.1	0	0.4	0.30	1.20	5.75
0.3	0.1	0	0.5	0.29	1.05	4.66

Table 11: (m/Q -scaling) Entropy relaxation times for the third ensemble of systems. The screening mass instead using (1.46).

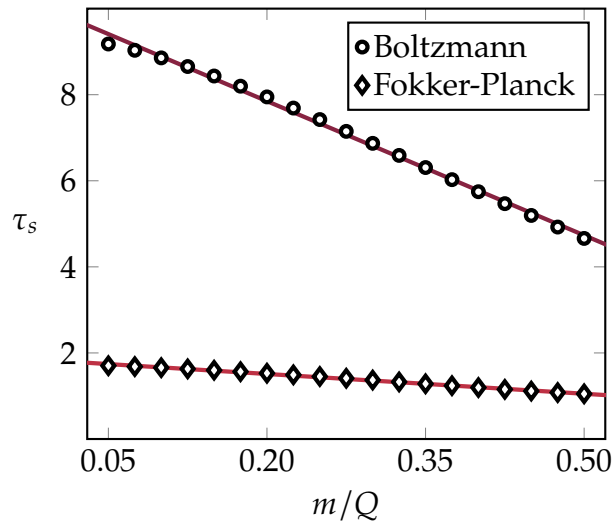


Figure 26: $\tau_s(m/Q)$ for the Fokker-Planck and Boltzmann models. The circular are data from Table 11. The solid lines are fits, $\tau_s = a(m/Q) + b$. For τ_s^{FP} , $a \simeq -1.5$ and $b \simeq 1.8$. For τ_s^{BE} , $a \simeq -10$ and $b \simeq 9.9$.

By calculating τ_l in the Fokker-Planck model and comparing these to the corresponding τ_s , we find the same pattern as before; $\tau_l/\tau_s \simeq 2$. This suggests the relationship between τ_s and τ_l is a general property which is robust against changes in the collision term.

Altogether, the scaling relations of the s -relaxation time against the coupling, population parameter and kinematic mass do not change form between Boltzmann and Fokker-Planck models. In this sense, the empirical scaling rule for τ_s presented in (2.5) remains valid in the approximation of small-scattering-angle. However, there is a general reduction of the equilibration time in the Fokker-Planck model relative to the Boltzmann model (in these examples, the relative ‘speed-up’ is by roughly a factor of five).

Condensation in the Fokker-Planck approximation

To further benchmark the small-scattering-angle approximation against the Boltzmann model, we investigate condensation behaviour. The condensate dynamics generated by the Fokker-Planck equation are qualitatively different from those in the Boltzmann model. We present an elementary discussion in this subsection.

The soft-scattering limit of the evolution equation is built on the assumption that small t and u exchanges are dominant; a feature automatically true for the square amplitude (1.39). Another assumption which enters in the development of the small-momentum expansion is that m^2 is negligible compared to $\langle s \rangle$ (cf. [11] or Appendix E). In this way, the Fokker-Planck model is best-suited to the description of systems where the gluon mass is ‘small’ compared to the characteristic energy of collisions (i.e. the $m \rightarrow 0$ limit). On the other hand, simple kinematics dictate that condensation cannot occur in a plasma consisting of massless particles. Therefore, the small-scattering-angle approximation seems conceptually unsuitable for the description of condensation.

Already in the calculation of s -relaxation times at finite mass, there was a large discrepancy between Fokker-Planck and Boltzmann models. This discrepancy is exacerbated in the condensate dynamics. For the initial conditions surveyed thus far, and with $0.1 \lesssim \alpha \lesssim 0.5$, condensation timescales in the Fokker-Planck approximation are reduced by as much as an order of magnitude compared to the Boltzmann model. If the coupling is limited $\alpha \rightarrow 0$, the leading-log approximation becomes more accurate and the discrepancy in equilibration timescales between the Boltzmann model and Fokker-Planck approximation shrinks.

The condensate dynamics in the leading-log approximation are not only ‘sped-up’, they are qualitatively different from those generated by Boltzmann collision term. Figure 27 shows an example of how the condensate density, n_c , and number density of ‘soft’ gluons, n_{soft} , behave in the leading-log approximation for a representative overpopulated initial condition. Comparing Figures 13 and 27, the n_c -evolutions are different around the time of onset. In the Boltzmann scheme, $\partial_\tau n_c$ is continuous. However, in the Fokker-Planck model, there is a discontinuity in $\partial_\tau n_c$ near $\tau = \tau_0$. The sudden jump in $\partial_\tau n_c$ around the time of onset persists in solutions generated with a finer time-resolution. Whether the discontinuity in $\partial_\tau n_c$ is related to the physics of the soft-scattering limit or is an idiosyncrasy of our numerical scheme is unclear.

One explanation for the jump in $\partial_\tau n_c$ is the exclusion of ‘up-scattering’ (processes where a condensate particle is scattered into a $k > 0$ mode) from our Fokker-Planck collision term (cf. Appendix E). Without explicit coupling to the condensate, kinematics where an incoming particle is in the zero-mode are effectively ignored. While net scattering out of the condensate remains small (known from our treatment of the Boltzmann collision term) we cannot dismiss that there may be strong cancellation of in- and out-scatterings during the period before onset. In any event, we can at least provide a lower bound on the condensation timescales using our scheme.

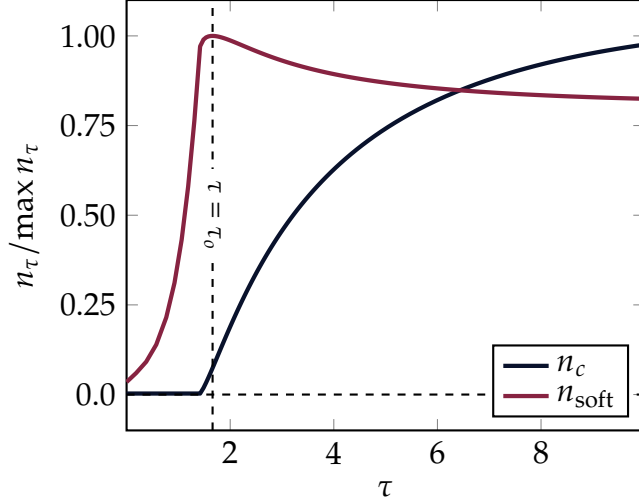


Figure 27: n_c and n_{soft} for an initial condition $\chi = 1$, $n_c(0) = 0$, $m/Q = 0.1$, $\alpha = 0.3$ and $\mu/Q \approx 1$. n_{soft} is the number density of gluons with ‘soft’ (but non-zero) momenta $0 < k \lesssim 0.14Q$. The evolution is computed on an equidistant momentum grid with resolution $\Delta_k \approx 0.1$ and cut-off $\Lambda_{\text{FP}}/Q \approx 2.9$. Number densities are normalised to their maxima on the sampled τ for legibility.

Preliminary results suggest that the scaling relations for condensation timescales in the Fokker-Planck approximation are the same as in the Boltzmann model. However, compared with our Boltzmann approach, the Fokker-Planck model consistently underestimates equilibration and condensation times. If one takes into account inelastic processes, the results of our Boltzmann approach suggest that condensation is not as likely as the Fokker-Planck approximation leads one to expect.

3 Summary and outlook

In this thesis, we studied the relativistic Boltzmann equation (using a discrete approximation) for glasma-inspired initial conditions. Similar to recent treatments of a scalar ϕ^4 -theory [8, 14], we express Boltzmann’s collision integral as a two-dimensional convolution of a kernel (1.25) and a statistical factor (1.6). The kernel for a weakly-screened QCD amplitude (1.41) was already known in [8] but no results were presented. Here, we numerically solved the Boltzmann equation using the weakly-screened kernel. Additionally, we presented as a novel result the kernel for a strongly-screened amplitude (1.42) and numerically solved the corresponding Boltzmann equation (by comparing to results for the weakly-screened case, we obtain a rough estimate of uncertainties due to the soft sector). For both kernels, we derived empirical scaling rules for the equilibration timescales against changes in the coupling, gluon mass, screening mass and initial gluon population. We found that the condensation timescales generated by either kernel follow qualitatively the same scaling rules. However, as expected, evolution generated by the weakly-screened kernel is faster than for the strongly-screened kernel. We verify the formation of a gluon condensate in overpopulated systems, using either the weakly- or strongly-screened kernel.

After studying the relativistic Boltzmann equation, we considered the ‘small-scattering-angle’ approximation of Boltzmann’s collision term. The resulting evolution equation takes the form of a Fokker-Planck equation, which is widely studied in the literature [11, 12, 17, 18, 19, 22]. We followed the analysis presented in [11] and confirmed their result for the so-called ‘Fokker-Planck kernel’ (shown in Appendix E) up to a prefactor. We then calibrated the Fokker-Planck collision term by comparison to our Boltzmann approach; by matching the initial rates of entropy production in either model (in the limit of small coupling), we constrained the auxiliary ultraviolet cut-off appearing in the Fokker-Planck approximation $\Lambda_{\text{FP}} \approx 5T$. Further uncertainties in the evolution reside in how we screen the QCD square amplitude.

At moderate values of the coupling, the equilibration timescales generated by the Fokker-Planck approximation are found to have qualitatively the same scaling behaviour as those in the Boltzmann approach. However, unless the coupling is very small, the leading-log approximation is found to reduce the equilibration timescales by as much as an order of magnitude compared to the Boltzmann model. Therefore, the mistreatment of hard processes in the small-scattering-angle approximation has a noticeable effect on the resulting evolution (it is accelerated).

More work needs to be done to better understand condensation in the Fokker-Planck approximation. In the small-scattering-angle approximation, condensates form relatively quickly. However, our treatment of the QCD Boltzmann equation suggests that equilibration is slower than suggested by the Fokker-Planck model, meaning that a gluon condensate is less likely to form (when taking into consideration number-changing processes). Future analyses would also do well to implement an explicit condensate coupling in the Fokker-Planck collision term (like in [22]). On the side of numerics, higher-fidelity results may be generated for the Fokker-Planck approach using a non-equidistant grid (for example, with an increased density of points in the infrared).

All timescales calculated in this thesis were reported in a dimensionless time, $\tau = Qt$ (with t being the physical time). As crude estimates for heavy ion phenomenology, with $Q \simeq 2 \text{ GeV}$ [12] and $0.1 \lesssim \alpha \lesssim 0.5$, we find s -relaxation times of a few fm/c and condensation-onset times that could be smaller than $1 \text{ fm}/c$. To further assess the possibility of gluon condensation in heavy ion collisions, we recommend considering elastic and inelastic processes together in future models of the evolution. This has already been done in a Fokker-Planck approximation [17] but an extension to the Boltzmann approach is desirable.

Acknowledgements

The author thanks André Peshier for the excellent supervision he provided during the completion of this M.Sc. We are grateful to the Department of Physics at the University of Cape Town for financially supporting this research. Additional thanks are extended to the Department of Physics and André Peshier for financially supporting the author's attendance at the 2025 Wolfram Summer School.

Appendix A: the relaxation time approximation

Boltzmann's collision integral (1.5) captures details of the interactions and statistics through $|\mathcal{M}|^2$ and $\mathcal{F}_{[f]}$ respectively. However, we cannot solve the evolution equation in closed-form if we use (1.5) as our collision term. Therefore, a numerical approach like the one described in Appendix X is usually necessary for treatments including (1.5).

The relaxation time approximation (otherwise the 'RTA') is a simplification of the collision term which admits a closed-form solution for f . The RTA collision term is proportional to the difference between the current and equilibrium distribution functions

$$C_{[f]}^{\text{RTA}} = \frac{f_{\text{eq}} - f}{\tau^*}, \quad (\text{A.1})$$

where τ^* , the 'relaxation time', is an independently determined parameter. The simple structure of the RTA collision term obviates many of the issues we tangle with when using Boltzmann's collision integral. Firstly, (A.1) is not an integral quantity; the net scattering into a mode is determined only by the surplus or deficit of particles in that mode relative to the expected equilibrium occupation. The solution for f is given by

$$f(k; t) = f_{\text{eq}}(k) [1 - e^{-t/\tau^*}] + f_{\text{in}}(k) e^{-t/\tau^*}, \quad (\text{A.2})$$

where t is time, f_{in} is the distribution function at $t = 0$ and f_{eq} is the equilibrium distribution (i.e. the limiting form of f as $t \rightarrow \infty$). As discussed in section 1.4, the equilibrium state in our model is described by a Bose function. The procedure for finding f_{eq} is discussed in section 1.5. In the RTA, f_{eq} must be specified manually (unlike the Boltzmann collision term, which dynamically generates the correct equilibrium state). The solution (A.2) is k -wise monotonic.

By substituting (A.2) back into (A.1), we obtain an expression for the RTA collision term at arbitrary times

$$C_t^{\text{RTA}} = e^{-t/\tau^*} \frac{f_{\text{eq}} - f_{\text{in}}}{\tau^*}. \quad (\text{A.3})$$

With a closed form expression for the collision term, we study the entropy evolution in the RTA model. The entropy, s , is defined in section 1.4. By substituting (A.3) into our expression for $\partial_t s$ (B.7) and expanding the result in powers of small $\exp[-t/\tau^*]$ (effectively exploring the large t limit) we find to second order in $\exp[-t/\tau^*]$

$$\partial_t s|_{t \gg \tau^*} \simeq e^{-t/\tau^*} \int \frac{d^3 k_1}{(2\pi)^3} (\varepsilon_k - \mu) C_{t=0}^{\text{RTA}} / T + e^{-t/(\tau^*/2)} \int \frac{d^3 k_1}{(2\pi)^3} \tau^* \{C_{t=0}^{\text{RTA}}\}^2 / f_{\text{eq}} \bar{f}_{\text{eq}}, \quad (\text{A.4})$$

where $\varepsilon_k = \sqrt{k^2 + m^2}$ and μ and T are the equilibrium parameters defining f_{eq} . The integrals appearing in (A.4) are named $\mathcal{I}_1$ and $\mathcal{I}_2$ respectively. The first integral, $\mathcal{I}_1$ is zero due to number and energy conservation (discussed in section 1.3). To see this, expand the copy of $C_{t=0}^{\text{RTA}}$ appearing in $\mathcal{I}_1$ as a difference between f_{eq} and f_{in} . Because the zeroth and first moments of the distribution function are conserved, integrals over $\varepsilon_k (f_{\text{eq}} - f_{\text{in}})$ and $f_{\text{eq}} - f_{\text{in}}$ vanish (cf. (1.10)).

Therefore, there is no contribution to (A.4) linear in $\exp[-t/\tau^*]$. The first non-zero contribution to the late-time entropy evolution enters quadratically in $\exp[-t/\tau^*]$ and is proportional to $\mathcal{I}_2$ (a constant determined entirely by the initial condition). To good approximation

$$\partial_t s|_{t \gg \tau^*} \propto e^{-t/(\tau^*/2)}. \quad (\text{A.5})$$

The result (A.5) is useful because it links the relaxation time τ^* (a parameter of the RTA collision term) to the entropy evolution. If the late-time entropy evolution generated by other models of the collision term is found to be exponential, one can estimate τ^* by means of a fit. In chapter 2 we indeed find the late-time entropy evolution is exponential and extract ‘entropy relaxation times’, τ_s which roughly correspond to $\tau^*/2$.

In the same chapter, we calculate so-called ‘linearisation times’, τ_l , which are defined independently of the entropy. These quantify how ‘quickly’ f approaches a Bose distribution and generally satisfy $\tau_l/\tau_s \simeq 2$. Therefore, we conjecture that the linearisation times calculated in chapter 2 are analogous to the relaxation time featuring in the RTA. With this connection, we can motivate values for τ^* from our Boltzmann approach.

Appendix B: details of the numerics

Here, we describe the basic numerical scheme, presented in [8, 14], which is used to compute and integrate the coupled evolution equations (1.36) and (1.37). The core idea is to discretise the energy domain and update the distribution function at these fixed energies. In addition to E_1 , E_3 and E_4 (which are considered ‘free’), the integrand of (1.26) samples the distribution function at a fixed energy $\underline{E}_2 = E_3 + E_4 - E_1$. Similarly, the condensate integrals require access to the distribution function at ‘conservative’ energies (albeit with one of the energies set to m). If $\underline{E}_2$ is not contained in the energy grid, errors arising from an interpolation of f_2 from known values of f spoils the conservation of n and ε guaranteed by the Boltzmann equation [14].

In a discrete representation, $\underline{E}_2$ is guaranteed to remain ‘on-grid’ only if the other energies are commensurable. Therefore, calculations are based on a fixed and equidistant energy grid, $E_i = m + i\Delta_E$ with $i = 1, \dots, M$ [8, 14]. Here, Δ_E is the fixed grid-spacing and i is a discrete index. Of course, practical applications require the grid is truncated at some finite cut-off, $\Lambda = m + M\Delta_E$. We avoid sampling the potentially singular distribution function at $E = m$ by maintaining $i \geq 1$ and tracking n_c separately. Without needing to interpolate f , the equidistant energy grid allows conservation of n and ε to within machine precision [14]. Recommendations on how to select the grid parameters (e.g. Λ and M) are described later in this appendix.

To compute the collision term (1.26) two quantities are needed at each point in the (discrete) integration region, the kernel, $\mathcal{K}$, and the statistical factor, $\mathcal{F}_{[f]}$. The kernel is generically a function of kinematic variables and is independent of the distribution function. Hence, the kernel is time-independent (assuming that the kinematic and screening masses are constant) and need only be calculated *once* at the beginning of a simulation. The statistical factor, however, must be updated at each time-step along with the distribution function ($\mathcal{F}_{[f]}$ is manifestly dependent on f). In discrete form, one can represent the kernel and statistical factor as 4-dimensional arrays

$$\mathcal{K}_{ijkl} = \mathcal{K}(k_i, k_j, k_k, k_l), \quad \mathcal{F}_{[f]}^{ijkl} = \mathcal{F}_{[f]}(k_i, k_j, k_k, k_l),$$

where the indices i, j, k and l are not yet constrained by energy conservation. In practice, one rather uses 3-dimensional arrays for the kernel and statistical factor with the energy constraint already ‘built-in’ (thereby eliminating the index j). In [14], the energy constraint is made explicit by writing the ‘discrete’ collision term, $C_{[f]}^i(E_i) = C_{[f]}^i$, as a triple sum including ‘conservative’ weights

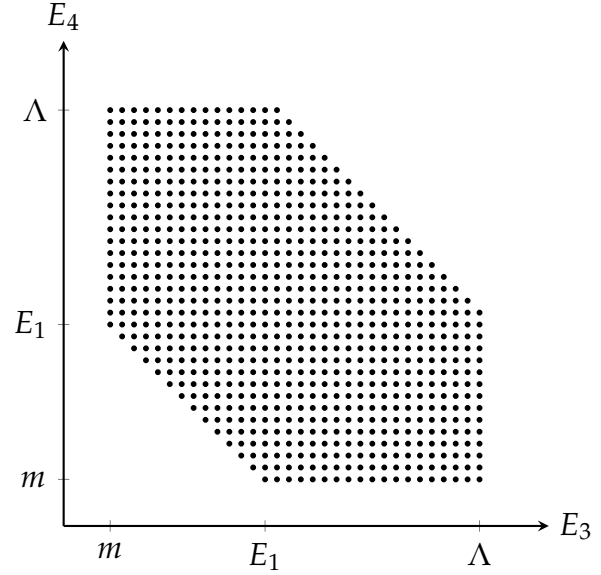
$$C_{[f]}^i = \frac{\Delta_E^2}{32(2\pi)^4 k_i E_i} \sum_{j,k,l=1}^M \mathcal{K}_{ijkl} \mathcal{F}_{[f]}^{ijkl} \delta_{k+l}^{i+j} w_{jkl}, \quad (\text{B.1})$$

where Kronecker- δ symbol is defined

$$\delta_n^m = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases},$$

and w_{jkl} are trapezoidal weights, $w_{jkl} = 2^{-\delta_j^1 - \delta_k^1 - \delta_l^1 - \delta_j^M - \delta_k^M - \delta_l^M}$. The Kronecker- δ therefore enforces ε -conservation by fixing $\underline{j} = k + l - i$ in all non-zero contributions to the sum. Therefore, (B.1) is effectively a double sum over the indices k and l where j is implicitly constrained by the Kronecker- δ (and within bounds $1 \leq j \leq M$). For a given external energy E_1 , the indices k and l which contribute to (B.1) typically form a ‘kite’ shape if arranged in the kl -lattice (equivalently, $E_3 E_4$ -lattice) [14], an arrangement shown in Figure 28.

Figure 28: Typical representation of the energies E_3 and E_4 (tracked with k and l in (B.1)) which contribute to the discrete collision term (B.1). The diagonal boundaries of the integration region correspond to extrema of E_2 (respectively j), with E_2 increasing linearly along the line $E_3 = E_4$. This structure was first observed in [14]. Note that the smallest energy obtainable on-grid is $m + \Delta_E$.



The condensate integrals (1.35) are computed similarly; they are effectively one-dimensional convolutions of their own statistical factors and the square amplitude (1.35)

$$[C_{[f]}^{12\circ 4}]^i = \frac{\Delta_E}{64\pi m E_i k_i} \sum_{j,l=1}^M |\underline{M}|_{ij0l}^2 [\underline{\mathcal{F}}_{[f]}^{12\circ 4}]^{ijl} \delta_l^{i+j} \tilde{w}_{jl}, \quad (\text{B.2})$$

where $E_3 = m$ translates to $k = 0$ (i.e. the condensate is regarded as a zeroth grid point) and the $\tilde{w}_{jl} = 2^{-\delta_j^1 - \delta_l^1 - \delta_j^M - \delta_l^M}$ are trapezoidal weights. The condensate integral $[C_{[f]}^{1\circ 34}]^i$ is computed in similar manner to (B.2).

With all the collision terms in hand, the evolution equations (1.36) and (1.37) are approximated

$$\partial_t f_i = C_{[f]}^i + n_c \left\{ [C_{[f]}^{1\circ 34}]^i + 2[C_{[f]}^{12\circ 4}]^i \right\}, \quad (\text{B.3})$$

$$\partial_t n_c = -\frac{n_c \Delta_E}{2\pi^2} \sum_{i=1}^M E_i k_i \left\{ [C_{[f]}^{1\circ 34}]^i + 2[C_{[f]}^{12\circ 4}]^i \right\}, \quad (\text{B.4})$$

where the combination $E_i k_i$ in the second equation is a Jacobian factor $\Delta_k k_i^2 \mapsto \Delta_E E_i k_i$.

We solve the Boltzmann equation by iterating (B.3) and (B.4)⁶. The update procedure used is detailed in Appendix B. Results are organised in a ‘dimensionless’ or ‘simulation’ time, $\tau = Qt$ (where t is the physical time).

⁶Evidently, the initial condition must be ‘seeded’ with a positive condensate to permit non-trivial n_c -dynamics. As shown in (B.4), the derivative $\partial_t n_c$ is proportional to n_c . Therefore, if $n_c = 0$ initially n_c will remain zero for all time, regardless of the scatterings described by $C_{[f]}^{1\circ 34}$ and $C_{[f]}^{12\circ 4}$.

Constructing the energy grid

Assuming that the gluon mass is already chosen, the energy grid is specified by just two parameters. For example, the grid can be defined by choosing a cut-off Λ and a grid size M (with corresponding spacing $\Delta_E = (\Lambda - m)/M$). Here, we present some guidelines for choosing Λ and M .

Boltzmann's collision integral (1.5) is ultraviolet-convergent. Therefore, by maintaining $\Lambda < \infty$, the collision integrals pick up a residual Λ -dependence (the energy integrals in (1.26) now have a finite upper bound). Cut-off effects also enter the constraint equations determining the critical and equilibrium parameters (the moments (1.16) are integrated up to $k \leq \sqrt{\Lambda^2 - m^2}$, not $k \rightarrow \infty$). To keep cut-off effects 'small', we expect that Λ ought to be larger than the average energy of collisions in a Bose gas; $\Lambda \gtrsim \langle \sqrt{s} \rangle$. First, we develop an intuition for Λ by estimating $\langle \sqrt{s} \rangle$ and the standard deviation in $\sqrt{s}$.

$$\langle \sqrt{s} \rangle = \frac{2\pi^4}{45\zeta(3)} T \simeq 3.6T, \quad \text{standard deviation } (\sqrt{s}) \simeq 2.8T, \quad (\text{B.5})$$

where the expectation is taken with respect to a massless, critically populated Bose gas and is normalised *per particle*. At equilibrium, many bosons naturally obtain momenta greater than $\langle \sqrt{s} \rangle$ (cf. Figure 29). Therefore, to capture the most significant ultraviolet behaviour, we recommend setting the hard scale to $\langle \sqrt{s} \rangle$ plus at least one standard deviation; $\Lambda \gtrsim 6T$.

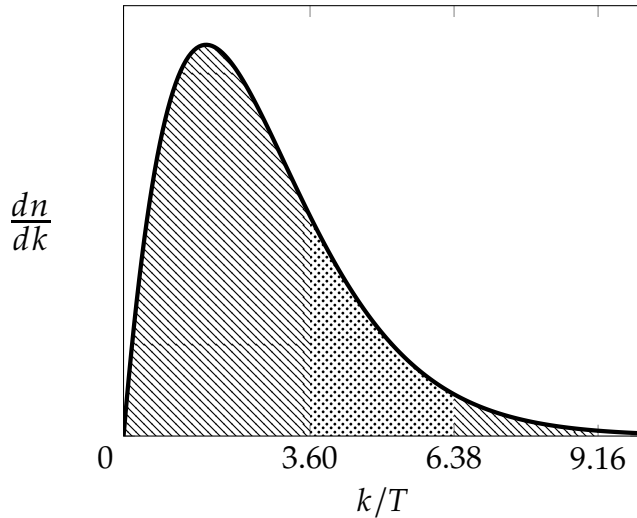
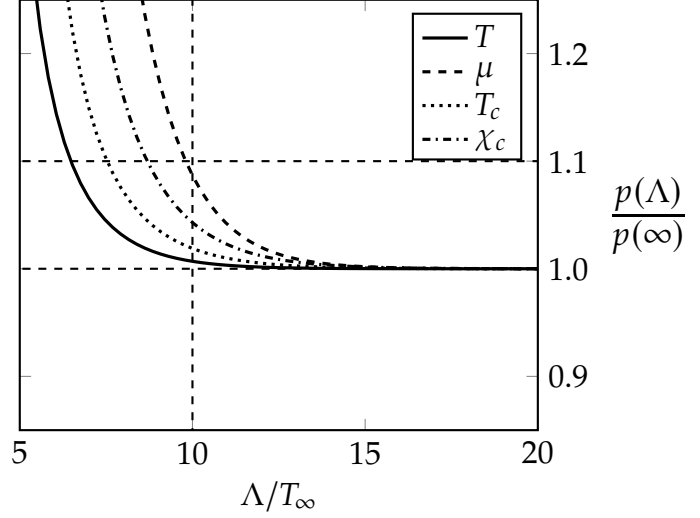


Figure 29: The differential number of bosons, $dn/dk \propto k^2 f$, with momentum k for a massless, critically populated Bose gas. The region $0 \leq k/T \leq 3.6$ contains roughly 75% of all particles. The first standard deviation above $\langle \sqrt{s} \rangle$, which spans $3.6 \leq k/T \leq 6.38$, accounts for an additional 22% of the Bose gas. The second standard deviation above $\langle \sqrt{s} \rangle$, which spans $6.38 \leq k/T \leq 9.16$, only holds 3% of the total particle number.

Whilst the bound $\Lambda \gtrsim 6T$ argued by Figure 29 and (B.5) seems reasonable, we gain more insight into the cut-off effects by solving the conservation laws with an explicit Λ -dependence incorporated into the moments n and ε . The data in Figure 30 show the relative error in the critical and equilibrium parameters at different Λ but with E still being continuous. To keep the realised equilibrium or critical parameters within 10% the values they obtain on an unbounded energy domain, $\Lambda \gtrsim 10T$ is a 'safer' bound. The temperature used to set Λ is the temperature estimated on an unbounded and continuous energy domain.

In practice, we specify the initial condition by choosing χ (as it appears in (1.21)). Unless otherwise stated, $\sigma/Q = 0.05$ always. We then estimate the equilibrium parameters as demonstrated in section 1.5. Once we know the temperature obtained on a continuous and unbounded energy domain, T_∞ , the ultraviolet cut-off is set $\Lambda \gtrsim 10T_\infty$.

Figure 30: p is a placeholder for any one of μ , T , χ_c or T_c . The quantity $p(\Lambda)/p(\infty)$ is therefore the ratio of p at particular Λ to its value as $\Lambda \rightarrow \infty$. For μ and T , we consider the special case with $m/Q = 0.1$ and $\chi = 0.1$. For χ_c and T_c , we use the same m/Q . The energy domain is treated as continuous ($M \rightarrow \infty$). The temperature as $\Lambda \rightarrow \infty$ is denoted T_∞ .



Once the initial condition is chosen and a suitable value for Λ has been determined, it remains to adjust the grid spacing, Δ_E , or number of cells, M . It's important to remember the computation time for the double sum (B.1) scales like $O(M^2)$. Calculating the collision term at every grid point lengthens computation by an additional factor of M . If the system is evolved over N time-steps, the full computation time scales like $O(M^3N)$.

Therefore, M must be chosen small enough for calculations to remain practical (in our case, we use a typical laptop running *Mathematica*). At the same time, the grid must resolve sufficient detail in the infrared and in the region $k \simeq Q$. Detail in the infrared is necessary to correctly describe Bose functions (which have most of their support at $k \lesssim Q$). Detail in the region $k \simeq Q$ is required to resolve the step in the initial condition (1.21). Whether or not the step in (1.21) is properly resolved is also subject to the choice of σ/Q .

With $M < \infty$, the critical and equilibrium parameters realised 'on-grid' are fixed by discretised versions of the conservation laws. The moments (1.16) are traded for sums

$$n_{[f]} = n_c + \frac{\Delta_E}{2\pi^2} \sum_{i=1}^M E_i k_i f_i w_i, \quad \varepsilon_{[f]} = m n_c + \frac{\Delta_E}{2\pi^2} \sum_{i=1}^M E_i^2 k_i f_i w_i, \quad (\text{B.6})$$

where the $w_i = 2^{-\delta_i^1 - \delta_i^M}$ are trapezoidal weights. The discrete moments (B.6) are then used in the conservation laws (1.10) used to solve for critical and equilibrium parameters. Expectedly, the critical and equilibrium parameters pick up a residual M -dependence. Figure 31 shows the sensitivity of the critical and equilibrium parameters against changes in M . The critical and equilibrium parameters are seemingly within 10% their continuum value when $M \gtrsim 256$.

We now study representative overpopulated initial conditions to understand the sensitivities of n_c (at equilibrium) against changes in Λ and M . If the energy domain is continuous and unbounded, and if $m/Q = 0.1$, critical population occurs when $\chi_c \simeq 0.32$ (cf. Figure 3). We investigate two systems which are near critical population, $\chi = 0.4, 0.5$, and one with stronger overpopulation, $\chi = 1.0$. In Figure 32, is the corresponding variation of n_c against changes in Λ and M . Evidently, $\Lambda \gtrsim 10T_\infty$ is enough to reduce cut-off effects to within 10%. However, discretisation effects seem to have a strong influence on n_c near criticality (e.g. in the example $\chi = 0.4$, one needs $M \gtrsim 4096$ grid points to reduce discretisation effects to within 10%).

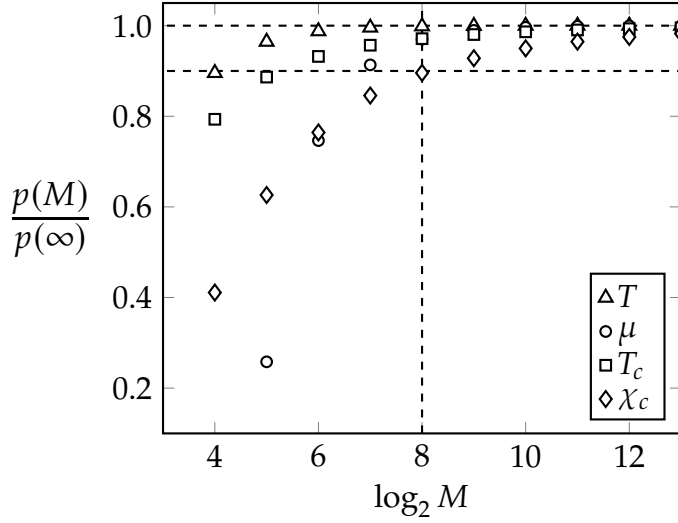


Figure 31: p is a placeholder for any one of μ , T , χ_c or T_c . The quantity $p(M)/p(\infty)$ is therefore the ratio of the parameter p at particular M to its value as $M \rightarrow \infty$. For μ and T , we consider the special case with $m/Q = 0.1$ and $\chi = 0.1$. For χ_c and T_c , we use the same m/Q . In each case, $\Lambda = 4Q$.

The strong sensitivity in n_c near criticality is simple to understand. (1.22) implies $n \propto \chi$ and, for critically populated systems, $n \propto \chi_c$. Therefore, $n_c \propto \chi - \chi_c$ for any overpopulated initial condition. Cut-off effects in χ_c are controllable (cf. Figure 30). However, substantial fluctuations in χ_c arise when discretising the energy domain. If $\chi - \chi_c$ is large compared to the change in χ_c arising from discretisation, there should be little effect on n_c . On the other hand, if the change in χ_c is comparable to $\chi - \chi_c$, the n_c obtained ‘on-grid’ will depart from its continuum value substantially.

The takeaway is that, on small grids, one requires $\chi \gg \chi_c$ in order to minimise uncertainties in n_c . Otherwise, studies of condensation behaviour near criticality (i.e. $\chi \simeq \chi_c$) would require ‘large’ grids to reduce the uncertainty in n_c . It may also be possible to use the known deviation in $\chi - \chi_c$ from its continuum value to ‘recalibrate’ certain aspects of the condensate dynamics. An alternative is to exchange the equidistant energy grid for a setup with higher resolution in the infrared (although this would spoil the exact n - and ε -conservation described in [14]).

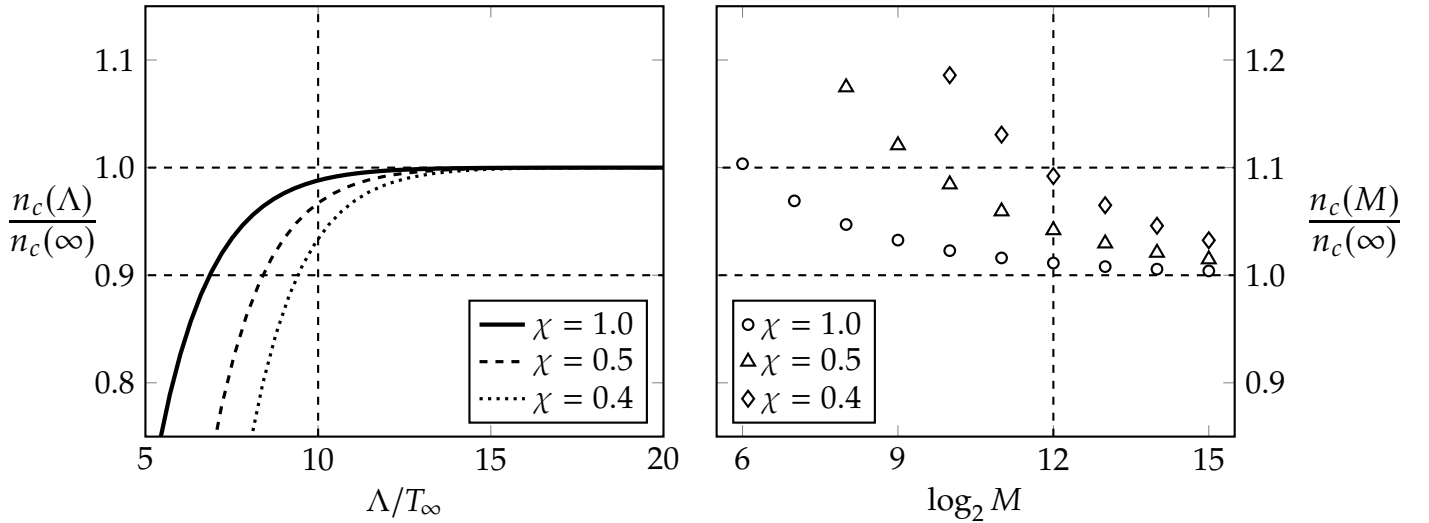


Figure 32: Residual variations in n_c as functions of Λ and M . When varying Λ , we assume $M = \infty$. When varying M , we fix $\Lambda = 4Q$.

Quantifying overall uncertainty

As shown in [chapter 2](#) of the main text, most dynamics typically happen during the early stages of equilibration. Therefore, we compute the initial rate of entropy production, $\partial_\tau s|_{\tau=0}$, which we abbreviate at $\dot{s}_0$, as a measure of agreement between our discrete approximation and the solution obtained on a continuous and unbounded energy domain. The continuum expression for $\dot{s}_0$ and its discrete approximation are

$$\dot{s}_0 = \int \frac{d^3 k_1}{(2\pi)^3} \ln(\bar{f}/f) C_{[f]} \Big|_{f=f_{\text{in}}}, \quad (\text{continuum}) \quad (\text{B.7})$$

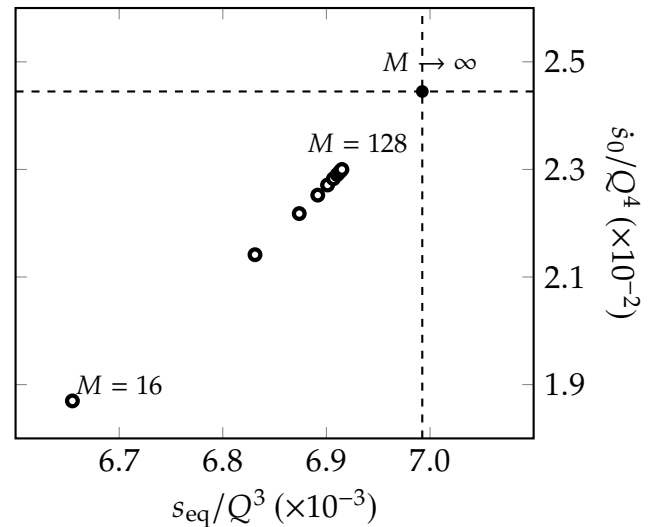
$$\dot{s}_0 = \frac{\Delta_E}{2\pi^2} \sum_{i=1}^M E_i k_i \ln(\bar{f}_i/f_i) C_{[f]}^i w_i \Big|_{f=f_{\text{in}}}, \quad (\text{discretised})$$

where the $w_i = 2^{-\delta_i^1 - \delta_i^M}$ are trapezoidal weights. In (B.7), we see that $\dot{s}_0$ is effectively a ‘weighted collision frequency’. Because $\partial_\tau s$ is maximal at $\tau = 0$ (cf. [Figures 11 and 13](#)) the relative error between the continuum and discretised values of $\dot{s}_0$ serves as an upper bound on the uncertainty of the collision term, $C_{[f]}$. We also compute the entropy obtained at equilibrium on the given grid

$$s_{\text{eq}} = \frac{\Delta_E}{2\pi^2} \sum_{i=1}^M E_i k_i \{ \bar{f}_i \ln \bar{f}_i - f_i \ln f_i \} w_i \Big|_{f=f_{\text{eq}}},$$

where, again, $w_i = 2^{-\delta_i^1 - \delta_i^M}$. To quantify agreement between discretised and continuum solutions of the evolution equation, we compare points they define in the $\dot{s}_0 s_{\text{eq}}$ -plane. [Figure 33](#) shows, for a representative initial condition, the convergence of numerical solutions to the continuum case as a function of the grid size M . Around $M = 32$ grid points, the relative uncertainty on $\dot{s}_0$ and s_{eq} is on the order of 10%. Note that the uncertainty due to E -discretisation is compounded with uncertainties due to cut-off effects and discretisation of time. We aim for an order 10% uncertainty, therefore $M \simeq 32$ is a good compromise between accuracy and computational efficiency. Unless indicated otherwise, all results in this thesis are generated using $M = 32$ grids.

Figure 33: s_{eq} and $\dot{s}_0$ for increasing M . The sampled values of M are $M = 16, 40, 44, 58, 72, 86, 100, 114$ and 128 . The dashed lines indicate the continuum values of s_{eq} and $\dot{s}_0$. The sequences of s_{eq} and $\dot{s}_0$ calculated for finite M converge to the continuum result. At $M = 128$, the relative errors in $\dot{s}_0$ and s_{eq} are approximately 6% and 1% respectively. The initial condition used has $\chi = 0.1$, $m/Q = 0.25$, $\alpha = 0.3$ and $\mu/Q \simeq 0.3$. The cut-off is consistently set $\Lambda/Q \simeq 2.5$.



The update procedure

There is, of course, a great diversity of methods we can choose for the numerical integration. Some points to keep in mind are that (i) the computation time for the collision term grows like $O(M^3)$ and (ii) the condensation is expected to be abrupt, with n_c increasing by many orders of magnitude in short time. Therefore, whatever numerical scheme is used must strike a balance between efficiency and stability. Drawing inspiration from previous work [8], we use a Fehlberg update procedure which is described in Algorithm 1.

Algorithm 1 Pseudocode for the Fehlberg method

```
1: provide the initial condition  $\{n_c, f\}$ ;
2: specify the basic time-step,  $\Delta\tau$ , and the tolerance, TOL;
3: specify the end-condition for the evolution, END;
4: initialise FLAG = false;
5: while END is false do
6:   while FLAG is false do
7:      $y_1 = \text{collision term}@f$ ;
8:      $y_2 = \text{collision term}@{f + \Delta\tau y_1}$ ;
9:      $y_3 = \text{collision term}@{f + \Delta\tau(y_1 + y_3)/4}$ ;
10:     $x_1 = -\text{integral of } y_1 \text{ over } k_1$ ;
11:     $x_2 = -\text{integral of } y_2 \text{ over } k_1$ ;
12:     $x_3 = -\text{integral of } y_3 \text{ over } k_1$ ;
13:     $\epsilon = \Delta\tau(x_1 + x_2 - 2x_3)/3$ ;
14:    if  $\epsilon \leq \text{TOL}$  then
15:      FLAG = True;
16:    else  $\Delta\tau \rightarrow 0.9(\text{TOL}/\epsilon)^{1/3}\Delta\tau$ ;
17:     $\{n_c, f\} \rightarrow \{n_c, f\} + \Delta\tau\{x_1 + x_2 + 4x_3, y_1 + y_2 + 4y_3\}/6$ ;
18:    update the end-condition END.
```

Motivations for using the Fehlberg method are that (i) its use of an adaptive time-step allows one to stably track the system's evolution across abrupt changes and (ii) among the procedures which use a moderated time-step, the Fehlberg method is relatively lightweight. In Algorithm 1 the tolerance, "TOL", is presented as a bound on the increment in n_c . One could exchange (or unify) this with a bound on the increment in f . Expectedly, there is much room for modification in the Fehlberg procedure; the pseudocode shown in Algorithm 1 is presented as a minimal working example.

Appendix C: the w -kernel

Here, we find the kernel (1.25) associated with the weakly-screened amplitude (??), which we name the ' w -kernel' and symbolise with $\mathcal{K}_W$. The w -kernel is an already-known result [8], whose derivation we summarise for convenience. Inserting (1.41) into (1.25), we expand the bracketed terms coming from $|\mathcal{M}_W|^2$ using the method of partial fractions

$$\frac{1}{t(t - \mu^2)} = \frac{1}{\mu^2} \left[\frac{1}{t - \mu^2} - \frac{1}{t} \right],$$

and similarly for the u -channel contribution. The w -kernel is then expressed in terms of a 'master integral', $\mathcal{I}$,

$$\mathcal{K}_W(k_1, \underline{k}_2, k_3, k_4) = \frac{72(4\pi\alpha)^2}{\mu^2} \left[\mathcal{I}(k_1, \underline{k}_2, k_3, k_4; \mu) - \mathcal{I}(k_1, \underline{k}_2, k_3, k_4; 0) \right],$$

where $\mathcal{I}$ is defined [8]

$$\mathcal{I}(k_1, \underline{k}_2, k_3, k_4; \mu) = \int_{s_-}^{s_+} ds \int_{t_-}^{t_+} dt \frac{s^2}{\sqrt{Q}} \left[\frac{1}{t - \mu^2} + \frac{1}{u - \mu^2} \right]. \quad (\text{C.2})$$

Due to crossing symmetry, the integral over t can be exchanged for an integral over u . Therefore, the t and u channels contribute equally to the master integral but are reflections of one another in the $k_3 k_4$ -plane. To illustrate, we define a functional, called 'sym' which symmetrises its argument

$$\text{sym}[f](\dots, k_3, k_4) = f(\dots, k_3, k_4) + f(\dots, k_4, k_3), \quad (\text{C.3})$$

where f is an arbitrary function of k_3 and k_4 . Using (C.3), the master integral (C.2) can be written

$$\mathcal{I}(k_1, \underline{k}_2, k_3, k_4; \mu) = \text{sym} \left[\int_{s_-}^{s_+} ds \int_{t_-}^{t_+} dt \frac{s^2}{\sqrt{Q}} \frac{1}{t - \mu^2} \right] (k_1, \underline{k}_2, k_3, k_4; \mu). \quad (\text{C.4})$$

Remembering that $Q = a_t t^2 + b_t t + c_t$, with coefficients documented in (1.23), one can easily find

$$\int_{t_-}^{t_+} dt \frac{1}{\sqrt{Q}} \frac{1}{t - \mu^2} = -\frac{\pi}{\sqrt{-a_t \mu^4 - b_t \mu^2 - c_t}}. \quad (3.2)$$

The quantity $-a_t \mu^4 - b_t \mu^2 - c_t$ on the righthand side is simply $Q|_{t=\mu^2}$ and is a quadratic function of the Mandelstam variable s . We name it $\tilde{Q}$ and write $\tilde{Q} = a_s s^2 + b_s s + c_s$ with

$$a_s = (E_3 - E_1)^2 - \mu^2,$$

$$b_s = -4m^2(E_1 - E_3)^2 + 2\mu^2(E_1 E_4 + E_2 E_3 + 2m^2) - \mu^4,$$

$$c_s = \mu^2(E_3 + E_4)^2(\mu^2 - 4m^2).$$

Importantly, the coefficients a_s , b_s and c_s remain functions of k_3 and k_4 .

By integrating $-\pi s^2/\sqrt{\tilde{Q}}$ against s , one finds [8]

$$\int_{s_-}^{s_+} ds \int_{t_-}^{t_+} dt \frac{1}{\sqrt{Q}} \frac{s^2}{t - \mu^2} = \frac{\pi}{4a_s^2} \left[(3b_s - 2a_s s) \sqrt{\tilde{Q}} + \frac{3b_s^2 - 4a_s c_s}{2\sqrt{-a_s}} \arcsin \left(\frac{b_s + 2a_s s}{\sqrt{b_s^2 - 4a_s c_s}} \right) \right]_{s_-}^{s_+}. \quad (\text{C.5})$$

The full master, $\mathcal{I}$, integral is found by symmetrising (C.5) in the manner of (C.4). We note that (C.5) is manifestly *real*; whenever the square roots or arcsine produce a result with non-zero imaginary part, there's always an exact cancellation arising from elsewhere in (C.5). Also noteworthy is that the $a_s \rightarrow 0$ limit of (C.5) necessarily exists (when $\mu > 0$) because the integrand on the lefthand side of (C.5) is a product of a finite quantity, $s^2/(t - \mu^2)$, and a quantity which diverges in an integrable way, $1/\sqrt{Q}$. However, in the $\mu \rightarrow 0$ limit, (C.5) diverges when $E_1 = E_3$

$$\int_{s_-}^{s_+} ds \int_{t_-}^{t_+} dt \frac{1}{\sqrt{Q}} \frac{s^2}{t} \rightarrow -\pi \frac{s_+^2 - s_-^2}{|E_1 - E_3|}.$$

As we see in (1.26), $\mathcal{K}_W$ is convolved with $\mathcal{F}_{[f]}$ when calculating the collision term, $C_{[f]}$. A sufficient condition for the collision integral to remain well-defined despite the singularities in the w -kernel is that $\mathcal{F}_{[f]}$ is at least $\mathcal{O}(E_1 - E_3)$ in the region $E_1 \simeq E_3$ (with a similar argument holding for the line $E_1 = E_4$). Supposing that $E_3 = E_1 + \epsilon$, we expand $\mathcal{F}_{[f]}$ in powers of small ϵ and indeed find

$$\mathcal{F}_{[f]} \simeq \epsilon(f'_1 f_4 \bar{f}_4 - f_1 f'_4 \bar{f}_4) + \mathcal{O}(\epsilon^2),$$

where f'_i is the energy-derivative of the distribution function at E_i ; $f'_i = f'(E_i)$. Ignoring higher order terms, $\mathcal{I}(\dots; 0) \mathcal{F}_{[f]} \sim \text{sign}(E_1 - E_3)$ when $E_1 \simeq E_3$. Similarly, such a step discontinuity exists along the line $E_4 = E_1$. Therefore, the singularities in $\mathcal{K}_W$ manifest as integrable discontinuities in $\mathcal{K}_W \mathcal{F}_{[f]}$. We handle the steps along $E_3 = E_1$ and $E_4 = E_1$ by setting the integrand in (1.26) to zero along these lines (which is the integrand's mean value within a small neighbourhood about the step). In the discrete approximation (B.1) we set $\mathcal{K}_{ijkl} \mathcal{F}_{[f]}^{ijkl} = 0$ when $i = k$ or $i = l$.

What remains is to reduce the condensate integrals for the weakly-screened model. Knowing (1.35), we substitute $|\mathcal{M}|^2 \mapsto |\mathcal{M}_W|^2$ and find the following expressions for $C_{[f]}^{1\circ 34}$ and $C_{[f]}^{12\circ 4}$

$$C_{[f]}^{1\circ 34} = \frac{18\pi\alpha^2(m + E_1)^2}{mE_1k_1} \int_m^{E_1} dE_3 \left\{ \frac{1}{(E_1 - E_3)(E_1 - E_3 + \mu^2/2m)} \cdots \right. \\ \left. \cdots + \frac{1}{(E_3 - m)(E_3 - m + \mu^2/2m)} \right\} \mathcal{F}_{[f]}^{1\circ 34}, \quad (\text{C.6})$$

and

$$C_{[f]}^{12\circ 4} = \frac{18\pi\alpha^2}{mE_1k_1} \int_{E_1}^{\infty} dE_4 (m + E_4)^2 \left\{ \frac{1}{(E_4 - E_1)(E_4 - E_1 + \mu^2/2m)} \right. \\ \left. \cdots + \frac{1}{(E_1 - m)(E_1 - m + \mu^2/2m)} \right\} \mathcal{F}_{[f]}^{12\circ 4}. \quad (\text{C.7})$$

Similar to $\mathcal{K}_W$, the kernels in (C.6) and (C.7) have linear singularities along the lines $E_3 = E_1$ and $E_4 = E_1$. For the $1/(E_1 - E_3)$ singularity appearing in the kernel of (C.6), we suppose $E_3 = E_1 + \epsilon$ and expand $\mathcal{F}_{[f]}^{1\circ 34}$ in powers of small ϵ . By doing so, one finds

$$\mathcal{F}_{[f]}^{1\circ 34} \simeq \epsilon f_1'(f_4 - f_1) + O(\epsilon^2).$$

Similarly, for the $1/(E_4 - E_1)$ singularity appearing in the kernel of (C.7), we set $E_4 = E_1 + \epsilon$ and expand $\mathcal{F}_{[f]}^{12\circ 4}$ in powers of small ϵ . The result is

$$\mathcal{F}_{[f]}^{12\circ 4} \simeq \epsilon f_1'(f_1 + f_2 + 1) + O(\epsilon^2).$$

In both cases, the ‘ $1/\epsilon$ ’ divergence appearing in the relevant kernel is regulated by a linear ϵ -dependence in the statistical factor. The divergences are again handled by setting the integrand to zero in these regions (which is, again, the average value of the integrand in a small neighbourhood about the singularities).

Appendix D: the s -kernel

Unlike the weakly-screened square amplitude, the strongly-screened square amplitude is completely *regular*. As such, there is no need to handle singularities in the development of the kernel or collision terms. We call the kernel associated with $|\mathcal{M}_S|^2$ the ‘ s -kernel’ and symbolise it with $\mathcal{K}_S$. As far as we’re aware, the s -kernel is a novel result of this thesis. Like before, we substitute the strongly-screened square amplitude (1.42) into (1.25), calculate the t -channel contribution and then symmetrise the result to obtain $\mathcal{K}_S$. As a first step, we compute

$$\int_{t_-}^{t_+} dt \frac{1}{\sqrt{Q}} \frac{1}{(t - \mu^2)^2} = -\frac{\pi}{2} \frac{s^2 + d_s s + e_s}{(a_s s^2 + b_s s + c_s)^{3/2}}, \quad (\text{D.1})$$

with

$$d_s = -2(E_1 E_4 + E_2 E_3 + 2m^2 - \mu^2), \quad e_s = 2(2m^2 - \mu^2)(E_3 + E_4)^2.$$

The result in (D.1) can be easily obtained by noticing

$$\frac{1}{(t - \mu^2)^2} = \frac{1}{2\mu} \frac{\partial}{\partial \mu} \left[\frac{1}{t - \mu^2} \right],$$

and appropriately adapting (3.2). The result of the s -integration is most simply expressed in terms of a family of integrals

$$\mathcal{I}_4(k_1, \underline{k}_2, k_3, k_4; \mu) = \int_{s_-}^{s_+} ds \frac{s^4}{\sqrt{\tilde{Q}}^3} = \frac{s^3}{2a_s \sqrt{\tilde{Q}}} \Big|_{s_-}^{s_+} - \frac{5b_s}{4a_s} \mathcal{I}_3(k_1, \underline{k}_2, k_3, k_4; \mu) - \frac{3c_s}{2a_s} \mathcal{I}_2(k_1, \underline{k}_2, k_3, k_4; \mu),$$

$$\begin{aligned} \mathcal{I}_3(k_1, \underline{k}_2, k_3, k_4; \mu) &= \int_{s_-}^{s_+} ds \frac{s^3}{\sqrt{\tilde{Q}}^3} = \frac{a_s(4a_s c_s - b_s^2)s^2 + b_s(10a_s c_s - 3b_s^2)s + c_s(8a_s c_s - 3b_s^2)}{a_s^2(4a_s c_s - b_s^2)\sqrt{\tilde{Q}}} \Big|_{s_-}^{s_+} \dots \\ &\dots - \frac{3b_s}{2a_s^2} \mathcal{I}_0(k_1, \underline{k}_2, k_3, k_4; \mu), \end{aligned}$$

$$\mathcal{I}_2(k_1, \underline{k}_2, k_3, k_4; \mu) = \int_{s_-}^{s_+} ds \frac{s^2}{\sqrt{\tilde{Q}}^3} = -\frac{(4a_s c_s - 2b_s^2)s - 2b_s c_s}{a_s(4a_s c_s - b_s^2)\sqrt{\tilde{Q}}} \Big|_{s_-}^{s_+} + \frac{1}{a_s} \mathcal{I}_0(k_1, \underline{k}_2, k_3, k_4; \mu),$$

$$\mathcal{I}_0(k_1, \underline{k}_2, k_3, k_4; \mu) = \int_{s_-}^{s_+} ds \frac{1}{\sqrt{\tilde{Q}}} = -\frac{1}{\sqrt{-a_s}} \arcsin \left(\frac{b_s + 2a_s s}{\sqrt{b_s^2 - 4a_s c_s}} \right) \Big|_{s_-}^{s_+},$$

where $\mathcal{I}_i$ is named for the i^{th} power of s appearing in its integrand. Using these, we compactly express the s -kernel

$$\mathcal{K}_S(k_1, \underline{k}_2, k_3, k_4; \mu) = -36\pi(4\pi\alpha)^2 \text{sym}[\mathcal{I}_4 + d_s \mathcal{I}_3 + e_s \mathcal{I}_2](k_1, \underline{k}_2, k_3, k_4; \mu).$$

Using (1.35) and (??), we find the condensate integrals

$$C_{[f]}^{1\circ 34} = \frac{18\pi\alpha^2(m+E_1)^2}{mE_1k_1} \int_m^{E_1} dE_3 \left\{ \frac{1}{(E_3-E_1-\mu^2/2m)^2} + \frac{1}{(m-E_3-\mu^2/2m)^2} \right\} \mathcal{F}_{[f]}^{1\circ 34}, \quad (\text{D.2})$$

and

$$C_{[f]}^{12\circ 4} = \frac{18\pi\alpha^2}{mE_1k_1} \int_{E_1}^{\infty} dE_4 (m+E_4)^2 \left\{ \frac{1}{(m-E_1-\mu^2/2m)^2} + \frac{1}{(E_1-E_4-\mu^2/2m)^2} \right\} \mathcal{F}_{[f]}^{12\circ 4}. \quad (\text{D.3})$$

Appendix E: the Fokker-Planck kernel

The small-scattering-angle approximation, which is briefly discussed in [section 1.10](#) is widely used in the literature [[19, 11, 22, 17, 18, 12](#)] as a lightweight alternative to Boltzmann’s collision term (1.5). In this appendix, we summarise for convenience the derivation of the ‘leading-log’ approximation shown in [[11](#)]. We also show our result for the ‘Fokker-Planck kernel’, which corroborates the result shown in [[11](#)] up to a prefactor. We also comment on the numerical scheme used to simulate the Fokker-Planck model.

The small- q expansion

We follow the procedure of [[11](#)], which we now describe in broad strokes. Firstly, recall that the collision term (1.5) is an integral over two-body phase space. Then, re-parametrise the integral over $\mathbf{k}_3$ as an integral over $\mathbf{q} = \mathbf{k}_3 - \mathbf{k}_1$. The pre-existing constraint for momentum conservation is rephrased like $\delta^{(3)}(\mathbf{k}_2 - \mathbf{q} - \mathbf{k}_4)$. By integrating over $\mathbf{k}_4$, we set $\mathbf{k}_4 = \mathbf{k}_2 - \mathbf{q}$ in the remainder of the integrand. There are six remaining integrals

$$C_{[f]}(\mathbf{k}_1) = \int \frac{d^3 k_2}{(2\pi)^3} \int \frac{d^3 q}{(2\pi)^3} \frac{\pi \delta(E_1 + E_2 - E_3 - E_4)}{16E_1 E_2 E_3 E_4} |\mathcal{M}|^2 \mathcal{F}_{[f]}. \quad (\text{E.1})$$

Keep in mind that $\mathbf{q}$ is directly related to Mandelstam t (which is small whenever $\mathbf{q}$ is small). Otherwise, the small- $\mathbf{q}$ expansion is equivalent, through crossing, to a small- $\mathbf{q}'$ expansion where $\mathbf{q}' = \mathbf{k}_4 - \mathbf{k}_1$ (which now captures the influence of small u). Because t and u channels contribute equally to $C_{[f]}$, we simply double the t -contribution to $|\mathcal{M}|^2$ and discard the u -processes. If soft exchanges are dominant, it’s justified to expand the collision term in powers of small $\mathbf{q}$ and keep only the leading contribution. Expanding the statistical factor

$$\mathcal{F}_{[f]}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4) \mapsto \mathbf{q} \cdot \{f_2 \bar{f}_2 \mathbf{grad}_k f_1 - f_1 \bar{f}_1 \mathbf{grad}_k f_2\}, \quad (\text{E.2})$$

where terms $O(q^2)$ are ignored and $f_i = f(\mathbf{k}_i)$, as usual. The small- $\mathbf{q}$ expansion has greatly simplified the statistical factor’s structure. Originally, $\mathcal{F}_{[f]}$ is a function of *all* participating momenta. In the limit of soft scattering, the statistical factor only depends on two internal momenta, $\mathbf{k}_2$ and $\mathbf{q}$ and the external momentum $\mathbf{k}_1$. Moreover, the $\mathbf{q}$ -dependence is easily separable from the $\mathbf{k}_1$ - and $\mathbf{k}_2$ -dependences. The statistical factor’s hanging $\mathbf{q}$ -dependence is packaged into the ‘kernel’.

In the regime of ‘small’ $\mathbf{q}$, collisions cause particles to move between adjacent modes. On the other hand, ‘large’ $\mathbf{q}$ exchanges describe a discontinuous shuffling of particles between modes (i.e. ingoing and outgoing modes aren’t adjacent). To emphasise the continuous flow of particles through momentum space, the collision term is written $C_{[f]} = -\text{div}_k \mathcal{J}$ with div_k being a $\mathbf{k}$ -divergence and $\mathcal{J}$ being the particle number current. By substituting (E.2) into (E.1) and integrating the collision term over a small $\mathbf{k}_1$ -cell, the i^{th} component of $\mathcal{J}$ in a Cartesian system is approximated

$$\mathcal{J}^i(\mathbf{k}_1) = \frac{1}{2} \int \frac{d^3 k_2}{(2\pi)^3} \{f_2 \bar{f}_2 \mathbf{grad}_k f_1 - f_1 \bar{f}_1 \mathbf{grad}_k f_2\} \cdot \int \frac{d^3 q}{(2\pi)^3} w(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}) q^i,$$

where $w(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}) = \pi \delta(E_1 + E_2 - E_3 - E_4) |\mathcal{M}|^2 / 16E_1 E_2 E_3 E_4$. As a first step to calculating the $\mathbf{q}$ -integral, we write E_3 and E_4 as functions of $\mathbf{q}$. Accordingly, we can rewrite $E_1 + E_2 - E_3 - E_4 \mapsto \mathbf{q} \cdot (\mathbf{v}_1 - \mathbf{v}_2) + O(q^2)$. The $\mathbf{v}$ s are 3-velocities. Neglecting higher-order terms, the energy constraint is modified to read $\delta(\mathbf{q} \cdot (\mathbf{v}_1 - \mathbf{v}_2))$. This unfortunately spoils exact energy conservation in the diffusion model [[11](#)]. In practice, however, energy fluctuations remain at the sub-percent level.

Also, the δ -function's modified argument facilitates upcoming integrations. In full, the combination $w(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q})$ is written

$$w(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}) = 36\pi(4\pi\alpha)^2\delta(\mathbf{q} \cdot (\mathbf{v}_1 - \mathbf{v}_2)) \left[\frac{1 - \mathbf{v}_1 \cdot \mathbf{v}_2}{(\mathbf{q} \cdot \mathbf{v})^2 - q^2 - \mu^2} \right]^2, \quad (\text{E.4})$$

where $\mathcal{O}(m^2)$ contributions to the Mandelstam variable s appearing in the square amplitude (1.42) are omitted. Otherwise, deriving (E.4) requires use of $\mathbf{v} = \mathbf{k}/E$ and an expansion of $E_1 E_2 / E_3 E_4$ in powers of small q . Assuming the system is isotropic, the full momentum gradients residing in the statistical factor are exchanged for energy derivatives; $\mathbf{grad}_k f(k) = \mathbf{v} f'(E)$. Passing a factor of $\mathbf{v}_1$ inside the $\mathbf{q}$ -integral, the statistical factor now depends only on E_1 and E_2 . An intermediate expression for $\mathcal{J}^i$ is

$$\mathcal{J}^i(k_1) = \frac{1}{2} \int \frac{d^3 k_2}{(2\pi)^3} \{f_2 \bar{f}_2 f'_1 - f_1 \bar{f}_1 f'_2\} \int \frac{d^3 q}{(2\pi)^3} w(\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}) (\mathbf{q} \cdot \mathbf{v}_1) q^i. \quad (\text{E.5})$$

Because $\mathbf{k}_1$ is the only distinguished direction, $\mathcal{J}$ must be parallel to $\mathbf{k}_1$. Therefore, the index i is dropped and we instead compute the magnitude of $\mathcal{J}$. The collision term $-\partial \mathcal{J} / \partial k_1$ will be phrased as a difference between first- and second-order derivatives of the distribution function f'_1 and f''_1 . The angular parts of the $\mathbf{k}_2$ - and $\mathbf{q}$ -integrals require the specification of a coordinate scheme, which is detailed in [11]. The statistical factor is obviously isotropic, which means these integrals can be passed into the kernel. All integrals except for those over $\mathbf{k}_2$ and $\mathbf{q}$ are organised into a 'proto-kernel', $\mathcal{Q}$ which is also separated from any constant factors

$$\mathcal{J}_{[f]}(k_1) = \frac{18\alpha^2}{\pi} \int dk_2 k_2^2 \{f_2 \bar{f}_2 f'_1 - f_1 \bar{f}_1 f'_2\} \int \frac{dq}{q} \mathcal{Q}(k_1, k_2, q). \quad (\text{E.6})$$

At this point, we see the emergence of a Fokker-Planck structure in (E.6)

$$\partial_t f = \frac{1}{k_1^2} \frac{\partial}{\partial k_1} \left[k_1^2 \left\{ \mathcal{I}_a f'_1 - \mathcal{I}_b f_1 \bar{f}_1 \right\} \right],$$

where

$$\mathcal{I}_a = \frac{18\alpha^2}{\pi} \int dk_2 k_2^2 \bar{f}_2 \int \frac{dq}{q} \mathcal{Q}, \quad \mathcal{I}_b = \frac{18\alpha^2}{\pi} \int dk_2 k_2^2 f'_2 \int \frac{dq}{q} \mathcal{Q}.$$

Perhaps the most important feature of the small- q expansion is that the q -integral is logarithmically divergent (cf. (E.6)). As such, we require meaningful bounds on the domain of the q -integration. In [11], Blaizot et al. verify that the q -integral is infrared finite. However, $q \geq 0$ is a deceptive lower limit for q . As the reader might verify or find in [11], the proto-kernel $\mathcal{Q}$ is strongly suppressed when $q < \mu$. Therefore, to leading-log accuracy, we may bound $q \geq \mu$.

On the other hand, there is no kinematic motivation for the q -integral's upper bound. Hard transfers are still permitted in the Fokker-Planck approximation of the transport equation; they are simply handled incorrectly due to the small- q expansion. Unlike the Boltzmann model (whose kernels are ultraviolet finite), the q -integral in (E.6) diverges logarithmically in Λ_{FP} .

However, the statistical factor provides some recourse. The systems we study naturally have compact support in phase space. Consequently, there exists some hard scale beyond which the distribution function is small and exchanges are statistically suppressed (hard exchanges typically

involve hard gluons). Accordingly, the upper bound of the q -integral ought to be some multiple of the temperature, $\Lambda_{\text{FP}} \propto T$. Estimating Λ_{FP}/T requires a calibration between Fokker-Planck and Boltzmann models.

Treating Λ_{FP} as a parameter which is currently unknown, the q -integration is performed to obtain the ‘Fokker-Planck kernel’, $\mathcal{K}_{\text{FP}}$

$$\mathcal{K}_{\text{FP}}(k_1, k_2; \mu, \Lambda) = \int_{\mu}^{\Lambda_{\text{FP}}} \frac{dq}{q} \mathcal{Q}(k_1, k_2, q), \quad (\text{E.7})$$

The divergence of the kernel against decreasing μ is unsurprising; by dialling in smaller values for the screening mass, we shield less of the singularity in $|\mathcal{M}|^2$. The key difference between the Boltzmann and Fokker-Planck kernels is that $\mathcal{K}_{\text{FP}}$ has an unphysical divergence for increasing Λ .

We now provide more details about the proto-kernel, $\mathcal{Q}$, and our result for the Fokker-Planck kernel (E.7). To extract the proto-kernel, one performs the angular integrals appearing in (E.5). To facilitate the angular integrations, we use the same coordinate convention as [11]. We measure the orientation of $\mathbf{q}$ relative to $\mathbf{k}_1$ using polar and azimuthal angles θ and ϕ . Therefore, $\mathbf{q} \cdot \mathbf{v}_1 = qv_1 \cos \theta$. We then measure the orientation of $\mathbf{k}_2$ relative to $\mathbf{q}$ with polar and azimuthal angles θ_2 and ϕ_2 . The azimuthal angle ϕ_2 is zero when $\mathbf{k}_2$ is in the plane spanned by $\mathbf{q}$ and $\mathbf{k}_1$. Therefore $\mathbf{q} \cdot \mathbf{v}_2 = qv_2 \cos \theta_2$ and $\mathbf{v}_1 \cdot \mathbf{v}_2 = v_1 v_2 (\cos \theta \cos \theta_2 + \sin \theta \sin \theta_2 \cos \phi_2)$ as were originally demonstrated in [11].

With the aforementioned coordinate scheme installed, some elementary integrations yield

$$\begin{aligned} \mathcal{Q}(k_1, k_2, q; \mu, \Lambda) = & \frac{1}{v_1^2 v_2} \left\{ v \left[v^2 - 6(1 - c_q) + (1 - v_1^2) + (1 - v_2^2) + \frac{(1 - c_q)^2}{c_q - v^2} + \frac{(c_q - v_1^2)(c_q - v_2^2)}{2(c_q - v^2)} \right] \dots \right. \\ & \left. \dots + \left[-\frac{(1 - c_q)^2}{c_q} - \frac{(c_q - v_1^2)(c_q - v_2^2)}{2c_q} - (1 - v_1^2) - (1 - v_2^2) + 6(1 - c_q) \right] \sqrt{c_q} \operatorname{artanh}(v/\sqrt{c_q}) \right\}, \end{aligned}$$

where $v = \min(v_1, v_2)$ and $c_q = 1 + \mu^2/q^2$. With $\mathcal{Q}$ specified, the motivation for bounding the q -integral like $\mu \leq q$ becomes clear. When $q < \mu$, the instances of c_q become large and suppress the proto-kernel. Indeed, if one limits $c_q \rightarrow \infty$ (which simulates the limit $q \rightarrow 0$ for finite μ) they should find that $\mathcal{Q} \rightarrow 0$. In other words, the proto-kernel is infrared finite. Because kinematics with $q < \mu$ are suppressed by $\mathcal{Q}$, it’s justified to maintain $\mu \leq q$ in (E.7).

In the limit that $q \rightarrow \infty$ the proto-kernel remains finite and non-zero. However, the integrand in (E.7) doesn’t decay quickly enough for the integral to converge; the q -integral is logarithmically divergent. Because there’s no kinematic reason to neglect hard exchanges, the upper bound of the q -integral is instead motivated by statistics. Hard exchanges typically involve hard particles, so any momentum scale beyond which f is ‘small’ effectively limits the realisable collisions. Hence, the manual installation of an ultraviolet cut-off $\Lambda_{\text{FP}} \propto T$.

Our full result for the Fokker-Planck kernel (E.7) is shown on the next page. Our result for (E.7) differs with the one presented in [11] by a prefactor but does reproduce the correct massless limit documented in [19] and [11]. The overall structure (e.g. location of roots and stationary points) of the collision term when using our result for $\mathcal{K}_{\text{FP}}$ corroborates the results of [11].

With a cut-off in place, the q -integral of (E.7) is performed to find

$$\begin{aligned}
v_1^2 v_2 \mathcal{K}_{\text{FP}}(k_1, k_2; \mu, \Lambda) = & \frac{\mathcal{I}_1}{4} \left\{ \text{artanh}(v/\sqrt{c_\Lambda}) \log(1 + 1/\sqrt{c_\Lambda}) - \frac{1}{2} \left[\log(1 + 1/\sqrt{c_\Lambda}) \log \left(\frac{(1+v)(1+v/\sqrt{c_\Lambda})}{(1-v)(1-v/\sqrt{c_\Lambda})} \right) \dots \right. \right. \\
& \dots + \text{Li}_2 \left(\frac{v(1+1/\sqrt{c_\Lambda})}{v-1} \right) - \text{Li}_2 \left(\frac{v(1+1/\sqrt{c_\Lambda})}{v+1} \right) \left. \right] \Bigg\} - \frac{\mathcal{I}_1}{4} \left\{ \text{artanh}(v/\sqrt{c_\Lambda}) \log(1 - 1/\sqrt{c_\Lambda}) \dots \right. \\
& \dots - \frac{1}{2} \left[\log(1 - 1/\sqrt{c_\Lambda}) \log \left(\frac{(1-v)(1+v/\sqrt{c_\Lambda})}{(1+v)(1-v/\sqrt{c_\Lambda})} \right) - \text{Li}_2 \left(\frac{v(1-1/\sqrt{c_\Lambda})}{v-1} \right) + \text{Li}_2 \left(\frac{v(1-1/\sqrt{c_\Lambda})}{v+1} \right) \right] \Bigg\} \dots \\
& \dots + \frac{1}{12} \left\{ -10v^3 + 9v(-7 + 2v_1^2 + 2v_2^2) - 3\mathcal{I}_1 \text{Li}_2 \left(\frac{v}{v-1} \right) \right\} + 3\mathcal{I}_1 \text{Li}_2 \left(\frac{v}{v+1} \right) + \frac{v}{4(v^2-1)} \left\{ 15(v^2-1)(c_\Lambda-1) \dots \right. \\
& \dots + \log(c_\Lambda-1)(-2v^4 - \mathcal{I}_2 - 2v^2(1-v_1^2-v_2^2)) + \log(c_\Lambda-v^2)(-3v^4 + (4+v_1^2+v_2^2)v^2 - v_1^2v_2^2 - 2) \Bigg\} \dots \\
& \dots + \frac{1}{4} \left\{ 2\text{arctanh}(v/\sqrt{c_\Lambda})\sqrt{c_\Lambda}(5c_\Lambda - 5v^2 + \mathcal{I}_3) + v\mathcal{I}_3 [5c_\Lambda + \log(1-v^2/c_\Lambda) + \log c_\Lambda] \right\},
\end{aligned}$$

where $\mathcal{I}_1 = 3v_1^2 - v_1^2v_2^2 + 3v_2^2 - 5$, $\mathcal{I}_2 = 5 - 3v_1^2 - 2v_2^2$ and $\mathcal{I}_3 = 5v^2 - 3(v_1^2 + v_2^2 - 1)$. The subscript ‘FP’ in Λ_{FP} has been suppressed for brevity.

Coupling to the condensate

If the continuity equation is valid, then the task of tracking n_c can be made very simple. We excavate a radius- ϵ hole in momentum space (with ϵ ‘small’) which is treated as a ‘condensate cell’. The evolution of the condensate, $\partial_t n_c$ is then directly computable from the current

$$\partial_t n_c = 4\pi\epsilon^2 \mathcal{J}_{[f]}(k_1 = \epsilon),$$

due to the continuity of gluon transport. Unlike the Boltzmann model, we don’t explicitly couple the collision term to the condensate. Although we keep track of the distribution function at all momenta, the k_2 -integral in (E.6) is taken over $\epsilon \leq k_2 < \infty$. This effectively excludes all processes where the external particle (with momentum k_1) scatters against a condensate particle (with momentum $k_2 = 0$). We therefore don’t capture the influence of ‘up-scatterings’ – exchanges that kick particles *out* of the zero mode. On the other hand, we certainly do capture processes where infrared particles, $k_1, k_2 \simeq \epsilon$, are pushed into the condensate cell during an exchange with some other thermal particle (k_3 or k_4 may, implicitly, be zero).

Details of the numerics

In our treatment of the Boltzmann model, we tracked a discretised version of the distribution function on an equidistant energy grid. Without using an equidistant energy grid, conservation of n and ε (to within machine precision) is typically violated [14]. The Fokker-Planck approximation, however, doesn't generate a ε -conservative evolution. Therefore, an equidistant energy grid is no longer required. In-line with [11], we use an equidistant momentum grid as the 'basis' to track a discretised distribution function. Up to the inclusion of n_c , we follow the same numerical scheme used in [11].

We begin by specifying a gluon mass, m , population parameter, χ and width σ (which, again, is fixed $\sigma/Q = 0.05$ unless stated otherwise). We then solve the n and ε conservation laws in order to determine the equilibrium state. Even though ε is not conserved by (E.4) [11], it remains within tight-enough bounds to justify the estimated equilibrium parameters. We then set the ultraviolet cut-off as some multiple of T_∞ (T_∞ is the equilibrium temperature obtained by the given initial condition on a continuous and unbounded momentum domain). Knowing the cut-off, and choosing a grid size M , the i^{th} grid point is defined

$$k_i = i\Delta_k \quad 1 \leq i \leq M,$$

where $\Delta_k = \Lambda_{\text{FP}}/M$. This is the so-called 'primary grid' where we keep record of the distribution function. At this point, our treatment of the Fokker-Planck approximation diverges significantly from that of the Boltzmann model. Instead of updating the distribution function directly, we calculate and update the numbers of gluons contained in each mode (i.e. we update n_k and not f_k). Updating n_k is made simple due to the continuity of gluon transport; the increment in the number of particles with $k = k_i$ is simply

$$\delta\tau\partial_\tau\Delta n_k = 4\pi \left\{ (k + \Delta_k/2)^2 \mathcal{J}_{k+\Delta_k/2} - (k - \Delta_k/2)^2 \mathcal{J}_{k-\Delta_k/2} \right\}. \quad (\text{E.8})$$

Once we know the updated value of n_k , we reconstruct the new f_k . Of course, this requires us to find the current at momenta which are *not* on the primary grid. For this purpose, we construct a 'secondary' grid, k'_i , which demarcates the boundaries of each momentum shell. The secondary grid contains $M + 1$ points which are defined

$$k'_i = \begin{cases} \epsilon & i = 1, \\ \Delta_k(i + 1/2) & 2 \leq i \leq M + 1, \end{cases}$$

where ϵ is 'small'. To compute (E.8), we estimate the distribution function and its derivative on the primary grid using b -lines [12]. Using b -lines is convenient because they correctly capture the divergent nature of Bose and Bose-like functions in the infrared. Moreover, they admit a compatible expression for finding the derivatives of f on the secondary grid.

Knowing f and f' on the secondary grid facilitates the calculation of $\mathcal{J}$ at all boundary points. To maintain overall number conservation, we stipulate the current is zero at k'_{M+1} . Otherwise, all particles which are pushed to momenta smaller than ϵ are regarded as being in the condensate. Using the prescription (E.8), we can easily update the numbers of particles contained in each momentum shell and reconstruct (in an n -conservative fashion) the updated distribution function. We continue to use the Fehlberg update procedure described in algorithm 1 to compute the evolution.

Like we see in (E.6), calculating the collision term in the Fokker-Planck approximation is computationally inexpensive compared to finding Boltzmann's collision integral. In the Boltzmann

model, the collision term (evaluated at a given energy or momentum) is approximated with a double sum whose computation time scales like $O(M^2)$. To then compute the collision term at all M energies included in the grid and then simulate N updates to the distribution function is a calculation which scales like $O(M^3N)$. In the Fokker-Planck approximation, we approximate (E.6) with a single sum which is calculated at each grid point. To then simulate N updates to the distribution function is a calculation which scales like $O(M^2N)$. The savings in computation time are significant even on our standard grid of $M = 32$ (more than an order of magnitude).

Appendix F: a dynamical μ

Being a ‘minus first’ moment of the distribution function, the Debye and screening masses (1.46) are subject to change throughout the plasma’s evolution. However, the calculations presented in this thesis do not incorporate a dynamical screening mass (i.e. μ is not updated during the τ -integration). Instead, we estimate μ for the initial condition and keep $\mu = \mu_{\text{in}}$ fixed throughout the evolution.

In this appendix, we quantify the typical variation of μ throughout the evolution. We consider a representative underpopulated initial condition which is time-evolved using a simple RTA (cf. Appendix A). By using the RTA, we obtain a closed-form expression for f at all times. Knowing f , it is straightforward to pick out some representative times and calculate the screening mass at those instants. Figure 34 shows the evolution of the screening mass throughout the evolution.

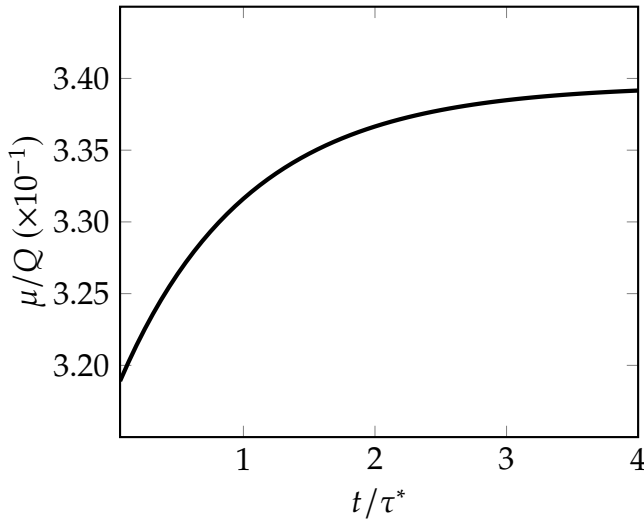


Figure 34: $\mu(\tau/\tau^*)$ for an RTA. The initial condition used in this example has $m/Q = 0.3$, $\chi = 0.1$, $n_c(0) = 0$ and $\alpha = 0.3$. The screening ratio is $\xi = 1$. The equilibrium state has chemical potential $\mu/Q \simeq 0.08$ and temperature $T/Q \simeq 0.25$. The initial screening mass is $\mu/Q|_{t=0} \simeq 0.32$. At equilibrium, the screening mass is larger, $\mu/Q|_{\text{eq}} \simeq 0.34$. Overall, the screening mass grows by some 6% between initial and final states.

Figure 34 indicates that the screening mass changes moderately over the plasma’s evolution, in this case growing by about 6%. In most cases, one finds that the variation in μ over the whole evolution is on the order of 10% (if not smaller). If the evolution of μ is generally monotonic, and within these very tight bounds, then we don’t lose much by omitting a dynamical screening mass from our calculations. In fact, keeping μ constant means that the collision kernels may be constructed just once at the beginning of a simulation. If μ were updated at every time-step (or every other time-step) we would have to re-construct the collision kernels at every such update, prolonging computation time.

There are also conceptual nuances associated with incorporating a dynamical screening mass. The kinematic mass, m , and screening mass, μ , are correlated quantities. If we evolve one, we also ought to evolve the other. However, in a practical application, changing m would also influence the energy or momentum grid upon which the numerical scheme is based. To properly update the energy or momentum domain whilst also preserving conservation of energy and particle number is, we expect, a non-trivial task. We do not target precisions finer than $\sim 10\%$, so we content ourselves with a ‘static’ screening mass.

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